



SAP Cloud Integration

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SAP Cloud Integration | Cloud

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SAP Integration Advisor

SAP Integration Advisor is a cloud application that helps you to simplify and streamline the implementation flow of your B2B/A2A and B2G integration process. It uses a crowd-based machine learning approach to help you create integration content easily. Tests indicate that you can speed up the content creation to deployment process by almost 60% using Integration Advisor. You can also manage and share your content, and leverage the content shared by other application users with similar business needs.

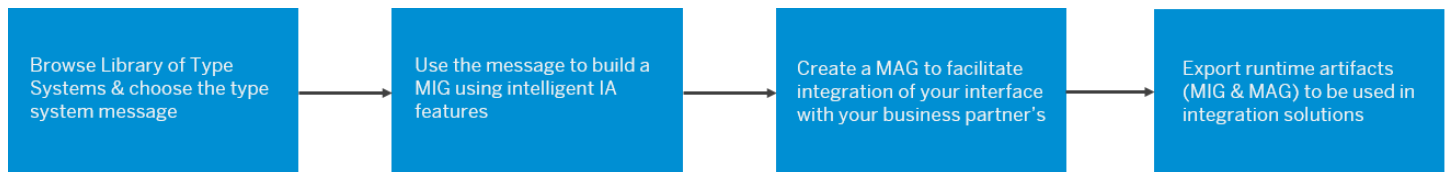
i Note

SAP Integration Advisor is available on the Cloud Foundry (CF) environment. The Interfaces and Mappings designed in Neo environment are also compatible in the CF environment.

Integration Advisor solves the biggest problem that you face in B2B/A2A/B2G integration: multiple business partners who use different industry standards like UN/EDIFACT, SAP IDoc, and ASC X12, to name a few. Each new standard creates a need for a new interface that facilitates this integration, and doing this manually is time consuming. With Integration Advisor, you have a library of type systems that you can use as a starting point. Use the messages in this type system library to create a new message implementation guideline (MIG), create mapping guidelines (MAG) to map the standard you use in your business system to that of your business partner's, and generate runtime artifacts that you can use in different integration solutions like SAP Cloud Integration and SAP Process Orchestration.

Here is a simple overview of your journey to implement a B2B/A2A/B2G integration:

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

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Overview of SAP Integration Advisor

Get an overview of SAP Integration Advisor and its components.

SAP Integration Advisor is a cloud application that simplifies and streamlines the implementation flow of your B2B/A2A and B2G integration process. The crowd-based machine learning approach helps you create integration content easily. Tests indicate that you can speed up the content creation to deployment process by almost 60% using SAP Integration Advisor. You can also manage and share your content, and leverage the content shared by other users of the application with similar business needs.

This is the landing page of the application:

Integration Advisor
Intelligently Simplify SAP Integration

Library of Type Systems

Overview of all available type systems, included messages, simple types, complex types and code lists.

Type System	Messages	Complex Types	Simple Types	Codelists
Accredited Standards Committee X12	325	1256	1626	866
commerce eXtensible Markup Language	78	694	0	247
GS1 EANCOM	48	212	364	169

[More](#)

The Application consists of the following sections:

Type System Overview

The library of type systems is a collection of message templates that are provided by agencies that maintain the B2B/A2A/B2G standards. Each of the type system is developed and maintained by the agency that owns it. For example, the type system SAP IDoc is developed and maintained by SAP SE. Similarly, the type system ASC X12 is maintained by ANSI ASC X12. This is a good starting point for implementing your B2B integration provided if you and your business partners are using these industry standards for your business messages.



Type Systems (12)			Search		
	SAP S/4HANA On Premise IDoc Standard SAP format for electronic data interchange between systems (Intermediate Document). Different message types (such as delivery notes or purchase orders) generally correspond to different special formats, known as IDoc types. Multiple message types with related content can, however, be assigned to a single IDoc type.	Responsible Agency: SAP Provisioned By: SAP License Type: Licensed			☆
	SAP S/4HANA On Premise SOAP SOAP is a standardized, messages-based protocol that is used by WEB Services for communication between the service consumer and the service provider. A SOAP message comprises a header part, which contains communication control parameters, and a body part which contains the payload.	Responsible Agency: SAP Provisioned By: SAP License Type: Licensed			☆
	UN/CEFACT – Trade Facilitation Recommendations UN/CEFACT's trade facilitation recommendations are made to improve the ability of business, trade and administrative organizations, from developed, developing and transitional economies, to exchange products and relevant services effectively. Its principal focus is on facilitating national and international transactions, through the simplification and harmonization of processes, procedures and...	Responsible Agency: UN/ECE Provisioned By: GEFEG License Type: Licensed			☆
	UN/EDIFACT – United Nations Electronic Data Interchange For Administration, Commerce and Transport United Nations/Electronic Data Interchange For Administration, Commerce and Transport (UN/EDIFACT) is the international EDI standard developed under the United Nations. The work of maintenance and further development of this standard is done through the United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT) under the UN Economic Commission for Europe, in the...	Responsible Agency: UN/ECE Provisioned By: GEFEG License Type: Licensed			☆

To know more, see [Library of Type Systems](#)

Custom Type Systems

The library of custom type systems allows you to upload custom messages – this enables you to create MIGs and MAGs based on your own message structures. Use cases for this could be proprietary messages, standard messages extended by you and messages from B2B standards not available in the SAP Integration Advisor. With this feature, you can:

- Upload custom interface xsd files
- Create MIG and MAG
- Customize your MIG
- View the list of your custom messages

To know more, see [Custom Type Systems](#)

Message Implementation Guidelines

Message implementation guideline, also referred to as MIG, is used as the source or target in a mapping guideline (MAG) of SAP Integration Advisor. This is created using one of the messages in the type system that is relevant to your scenario as the template.

The MIG contains all the information for implementing a customized message interface. SAP Integration Advisor uses the message in a type system as a starting point to ensure that you don't have to refer to any additional documents to implement the interface. By providing a MIG, you ensure that all users who are involved in the process of implementing the interface have a clear understanding of the guidelines.

Here are the next steps for you:

- [Creating Message Implementation Guidelines](#)
- [Working with a Message Implementation Guideline](#)

Mapping Guidelines

A mapping guideline is the runtime artifact that you use in the mapping step of an integration application like SAP Cloud Integration. This is used as a reference or guidance for implementing mapping in the integration application. By providing such an artifact, you simplify the mapping task for users who use messages that adhere to the A2A/B2B type system standards.

A Mapping Guideline (MAG) is based on a source and a target message implementation guideline. It demonstrates how the defined nodes at each side are mapped, describing all mapping elements in detail with definitions or notes and providing further instructions for the transformation, such as functions or code value mappings.

Here are the next steps for you:

- [Creating Standard Graphical Mapping Guidelines](#)
- [Working with Mapping Guidelines \(MAGs\)](#)

Related Information

[Blog: Integration Advisor - Overview of components for building B2B integration content and further reading](#) 

Terminology & Glossary for SAP Integration Advisor

Term	Definition
Type System	A "Type System" is an independent system of data description format of reusable message and data types, which is based on a combination of: • Responsible agency • Syntax rules for syntax representation • Schema notation and definition • Methodology for modeling, naming, and structuring of complete messages • Template library Only a single agency (organization ANSI ASC X12 for the Type System ASC X12 or organization SAP SE for the Type System SAP IDoc) is responsible for the development and maintenance of the syntax, concept (methodology) and library.
Message Implementation Guideline (MIG)	A Message Implementation Guideline (also called as Message Guideline or MIG) is a detailed documentation of a A2A/B2B message type considering just these aspects and constraints for fulfilling the needs of the user's business purpose in a given business context.
Mapping Guideline (MAG)	A Mapping Guideline (also called as a MAG) is an artifact that you can directly use in the mapping step of an integration application. A Mapping Guideline (MAG) is based on a source and a target message implementation guideline. It demonstrates how the defined nodes at each side are mapped, describing all mapping elements in detail with definitions or notes and providing further instructions for the transformation, such as functions or code value mappings.
Business Context	The formal description of a specific business circumstance as identified by the values of a set of context categories, allowing different business circumstances to be uniquely distinguished. The business context is a very important part of the MIG. It labels the MIG so that the system can give you an appropriate proposal based on the defined business context and will be used for statistical information, which will be provided in future releases of IA. Therefore, at least one business context value must be set. IA provides for setting the own business context and the business partner's business context five different business context categories with a list of business context values. These business context categories are:

Term	Definition
	• Business Process - In describing a business situation, generally the most important aspect of that situation is the business activity being conducted. Business process context provides a way to unambiguously identify the business activity. • Product Classification - It describes those aspects of a business situation related to the goods or services being exchanged by, or otherwise manipulated, or concerned, in the business process. • Industry Classification - It provides a description of the industry or sub-industry in which the business process takes place. • Geo Political - Allows the description of those aspects of the business context that are related to region, nationality, or geographically based cultural factors. The starting point of the geo political context category is country. • Business Process Role - It describes those aspects of a business situation that are specific to an actor or actors within the business process. The accuracy of the proposal depends on the selected number of business context categories and business context values. As more business context categories and values you selected as tighter will be the proposal.
Message	A message is used to describe business document transactions between two business partners in electronic manner.
Complex Types	Reusable complex structures that are used for assembling a message. Each complex type has a complex structure of child elements for expressing the properties of a complex object such as Partner , Location or Address .
Simple Types	Reusable simple types are defining the characteristics and properties of leaf elements that are used for carrying data in an instance
Codelists	A codelist is a stable collection of code values that's maintained by a responsible agency. Codelists can be used in several ways: 1. A fixed list of commonly used acronyms and abbreviations that provides a common understanding of the names, meanings, or objects that are represented by code values; for example, the country codelist. 2. Shorter representation and controlled list of methods, conditions, or rules, that provides the system further instruction for processing or validating data; for example, the status list, which tells if something is new or in progress. 3. List of qualifying values that provide a more precise meaning to semantically generic elements in an interface; for example, the qualifiers for the generic group Party might be seller or buyer. Codelists require consideration when you use them in MIGs, for the following reasons: • Code values must be mapped from the source side to the target side in a mapping guideline (MAG). • Code values are used as qualifiers that make MIG semantics more precise. • Code values are used when generating, processing, or validating payloads.
Qualifiers	The qualifier reference approach in IA supports the semantic qualification in a very efficient manner. The qualifier references bring both, the semantically generic elements and the leaf elements who provide the additional meaning by code values together. This new approach is very powerful, because it increases the understanding of customized interfaces in MIGs on business meaning level and furthermore, it significantly reduces the efforts in mappings.
Notes	Notes gives further details to the definition and meaning of a node. These details can be categorized by different kinds of notes, which describe usage scenarios, further constraints, a comment or a usage example. The Usage category is used for specification of possible usages of a node. The Constraint category is used to provide information about the qualifiers.

Term	Definition
	The Example category is used to provide example data.
	The Comment category is used to provide addition information.

Initial Setup

Related Information

[Initial Setup of SAP Integration Advisor in Cloud Foundry Environment](#)

[Initial Setup of SAP Integration Advisor in Neo Environment](#)

Initial Setup of SAP Integration Advisor in Cloud Foundry Environment

Related Information

[Creating a Subaccount and Enabling Cloud Foundry](#)

[Subscribing to SAP Integration Advisor](#)

[Assigning Users for SAP Integration Advisor](#)

[Managing Licence for Type Systems](#)

Creating a Subaccount and Enabling Cloud Foundry

Create subaccounts in your global account using the SAP BTP cockpit.

Context

The tenant administrator must perform the procedure mentioned here to enable the SAP Cloud Integration service.

Procedure

1. Log on to the SAP BTP cockpit and navigate to the global account.
2. In the overview page, choose **Subaccounts** from the left pane and choose **New Subaccount** to create a new subaccount.
3. Enter the details in the new subaccount dialog box and save. For more information about creating subaccounts, see [Create Subaccounts using the Cockpit](#).
4. Set the value of **Environment** field to **Cloud Foundry**.
5. After creating the subaccount, choose **Enable Cloud Foundry** to create a **Cloud Foundry Organization** from the subaccount **Overview** menu.

Subscribing to SAP Integration Advisor

Prerequisites

You have obtained a license for SAP Cloud Integration Enterprise Edition.

Context

SAP Integration Advisor is available with the Standard edition and Premium edition of Integration Suite. However, you need to provision the application from SAP BTP cockpit. This procedure guides you on subscribing to SAP Integration Advisor application.

Procedure

1. Login to your SAP BTP cockpit and navigate to your subaccount in Cloud Foundry Environment.
2. In the navigation area of the subaccount, choose **Subscriptions**.

The following information is displayed for the business applications to which your global account is entitled in the Cloud Foundry environment:

- The name and short description of the application.
- **Subscribed / Not subscribed**: The status of the application, indicating whether the subscription is active in your subaccount in the current region.

3. Choose the **Integration Suite** tile to open its **Overview** page.
4. Choose **Subscribe**. Wait for the subscription to complete and once done successfully, you will see the **Integration Suite** tile is shown as **Subscribed**.

i Note

To login to the SAP Cloud Integration application, you have to assign the relevant roles first.

5. Choose **Go to Application** to provision the tenant. If this option is not enabled, refer the chapter [Configuring User Access](#) and assign the necessary role collection.
6. In the resulting **Activate Capabilities** dialog, select the checkbox for **Implement Interfaces and Mapping** to activate Integration Advisor and choose **Next**.
7. Set the necessary values in step 2 **Cloud Integration** and choose **Next**.
8. Review the summary of the capability and choose **Finish**. This will activate Integration Advisor in your tenant.

i Note

You can also **Deactivate** the capability when you no longer need the service, but you will lose all the data uploaded and used in this application along with the deactivation.

Assigning Users for SAP Integration Advisor

Prerequisites

- You have subscribed to SAP Integration Advisor. For more information, see [Subscribing to SAP Integration Advisor](#).
- You are assigned to the **User & Role Administrator** role in the **Security** section of your subaccount.
- The users are stored in identity providers that are connected to SAP BTP.
 - Default identity provider (SAP ID service). For more information, see [SAP ID Service](#).
 - Custom identity provider. For more information, see [User Management](#) in SAP Cloud Identity Services - Identity Authentication.

Context

After subscribing to Integration Advisor from the SAP BTP cockpit, you have to assign the role collection **iadv-content-developer** to the users who want to access the application. This will allow them to use SAP Integration Advisor with the following access rights:

- Discover the different Type systems enabled for the B2B standards
- Design your own custom type system
- Create Message Implementation Guidelines (MIGs) using the type systems
- Simulate and activate your MIGs
- Create interfaces and mappings from these MIGs using Mapping Guidelines (MAGs)
- Simulate and activate your MAGs
- Inject or export these data as Runtime Artifacts
- Export and import your MIGs and MAGs

To know more about managing the role collection, see [Working with Role Collections](#).

Here's how you can do it:

Procedure

1. Log in to the SAP BTP cockpit and navigate to your subaccount.
2. Choose **▮ Security > Role Collections ▮**.
3. Choose the role collection **iadv-content-developer**.
4. Choose **Edit** and navigate to the **Users** section.
5. Enter the ID of the user that you want to assign to the role collection.

i Note

You can assign users from default identity providers, and from custom identity providers, to a role collection. Once you enter the user ID, choose the origin key of the identity provider and the e-mail address. The origin key tells you in which identity provider the user is stored. You can also find that detail in **▮ Security > Trust Configuration ▮**.

If the user only exists in a connected identity provider, you must choose the identity provider and type in the e-mail address.

6. (Optional) To add more users, choose **+**.
7. Choose **Save**.

i Note

If you have logged in to the provisioning application before assigning the role collection, log out and log in again for the changes to reflect in your tenant.

8. You can also add the following role collections as needed by following the steps mentioned above:

→ Remember

For read-only access, assign only the **iadv-content-read** role and remove any existing roles.

Role Collection	Description
iacd-content-administrator	- o iadv-content-developer tasks - o Unlock MIGs and MAGs locked by other users - o Disable Proposal service
iadv-content-read i Note This role is designated for read-only privileges and must be assigned standalone.	- o Read-only access to view the following artefacts: ■ Type system library ■ MIGs ■ MAGs ■ Custom Type Systems - o Simulate MIGs and MAGs - o Export artefacts i Note Users with this role can view and browse these objects, but cannot edit or modify them in any way. This role does not include importing artefacts (MIGs/MAGs).

Managing Licence for Type Systems

Prerequisites

- You have purchased the Integration Advisor B2B Libraries.
- The Integration Advisor B2B Library for the purchased Type System quota has been enabled for the global account.

Context

Follow the procedure to assign the B2B Library Entitlement in your subaccount:

i Note

This procedure is not relevant for SAP Integration Suite users as the licensing of the type systems is provided along with the software subscription.

Procedure

1. Log on to the SAP BTP Cockpit and navigate to the Global account.
2. In the overview page, navigate to **Subaccounts** from the left pane and choose your subaccount.
3. Navigate to **Entitlements** in your subaccount.
4. Choose **Configure Entitlements**.

5. In the resulting page, choose [Add Service Plans](#).
6. Search for and choose [SAP Integration Advisor B2B Library](#).
7. Select your purchased Type Systems under the [Available Plans](#) and choose [Add Service Plan](#).
8. Choose [Save](#).

Initial Setup of SAP Integration Advisor in Neo Environment

Related Information

[Subscribing to SAP Integration Advisor](#)
[Assigning Users for SAP Integration Advisor](#)
[Managing License for Type Systems](#)

Subscribing to SAP Integration Advisor

Prerequisites

You have obtained a license for SAP Cloud Integration Enterprise Edition.

Context

Integration Advisor is available with the enterprise edition of SAP Cloud Integration. However, you need to provision the application from SAP BTP cockpit. This procedure guides you on subscribing to Integration Advisor application.

→ Remember

Please assign user roles after subscribing to SAP Integration Advisor by following the guidance here: [Assigning Users for SAP Integration Advisor](#).

Procedure

1. Log on to the SAP BTP cockpit.

→ Tip

You can find the URL to the SAP BTP cockpit in the welcome mail that you received from SAP after purchasing the license for SAP Cloud Integration.

2. Choose **Services** > **Process Integration** >

If you do not see the tile **Process Integration**, perform steps 1 and 2 in [Activating Enterprise Messaging](#).

3. If the service is not enabled, choose **Enable**. If it is already enabled, you can proceed to the next step.
4. Choose **Configure Process Integration** in the **Take Action** screen area of the **Process Integration** service page.

You see the provisioning self-service application launched in another browser window or tab.

If you encounter 403 HTTP error, please double-check if you have performed all the instructions in step 2 of [Activating Enterprise Messaging](#).

5. Choose **Integration Advisor**.

6. Choose **Activate** to start the service.

You see a URL in the **Details** section. You can use this URL to launch Integration Advisor application. This completes the subscription to **Integration Advisor**.

Next Steps

As mentioned earlier, please assign the user roles based on your needs as described here: [Assigning Users for SAP Integration Advisor](#).

Assigning Users for SAP Integration Advisor

Prerequisites

You have subscribed to SAP Integration Advisor. For more information, see [Subscribing to SAP Integration Advisor](#).

You have the authgroup, **Authgroup.Administrator** assigned to your user. This is mandatory for being able to assign users for the application. If you do not have this role, please contact your administrator for assigning users or obtaining the administrator role.

Context

After subscribing to Integration Advisor from the SAP BTP cockpit, you have to assign the roles **Guidelines.ReadWrite** and **TypeSystem.Read** to the users who want to access the application. This will allow them to use SAP Integration Advisor.

Here's how you can do it:

Procedure

1. Log in to the SAP BTP cockpit.
2. Choose **Subscriptions**.

You see a list of the Java applications and HTML5 applications that you have subscribed to.

3. In the **Subscribed Java Applications** table, choose **icaapp**.
4. Perform the following substeps to assign a role to a user:

- a. Choose the role that you want to assign to a user or a group.

You can choose the role **Guidelines.ReadWrite** or **TypeSystem.Read** from the **Roles** table.

i Note

- If you want to assign admin role to an user, then add the role **Guidelines.Administrator** to the user.
 - If you want to assign read-only access to an user, then add the role **Guidelines.Read** to the user. This role allows user to view, simulate and export all artefacts (MIGs, MAGs, Type Systems, Custom Type Systems) with no editing and importing privileges.
- b. If you want to assign the role to an individual user, choose **Assign** in the **Individual Users** screen area. Enter the ID of the user and choose **Assign**.
 - c. If you want to assign the role to a group, choose **Assign** in the **Groups** screen area. Select the group from the dropdown list and choose **Assign**.

You can manage groups by choosing **Security > Authorizations > Groups** after logging into the SAP BTP cockpit. For more information on managing users and groups, see [Authorizing Users or User Groups](#).

Managing License for Type Systems

i Note

This procedure is not relevant for SAP Integration Suite users as the licensing of the type systems is provided along with the software subscription.

You can access Integration Advisor when you purchase an Enterprise Edition license for SAP Cloud Integration. However, you need to purchase licenses for the type systems used in SAP Integration Advisor, like [X12](#) or [EDIFACT](#) separately.

For information about purchasing these type system licenses, please raise an ticket under the component [LOD-HCI-PI-SUG](#).

Update or verify the license you have purchased on SAP store as follows:

1. Log on to Integration Advisor and access the [Profile](#) section.
2. The [Licenses](#) tab list the type systems and their license information.

i Note

To update the license information, choose  *Synchronize license purchases with your profile*.

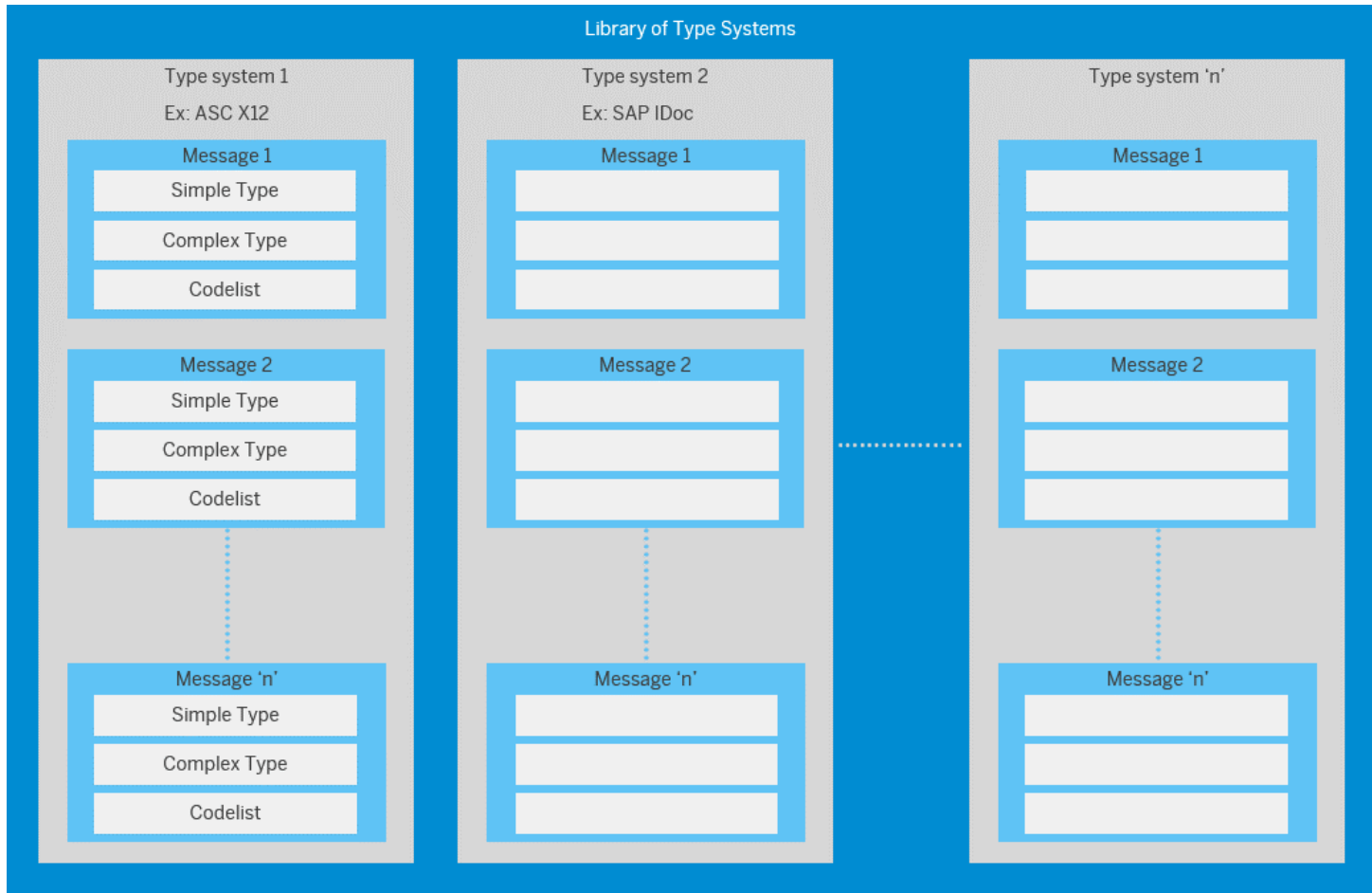
i Note

In case of a subaccount move, check for the type system licenses and raise an incident under the component [LOD-HCI-PI-SUG](#) to move the licenses to your new subaccount.

Library of Type Systems

The library of type systems is a collection of message templates that are provided by agencies that maintain the B2B/A2A/B2G standards. Each of the type system is developed and maintained by the agency that owns it. For example, the type system [SAP IDoc](#) is developed and maintained by SAP SE. Similarly, the type system ASC X12 is maintained by ANSI ASC X12. This is a very good starting point for implementing your B2B integration provided if you and your business partners are using these industry standards for your business messages.

A typical type system in the library is comprised of a collection of messages, and associated complex types, simple types, and code lists. Refer to [Terminology & Glossary for SAP Integration Advisor](#) to get an understanding of Messages, Complex Types, Simple Types and Codelists.



These agencies provide message templates that you can use as the starting point for creating your own message implementation guideline (MIG). You can tailor the MIG to suit your scenario-specific requirements. You can then use the MIG in a mapping guideline (MAG) and create a mapping that allows integration with your business partners.

For more information on library of type systems, their representation and uses in the application, see the blog [Integration Content Advisor: Discover B2B/A2A standard libraries](#).

Overview Of B2B Standards

Get to know the list of available B2B standards, the supported message types and versions in SAP Integration Advisor.

The B2B standards are grouped as follows:

- EDI standards
 - [ASC X12](#)
 - [Automotive EDIFACT](#)
 - [EANCOM](#)
 - [Odette](#)
 - [Tradacoms](#)
 - [UN/EDIFACT](#)
 - [VDA](#)
- XML standards

- [cXML](#)
 - [GS1 XML](#)
 - [xCBL](#)
- Codelist standards
 - [Automotive Codelists](#)
 - [GS1 Codelists](#)
 - [ISO Codelists](#)
 - [UN/CEFACT Recommendations](#)

ASC X12

ASC X12 (also known as ANSI ASC X12) is an Electronic Data Interchange (EDI) standard and the message types are created and maintained by the Accredited Standards Committee (ASC) X12.

Versions List (40)

Version
002001
002002
002003
002040
002042
003010
003020
003030
003040
003050
003060
003070
004010
004020
004030
004040
004050
004060
005010

Version
005020
005030
005040
005050
006010
006020
006030
006040
006050
007010
007020
007030
007040
007050
007060
008010
008020
008030
008040
008050
008060

Message Types (326)

Message Type	Available from Version	Last Version (if discontinued)
100 - Insurance Plan Description	004010	
101 - Name and Address Lists	003070	
102 - Associated Data	004020	
103 - Abandoned Property Filings	004020	
104 - Air Shipment Information	003020	
105 - Business Entity Filings	004010	
106 - Motor Carrier Rate Proposal	004010	

Message Type	Available from Version	Last Version (if discontinued)
107 - Request for Motor Carrier Rate Proposal	004010	
108 - Response to a Motor Carrier Rate Proposal	004010	
109 - Vessel Content Details	004010	
110 - Air Freight Details and Invoice	003020	
	004020	
112 - Property Damage Report	004010	
113 - Election Campaign and Lobbyist Reporting	004030	
114 - Air Shipment Status Message111 - Individual Insurance Policy and Client Information	003020	003030
120 - Vehicle Shipping Order	003050	
121 - Vehicle Service	003050	
124 - Vehicle Damage	003050	
125 - Multilevel Railcar Load Details	003040	
126 - Vehicle Application Advice	003040	
127 - Vehicle Baying Order	003040	
128 - Dealer Information	003040	
129 - Vehicle Carrier Rate Update	003040	
130 - Student Educational Record (Transcript)	003030	
111 - Individual Insurance Policy and Client131 - Student Educational Record (Transcript) Acknowledgment	003030	
132 - Human Resource Information	004040	
133 - Educational Institution Record	004040	
135 - Student Aid Origination Record	003030	
138 - Educational Testing and Prospect Request and Report	004010	
139 - Student Loan Guarantee Result	003030	
140 - Product Registration	003030	
141 - Product Service Claim Response	003030	
142 - Product Service Claim	003030	

Message Type	Available from Version	Last Version (if discontinued)
143 - Product Service Notification	003030	
144 - Student Loan Transfer and Status Verification	003030	
146 - Request for Student Educational Record (Transcript)	003030	
147 - Response to Request for Student Educational Record (Transcript)	003030	
148 - Report of Injury, Illness or Incident	003040	
149 - Notice of Tax Adjustment or Assessment	003060	
150 - Tax Rate Notification	003050	
151 - Electronic Filing of Tax Return Data Acknowledgment	003030	
152 - Statistical Government Information	003030	
153 - Unemployment Insurance Tax Claim or Charge Information	003070	
154 - Secured Interest Filing	003040	
155 - Business Credit Report	003070	
157 - Notice of Power of Attorney	004010	
158 - Tax Jurisdiction Sourcing	004060	
159 - Motion Picture Booking Confirmation	003060	
160 - Transportation Automatic Equipment Identification	003070	
161 - Train Sheet	003030	
163 - Transportation Appointment Schedule Information	003060	
170 - Revenue Receipts Statement	003020	
175 - Court and Law Enforcement Notice	003050	
176 - Court Submission	003050	
179 - Environmental Compliance Reporting	004040	
180 - Return Merchandise Authorization and Notification	003030	
185 - Royalty Regulatory Report	003050	
186 - Insurance Underwriting Requirements Reporting	003040	
187 - Premium Audit Request and Return	004020	

Message Type	Available from Version	Last Version (if discontinued)
188 - Educational Course Inventory	003060	
189 - Application for Admission to Educational Institutions	003060	
190 - Student Enrollment Verification	003030	
191 - Student Loan Pre-Claims and Claims	003050	
194 - Grant or Assistance Application	003060	
195 - Federal Communications Commission (FCC) License Application	003050	
196 - Contractor Cost Data Reporting	003040	
197 - Real Estate Title Evidence	003060	
198 - Loan Verification Information	003070	
199 - Real Estate Settlement Information	003060	
200 - Mortgage Credit Report	003050	
201 - Residential Loan Application	003050	
202 - Secondary Mortgage Market Loan Delivery	004010	
203 - Secondary Mortgage Market Investor Report	003050	
204 - Motor Carrier Load Tender	003020	
205 - Mortgage Note	003070	
206 - Real Estate Inspection	003060	
210 - Motor Carrier Freight Details and Invoice	003020	
211 - Motor Carrier Bill of Lading	003070	
212 - Motor Carrier Delivery Trailer Manifest	003070	
213 - Motor Carrier Shipment Status Inquiry	003020	
214 - Transportation Carrier Shipment Status Message	003020	
215 - Motor Carrier Pickup Manifest	003070	
216 - Motor Carrier Shipment Pickup Notification	003070	
217 - Motor Carrier Loading and Route Guide	003020	
218 - Motor Carrier Tariff Information	003020	004040

Message Type	Available from Version	Last Version (if discontinued)
219 - Logistics Service Request	004010	
220 - Logistics Service Response	004010	
222 - Cartage Work Assignment	004010	
223 - Consolidators Freight Bill and Invoice	004010	
224 - Motor Carrier Summary Freight Bill Manifest	004010	
225 - Response to a Cartage Work Assignment	004010	
227 - Trailer Usage Report	004030	
228 - Equipment Inspection Report	005010	
240 - Motor Carrier Package Status	004020	
242 - Data Status Tracking	003050	
244 - Product Source Information	003070	
245 - Real Estate Tax Service Response	004030	
248 - Account Assignment/Inquiry and Service/Status	003070	
249 - Animal Toxicological Data	003070	
250 - Purchase Order Shipment Management Document	003040	
251 - Pricing Support	003040	
252 - Insurance Producer Administration	003070	
255 - Underwriting Information Services	003060	
256 - Periodic Compensation	003060	
259 - Residential Mortgage Insurance Explanation of Benefits	004060	
260 - Application for Mortgage Insurance Benefits	003040	
261 - Real Estate Information Request	003060	
262 - Real Estate Information Report	003050	
263 - Residential Mortgage Insurance Application Response	003030	
264 - Mortgage Loan Default Status	003040	
265 - Real Estate Title Insurance Services Order	003050	

Message Type	Available from Version	Last Version (if discontinued)
266 - Mortgage or Property Record Change Notification	003050	
267 - Individual Life, Annuity and Disability Application	004010	
268 - Annuity Activity	003070	
269 - Health Care Benefit Coordination Verification	004040	
270 - Eligibility, Coverage or Benefit Inquiry	003040	
271 - Eligibility, Coverage or Benefit Information	003040	
272 - Property and Casualty Loss Notification	003040	
273 - Insurance/Annuity Application Status	003060	
274 - Healthcare Provider Information	004020	
275 - Patient Information	003060	
276 - Health Care Claim Status Request	003040	
277 - Health Care Information Status Notification	003040	
278 - Health Care Services Review Information	003050	
280 - Voter Registration Information	003070	
283 - Tax or Fee Exemption Certification	004020	
284 - Commercial Vehicle Safety Reports	004020	
285 - Commercial Vehicle Safety and Credentials Information Exchange	003070	
286 - Commercial Vehicle Credentials	004010	
288 - Wage Determination	003060	
290 - Cooperative Advertising Agreements	003040	
300 - Reservation (Booking Request) (Ocean)	003020	
301 - Confirmation (Ocean)	003020	
303 - Booking Cancellation (Ocean)	003020	
304 - Shipping Instructions	003020	
309 - Customs Manifest	003020	
310 - Freight Receipt and Invoice (Ocean)	003020	

Message Type	Available from Version	Last Version (if discontinued)
311 - Canada Customs Information	003030	
312 - Arrival Notice (Ocean)	003020	
313 - Shipment Status Inquiry (Ocean)	003020	
315 - Status Details (Ocean)	003020	
317 - Delivery/Pickup Order	003030	
319 - Terminal Information	003030	
322 - Terminal Operations and Intermodal Ramp Activity	003020	
323 - Vessel Schedule and Itinerary (Ocean)	003020	
324 - Vessel Stow Plan (Ocean)	003020	
325 - Consolidation of Goods In Container	003030	
326 - Consignment Summary List	003030	
350 - Customs Status Information	003020	
352 - Customs Carrier General Order Status	003030	
353 - Customs Events Advisory Details	003020	
354 - Customs Automated Manifest Archive Status	003020	
355 - Customs Acceptance/Rejection	003030	
356 - Customs Permit to Transfer Request	003030	
357 - Customs In-Bond Information	003050	
358 - Customs Consist Information	003050	
359 - Customs Customer Profile Management	005040	
361 - Carrier Interchange Agreement (Ocean)	003020	
362 - Cargo Insurance Advice of Shipment	003050	
404 - Rail Carrier Shipment Information	003020	
410 - Rail Carrier Freight Details and Invoice	003020	
412 - Trailer or Container Repair Billing	004030	
414 - Rail Carhire Settlements	003020	
417 - Rail Carrier Waybill Interchange	003020	
418 - Rail Advance Interchange Consist	003020	

Message Type	Available from Version	Last Version (if discontinued)
419 - Advance Car Disposition	003030	
420 - Car Handling Information	003030	
421 - Estimated Time of Arrival and Car Scheduling	003030	
422 - Equipment Order	003030	
423 - Rail Industrial Switch List	003050	
424 - Rail Carrier Services Settlement	004050	
425 - Rail Waybill Request	003030	
426 - Rail Revenue Waybill	003020	
429 - Railroad Retirement Activity	003020	
431 - Railroad Station Master File	003020	
432 - Rail Car Hire Rate Negotiation	003050	
433 - Railroad Reciprocal Switch File	003050	
434 - Railroad Mark Register Update Activity	003070	
435 - Standard Transportation Commodity Code Master	003050	
436 - Locomotive Information	003060	
437 - Railroad Junctions and Interchanges Activity	003070	
440 - Shipment Weights	003030	
451 - Railroad Event Report	003050	
452 - Railroad Problem Log Inquiry or Advice	003050	
453 - Railroad Service Commitment Advice	003050	
455 - Railroad Parameter Trace Registration	003050	
456 - Railroad Equipment Inquiry or Advice	003050	
460 - Railroad Price Distribution Request or Response	003070	
463 - Rail Rate Reply	003070	
466 - Rate Request	003020	
468 - Rate Docket Journal Log	003020	
470 - Railroad Clearance	003070	
475 - Rail Route File Maintenance	003050	

Message Type	Available from Version	Last Version (if discontinued)
485 - Ratemaking Action	003020	
486 - Rate Docket Expiration	003070	
490 - Rate Group Definition	003020	
492 - Miscellaneous Rates	003020	
494 - Rail Scale Rates	003020	
500 - Medical Event Reporting	003070	
501 - Vendor Performance Review	003050	
503 - Pricing History	003050	
504 - Clauses and Provisions	003050	
511 - Requisition	003030	
517 - Material Obligation Validation	003030	
521 - Income or Asset Offset	003070	
527 - Material Due-In and Receipt	003030	
536 - Logistics Reassignment	003030	
540 - Notice of Employment Status	003070	
561 - Contract Abstract	003030	
567 - Contract Completion Status	003030	
568 - Contract Payment Management Report	003040	
601 - Customs Export Shipment Information	003030	
602 - Transportation Services Tender	003030	005050
603 - Transportation Equipment Registration	006030	
620 - Excavation Communication	004010	
622 - Intermodal Ramp Activity	003030	003050
625 - Well Information	004010	
650 - Maintenance Service Order	003070	
715 - Intermodal Group Loading Plan	003060	
753 - Request for Routing Instructions	004030	
754 - Routing Instructions	004030	
805 - Contract Pricing Proposal	003030	

Message Type	Available from Version	Last Version (if discontinued)
806 - Project Schedule Reporting	003030	
810 - Invoice	002001	
811 - Consolidated Service Invoice/Statement	003020	
812 - Credit/Debit Adjustment	003020	
813 - Electronic Filing of Tax Return Data	003030	
814 - General Request, Response or Confirmation	003070	
815 - Cryptographic Service Message	003020	
816 - Organizational Relationships	003040	
818 - Commission Sales Report	003030	
819 - Joint Interest Billing and Operating Expense Statement	002003	
820 - Payment Order/Remittance Advice	002001	
821 - Financial Information Reporting	002040	
822 - Account Analysis	002003	
823 - Lockbox	002002	
824 - Application Advice	002042	
826 - Tax Information Exchange	002042	
827 - Financial Return Notice	002042	
828 - Debit Authorization	003020	
829 - Payment Cancellation Request	002042	
830 - Planning Schedule with Release Capability	002001	
831 - Application Control Totals	003040	
832 - Price/Sales Catalog	002001	
833 - Mortgage Credit Report Order	003030	
834 - Benefit Enrollment and Maintenance	003030	
835 - Health Care Claim Payment/Advice	003020	
836 - Procurement Notices	003010	
837 - Health Care Claim	003030	
838 - Trading Partner Profile	003020	
839 - Project Cost Reporting	003030	

Message Type	Available from Version	Last Version (if discontinued)
840 - Request for Quotation	002001	
841 - Specifications/Technical Information	003010	
842 - Nonconformance Report	003010	
843 - Response to Request for Quotation	002001	
844 - Product Transfer Account Adjustment	002003	
845 - Price Authorization Acknowledgment/Status	002003	
846 - Inventory Inquiry/Advice	002001	
847 - Material Claim	003040	
848 - Material Safety Data Sheet	003020	
849 - Response to Product Transfer Account Adjustment	002003	
850 - Purchase Order	002001	
851 - Asset Schedule	003020	
852 - Product Activity Data	003010	
853 - Routing and Carrier Instruction	003030	
854 - Shipment Delivery Discrepancy Information	003030	
855 - Purchase Order Acknowledgment	002001	
856 - Ship Notice/Manifest	002001	
857 - Shipment and Billing Notice	003020	
858 - Shipment Information	002040	
859 - Freight Invoice	002042	
860 - Purchase Order Change Request - Buyer Initiated	002001	
861 - Receiving Advice/Acceptance Certificate	002001	
862 - Shipping Schedule	002002	
863 - Report of Test Results	002003	
864 - Text Message	002040	
865 - Purchase Order Change Acknowledgment/Request - Seller Initiated	002001	
866 - Production Sequence	002040	
867 - Product Transfer and Resale Report	002001	

Message Type	Available from Version	Last Version (if discontinued)
868 - Electronic Form Structure	002042	
869 - Order Status Inquiry	002002	
870 - Order Status Report	002002	
871 - Component Parts Content		005010
872 - Residential Mortgage Insurance Application	003020	
873 - Commodity Movement Services	004030	
874 - Commodity Movement Services Response	004040	
875 - Grocery Products Purchase Order	003040	
876 - Grocery Products Purchase Order Change	003040	
877 - Manufacturer Coupon Family Code Structure	003060	
878 - Product Authorization/De-authorization	003020	
879 - Price Information	003020	
880 - Grocery Products Invoice	003040	
881 - Manufacturer Coupon Redemption Detail	003060	
882 - Direct Store Delivery Summary Information	003030	
883 - Market Development Fund Allocation	003050	
884 - Market Development Fund Settlement	003050	
885 - Retail Account Characteristics	003050	
886 - Customer Call Reporting	003050	
887 - Coupon Notification	003060	
888 - Item Maintenance	003020	
889 - Promotion Announcement	003020	
890 - Contract & Rebate Management Transaction	006010	
891 - Deduction Research Report	003050	
892 - Trading Partner Performance Measurement	007020	
893 - Item Information Request	003030	

Message Type	Available from Version	Last Version (if discontinued)
894 - Delivery/Return Base Record	003020	
895 - Delivery/Return Acknowledgment or Adjustment	003020	
896 - Product Dimension Maintenance	003020	
897 - Data Synchronization	007060	
920 - Loss or Damage Claim - General Commodities	003040	
924 - Loss or Damage Claim - Motor Vehicle	003030	
925 - Claim Tracer	003030	
926 - Claim Status Report and Tracer Reply	003030	
928 - Automotive Inspection Detail	003030	
940 - Warehouse Shipping Order	003020	
943 - Warehouse Stock Transfer Shipment Advice	003020	
944 - Warehouse Stock Transfer Receipt Advice	003020	
945 - Warehouse Shipping Advice	003020	
947 - Warehouse Inventory Adjustment Advice	003020	
980 - Functional Group Totals	003020	
990 - Response to a Load Tender	003020	
993 - Secured Receipt or Acknowledgment	004020	
996 - File Transfer	003020	
997 - Functional Acknowledgment	002001	
998 - Set Cancellation	003030	
999 - Implementation Acknowledgment	005010	
Envelope – Envelope with Functional Groups of Messages i Note This message is specifically designed to capture and define the envelope structure.	002001	

Automotive EDIFACT

JAIF EDIFACT

JAIF EDIFACT is a subset of the UN/EDIFACT standard from the JAIF organization (Joint Automotive Initiative Forum) and is used in the global automotive industry.

Versions List (7)

Version	Based on Edifact Version
D.03A S3	D.03A S3
D.04A S3	D.04A S3
D.04B S3	D.04B S3
D.07A S3	D.07A S3
D.09B S3	D.09B S3
D.13A S3	D.13A S3
D.16A S3	D.16A S3

Message Types

The JAIF EDIFACT Message Standard contains the same list of Messages Types as defined by the underlying UN/EDIFACT version. In difference to EANCOM, the JAIF organization did not explicitly distinguish between UN/EDIFACT messages they use and don't use.

To look up the available Message Types for a specific version, you can directly look it up in the according section for UN/EDIFACT.

Codelists

Codelists for this Automotive EDIFACT subset were not defined within the Message Standard. Instead you will find 5 separate Automotive Codelist standards which contain all the required codelists from these organizations.

Reason for providing these codelists as separate libraries was to allow the mixed usage of them and the benefit of separating the lifecycle (versioning) of message standard and codelists.

Message Guidelines

JAIF EDIFACT (like many other UN/EDIFACT subsets) distinguishes between two types of libraries:

- Message Standards: The library of Messages, Segments, Data Elements and Codelists which are required for the subset.
- Message Guidelines: Specific Guidelines for countries and business processes which are defined in the context of the JAIF EDIFACT subset.

The application currently provides the JAIF EDIFACT Message Standard upon which you can build your own Message Implementation Guidelines (MIGs).

VDA EDIFACT

VDA EDIFACT is a subset of the UN/EDIFACT standard from the VDA organization (Verband der Automobilindustrie Germany) and is used in the German automotive industry.

Versions List (13)

Version	Based on Edifact Versio
D.96A S3	D.96A S3
D.97B S3	D.97B S3
D.03A S3	D.03A S3
D.04A S3	D.04A S3
D.05B S3	D.05B S3
D.06A S3	D.06A S3
D.07A S3	D.07A S3
D.08A S3	D.08A S3
D.13A S3	D.13A S3
D.14B S3	D.14B S3
D.16A S3	D.14B S3
D.18A S3	D.18A S3
D.20B S3	D.20B S3

Message Types

The VDA EDIFACT Message Standard contains the same list of Messages Types as defined by the underlying UN/EDIFACT version. In difference to EANCOM, the VDA organization did not explicitly distinguish between UN/EDIFACT messages they use and don't use.

To look up the available Message Types for a specific version, you can directly look it up in the according section for UN/EDIFACT.

Codelists

Codelists for this Automotive EDIFACT subset were not defined within the Message Standard. Instead you will find 5 separate Automotive Codelist standards which contain all the required codelists from these organizations.

Reason for providing these codelists as separate libraries was to allow the mixed usage of them (such as to using an ODETTE Codelist in a VDA Guideline) and the benefit of separating the lifecycle (versioning) of message standard and codelists.

Message Guidelines

VDA EDIFACT (like many other UN/EDIFACT subsets) distinguishes between two types of libraries:

- Message Standards: The library of Messages, Segments, Data Elements and Codelists which are required for the subset.
- Message Guidelines: Specific Guidelines for countries and business processes which are defined in the context of the VDA EDIFACT subset.

The application currently provides the VDA EDIFACT Message Standard upon which you can build your own Message Implementation Guidelines (MIGs).

ODETTE EDIFACT

Odette EDIFACT is a subset of the UN/EDIFACT standard from the Odette organization and is used in the European automotive industry.

Versions List (7)

Version	Based on Edifact Version
D.96A	D.96A S3
D.98B	D.98B S3
D.03A	D.03A S3
D.05B	D.05B S3
D.07A	D.07A S3
D.16A	D.16A S3
D.18B	D.18B S3

Message Types

The ODETTE EDIFACT Message Standard contains the same list of Messages Types as defined by the underlying UN/EDIFACT version. In difference to EANCOM, the Odette organization did not explicitly distinguish between UN/EDIFACT messages they use and don't use.

To look up the available Message Types for a specific version, you can directly look it up in the according section for UN/EDIFACT.

Codelists

Codelists for this Automotive EDIFACT subset were not defined within the Message Standard. Instead you will find 5 separate Automotive Codelist standards which contain all the required codelists from these organizations.

Reason for providing these codelists as separate libraries was to allow the mixed usage of them (such as to using an ODETTE Codelist in a VDA Guideline) and the benefit of separating the lifecycle (versioning) of message standard and codelists.

Message Guidelines

ODETTE EDIFACT (like many other UN/EDIFACT subsets) distinguishes between two types of libraries:

- Message Standards: The library of Messages, Segments, Data Elements and Codelists which are required for the subset.
- Message Guidelines: Specific Guidelines for countries and business processes which are defined in the context of the ODETTE EDIFACT subset.

The application currently provides the ODETTE EDIFACT Message Standard upon which you can build your own Message Implementation Guidelines (MIGs).

EANCOM

GS1 EANCOM® is a GS1 subset of the UN/EDIFACT standard (United Nations Electronic Data Interchange for Administration, Commerce and Transport). It contains the message elements required by business applications especially for domains supported by the GS1 organizations such as retail, consumer products, transport and logistics.

Versions List (11)

Version	Based on Edifact Version
1995 S3	D.93A S3
1997 S3	D.96A S3
1997 S3 Edition 1999	D.96A S3
1997 S3 Edition 2001	D.96A S3
2002 S3	D.01B S3
2002 S3 Edition 2008	D.01B S3
2002 S3 Edition 2010	D.01B S3
2002 S3 Edition 2012	D.01B S3
2002 S3 Edition 2014	D.01B S3
2002 S3 Edition 2016	D.01B S3
2002 S3 Edition 2016 Update 2021	D.01B S3

Message Types (48)

Message	Available from Version	Last Version (if discontinued)
APERAK - Application error and acknowledgement message	1997 S3 Edition 1999	
BANSTA - Banking status message	1997 S3	
CNTCND - Contractual conditions message	1997 S3 Edition 1999	
COACSU - Commercial account summary message	1997 S3	
COMDIS - Commercial dispute message	1997 S3	
CONDRA - Drawing administration message	1997 S3 Edition 1999	
CONTRL - Syntax and service report message	1995 S3	
CREMUL - Multiple credit advice message	1997 S3	
DEBMUL - Multiple debit advice message	1997 S3	
DELFOR - Delivery schedule message	1995 S3	
DESADV - Despatch advice message	1995 S3	
DIRDEB - Direct debit message	1997 S3 Edition 1999	

Message	Available from Version	Last Version (if discontinued)
EnvelopeForMessagesS3 - Envelope for Messages (Syntax Versions 1/2/3) i Note This message is specifically designed to capture and define the envelope structure.	1995 S3	
EnvelopeWithFunctionalGroupsS3 – Envelope with Functional Groups of Messages (Syntax Versions 1/2/3) i Note This message is specifically designed to capture and define the envelope structure.	1995 S3	
FINCAN - Financial cancellation message	1997 S3	
FINSTA - Financial statement of an account message	1997 S3	
GENRAL - General purpose message	1995 S3	
HANMOV - Cargo/goods handling and movement message	1997 S3	
IFCSUM - Forwarding and consolidation summary message	1997 S3	
IFTMAN - Arrival notice message	1997 S3	
IFTMBC - Booking confirmation message	1997 S3	
IFTMBF - Firm booking message	1997 S3	
IFTMIN - Instruction message	1995 S3	
IFTSTA - International multimodal status report message	1997 S3	
INSDDES - Instruction to despatch message	1997 S3	
INVOIC - Invoice message	1995 S3	
INVRPT - Inventory report message	1995 S3	
MSCONS - Metered services consumption report message	1997 S3	
ORDCHG - Purchase order change request message	1995 S3	
ORDERS - Purchase order message	1995 S3	
ORDRSP - Purchase order response message	1995 S3	

Message	Available from Version	Last Version (if discontinued)
OSTENQ - Order status enquiry message	1997 S3	
OSTRPT - Order status report message	1997 S3	
PARTIN - Party information message	1995 S3	
PAYDUC - Payroll deductions advice message	1997 S3 Edition 1999	1997 S3 Edition 2001
PAYMUL - Multiple payment order message	1997 S3	
PRICAT - Price/sales catalogue message	1995 S3	
PRODAT - Product data message	1997 S3	
PROINQ - Product inquiry message	1997 S3	
QALITY - Quality data message	1997 S3	
QUOTES - Quote message	1995 S3	
RECADV - Receiving advice message	1995 S3	
REMADV - Remittance advice message	1995 S3	
REQOTE - Request for quote message	1995 S3	
RETANN - Announcement for returns message	1997 S3	
RETINS - Instruction for returns message	1997 S3	
SLSFCT - Sales forecast message	1995 S3	
SLSRPT - Sales data report message	1995 S3	
STATAC - Statement of account message	1995 S3	1995 S3
TAXCON - Tax control message	1995 S3	

The following table shows the EANCOM version codes that are supported for each message types and their respective release versions.

Message Types and EANCOM Codes

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
APERAK	EAN001	1997 S3 Edition 1999	1997 S3 Edition 1999
APERAK	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
APERAK	EAN003	2002 S3	
BANSTA	EAN001	1997 S3	1997 S3 Edition 1999
BANSTA	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
BANSTA	EAN003	2002 S3	
CNTCND	EAN001	1997 S3 Edition 1999	1997 S3 Edition 1999

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
CNTCND	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
CNTCND	EAN003	2002 S3	
COACSU	EAN002	1997 S3	1997 S3 Edition 1999
COACSU	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
COACSU	EAN004	2002 S3	
COMDIS	EAN001	1997 S3	1997 S3 Edition 1999
COMDIS	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
COMDIS	EAN003	2002 S3	
CONDRA	EAN001	1997 S3 Edition 1999	1997 S3 Edition 1999
CONDRA	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
CONDRA	EAN003	2002 S3	2002 S3
CONDRA	EAN004	2002 S3 Edition 2008	
CONTRL	EAN001	1995 S3	1995 S3
CONTRL	EAN002	1997 S3	1997 S3 Edition 1999
CONTRL	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
CONTRL	EAN004	2002 S3	
CREMUL	EAN001	1997 S3	1997 S3 Edition 1999
CREMUL	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
CREMUL	EAN003	2002 S3	
DEBMUL	EAN001	1997 S3	1997 S3 Edition 1999
DEBMUL	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
DEBMUL	EAN003	2002 S3	
DELFOR	EAN001	1995 S3	1995 S3
DELFOR	EAN002	1997 S3	1997 S3 Edition 1999
DELFOR	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
DELFOR	EAN004	2002 S3	2002 S3 Edition 2014
DELFOR	EAN005	2002 S3 Edition 2016	
DESADV	EAN004	1995 S3	1995 S3
DESADV	EAN005	1997 S3	1997 S3 Edition 1999
DESADV	EAN006	1997 S3 Edition 2001	1997 S3 Edition 2001
DESADV	EAN007	2002 S3	2002 S3 Edition 2014

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
DESADV	EAN008	2002 S3 Edition 2016	
DIRDEB	EAN001	1997 S3 Edition 1999	1997 S3 Edition 1999
DIRDEB	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
DIRDEB	EAN003	2002 S3	
FINCAN	EAN001	1997 S3	1997 S3 Edition 1999
FINCAN	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
FINCAN	EAN003	2002 S3	
FINSTA	EAN001	1997 S3	1997 S3 Edition 1999
FINSTA	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
FINSTA	EAN003	2002 S3	
GENRAL	EAN002	1995 S3	1997 S3
GENRAL	EAN003	1997 S3 Edition 1999	1997 S3 Edition 1999
GENRAL	EAN004	1997 S3 Edition 2001	1997 S3 Edition 2001
GENRAL	EAN005	2002 S3	
HANMOV	EAN002	1997 S3	1997 S3 Edition 1999
HANMOV	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
HANMOV	EAN004	2002 S3	
IFCSUM	EAN001	1997 S3	1997 S3 Edition 1999
IFCSUM	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
IFCSUM	EAN003	2002 S3	2002 S3
IFCSUM	EAN004	2002 S3 Edition 2008	
IFTMAN	EAN001	1997 S3	1997 S3 Edition 1999
IFTMAN	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
IFTMAN	EAN003	2002 S3	
IFTMBC	EAN001	1997 S3	1997 S3 Edition 1999
IFTMBC	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
IFTMBC	EAN003	2002 S3	
IFTMBF	EAN001	1997 S3	1997 S3 Edition 1999
IFTMBF	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
IFTMBF	EAN003	2002 S3	
IFTMIN	EAN001	1995 S3	1995 S3

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
IFTMIN	EAN002	1997 S3	1997 S3
IFTMIN	EAN003	1997 S3 Edition 1999	1997 S3 Edition 2001
IFTMIN	EAN004	2002 S3	
IFTSTA	EAN002	1997 S3	1997 S3 Edition 1999
IFTSTA	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
IFTSTA	EAN004	2002 S3	
INSDDES	EAN001	1997 S3	1997 S3 Edition 1999
INSDDES	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
INSDDES	EAN003	2002 S3	
INVOIC	EAN007	1995 S3	1995 S3
INVOIC	EAN008	1997 S3	1997 S3 Edition 1999
INVOIC	EAN009	1997 S3 Edition 2001	1997 S3 Edition 2001
INVOIC	EAN010	2002 S3	2002 S3
INVOIC	EAN011	2002 S3 Edition 2008	
INVRPT	EAN003	1995 S3	1995 S3
INVRPT	EAN004	1997 S3	1997 S3 Edition 1999
INVRPT	EAN005	1997 S3 Edition 2001	1997 S3 Edition 2001
INVRPT	EAN006	2002 S3	2002 S3
INVRPT	EAN007	2002 S3 Edition 2008	
MSCONS	EAN002	1997 S3	1997 S3 Edition 1999
MSCONS	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
MSCONS	EAN004	2002 S3	
ORDCHG	EAN004	1995 S3	1995 S3
ORDCHG	EAN005	1997 S3	1997 S3 Edition 1999
ORDCHG	EAN006	1997 S3 Edition 2001	1997 S3 Edition 2001
ORDCHG	EAN007	2002 S3	
ORDERS	EAN007	1995 S3	1995 S3
ORDERS	EAN008	1997 S3	1997 S3 Edition 1999
ORDERS	EAN009	1997 S3 Edition 2001	1997 S3 Edition 2001
ORDERS	EAN010	2002 S3	
ORDRSP	EAN004	1995 S3	1995 S3

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
ORDRSP	EAN005	1997 S3	1997 S3 Edition 1999
ORDRSP	EAN006	1997 S3 Edition 2001	1997 S3 Edition 2001
ORDRSP	EAN007	2002 S3	2002 S3
ORDRSP	EAN008	2002 S3 Edition 2008	2002 S3 Edition 2008
ORDRSP	EAN009	2002 S3 Edition 2010	
OSTENQ	EAN002	1997 S3	1997 S3 Edition 1999
OSTENQ	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
OSTENQ	EAN004	2002 S3	
OSTRPT	EAN002	1997 S3	1997 S3 Edition 1999
OSTRPT	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
OSTRPT	EAN004	2002 S3	2002 S3
OSTRPT	EAN005	2002 S3 Edition 2008	
PARTIN	EAN005	1995 S3	1995 S3
PARTIN	EAN006	1997 S3	1997 S3 Edition 1999
PARTIN	EAN007	1997 S3 Edition 2001	1997 S3 Edition 2001
PARTIN	EAN008	2002 S3	2002 S3 Edition 2010
PARTIN	EAN009	2002 S3 Edition 2012	
PAYDUC	EAN001	1997 S3 Edition 1999	1997 S3 Edition 1999
PAYDUC	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
PAYMUL	EAN001	1997 S3	1997 S3 Edition 1999
PAYMUL	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
PAYMUL	EAN003	2002 S3	
PRICAT	EAN005	1995 S3	1995 S3
PRICAT	EAN006	1997 S3	1997 S3 Edition 1999
PRICAT	EAN007	1997 S3 Edition 2001	1997 S3 Edition 2001
PRICAT	EAN008	2002 S3	2002 S3
PRICAT	EAN009	2002 S3 Edition 2008	
PRODAT	EAN002	1997 S3	1997 S3 Edition 1999
PRODAT	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
PRODAT	EAN004	2002 S3	
PROINQ	EAN002	1997 S3	1997 S3 Edition 1999

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
PROINQ	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
PROINQ	EAN004	2002 S3	
QALITY	EAN001	1997 S3	1997 S3 Edition 1999
QALITY	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
QALITY	EAN003	2002 S3	
QUOTES	EAN001	1995 S3	1995 S3
QUOTES	EAN002	1997 S3	
QUOTES	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
QUOTES	EAN004	2002 S3	
RECADV	EAN002	1995 S3	1995 S3
RECADV	EAN003	1997 S3	1997 S3 Edition 1999
RECADV	EAN004	1997 S3 Edition 2001	1997 S3 Edition 2001
RECADV	EAN005	2002 S3	2002 S3
RECADV	EAN006	2002 S3 Edition 2008	2002 S3 Edition 2008
RECADV	EAN007	2002 S3 Edition 2010	2002 S3 Edition 2010
RECADV	EAN008	2002 S3 Edition 2012	
REMADV	EAN002	1995 S3	1997 S3
REMADV	EAN003	1997 S3 Edition 1999	1997 S3 Edition 1999
REMADV	EAN004	1997 S3 Edition 2001	1997 S3 Edition 2001
REMADV	EAN005	2002 S3	
REQOTE	EAN001	1995 S3	1995 S3
REQOTE	EAN002	1997 S3	1997 S3 Edition 1999
REQOTE	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
REQOTE	EAN004	2002 S3	
RETANN	EAN001	1997 S3	1997 S3 Edition 1999
RETANN	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
RETANN	EAN003	2002 S3	
RETINS	EAN001	1997 S3	1997 S3 Edition 1999
RETINS	EAN002	1997 S3 Edition 2001	1997 S3 Edition 2001
RETINS	EAN003	2002 S3	
SLSFCT	EAN003	1995 S3	1995 S3

Message	EANCOM Version Code	Available from Version	Last Version (if discontinued)
SLSFCT	EAN004	1997 S3	1997 S3 Edition 1999
SLSFCT	EAN005	1997 S3 Edition 2001	1997 S3 Edition 2001
SLSFCT	EAN006	2002 S3	
SLSRPT	EAN003	1995 S3	1995 S3
SLSRPT	EAN004	1997 S3	1997 S3 Edition 1999
SLSRPT	EAN005	1997 S3 Edition 2001	1997 S3 Edition 2001
SLSRPT	EAN006	2002 S3	2002 S3
SLSRPT	EAN007	2002 S3 Edition 2008	
STATAC	No Code assigned by EANCOM	1995 S3	1995 S3
TAXCON	EAN001	1995 S3	1995 S3
TAXCON	EAN002	1997 S3	1997 S3 Edition 1999
TAXCON	EAN003	1997 S3 Edition 2001	1997 S3 Edition 2001
TAXCON	EAN004	2002 S3	

The EANCOM versioning scheme (as defined by the GS1 organisation) consists of three parts:

- EANCOM Releases/Version (e.g. 1997 S3 Edition 1999)
- Underlying Standard Edifact Version (e.g. D.96A S3)
- EANCOM Version Code (e.g. EAN008)

For example, the preceding table shows that ORDERS EAN008 can be implemented by either EANCOM Release **1997 S3** or **1997 S3 Edition 1999**. Both are defined on top of the underlying Standard Edifact Release D.96A S3.

i Note

The EANCOM version code like EAN008 is a message-specific version code. For example, ORDERS EAN008 belongs to EANCOM Release **1997 S3** (based on Edifact D.96A S3), while DESADV EAN008 is part of EANCOM Release **2002 S3 Edition 2016** (based on Edifact D.01B S3).

Odette

The Odette Message Standard consists of the early Odette messages and codelists which were defined by Odette in the 1980s and 1990s and are used in the European automotive industry.

The newer Odette offerings are based directly on Edifact and form a so called Edifact subset. You find these Odette Edifact messages under the section Automotive Edifact.

Version

The Odette Message Standard used a different versioning strategy - It did not publish complete release versions like other standards (see e.g. UN/EDIFACT). Odette versioned all their messages and segments separately.

Due to this versioning strategy and the fact that no updates to the Odette messages are expected, you will find all definitions together in one version called [All](#).

Message Types (28)

Message Types

Message Type
AVIEXP:3 - Despatch Advice (Version 3)
AVIEXP:4 - Despatch Advice
AVIGRU:2 - Grouping Centre Despatch Advice
AVIREX:2 - Receival Discrepancy Advice
BASDAT:2 - Basic Data
CONTRL:1 - Acknowledgement or Rejection Advice
CONTRL:2 - Acknowledgement or Rejection Advice
CREDIT:2 - Credit Note
DEBNOT:2 - Debit Note
DELINS:3 - Delivery Instruction
ENGDAT:1 - Engineering Data Message
ENQIRY:2 - Enquiry
EnvelopeForMessagesS3 - Envelope for Messages (Syntax Versions 1/2/3)
i Note This message is specifically designed to capture and define the envelope structure.
FORDIS:2 - Forecast of Despatch
INVOIC:3 - Invoice
KANBAN:2 - KANBAN
OFFERR:2 - Offer
ORDELI:1 - Delivery Order
ORDERR:1 - Order
ORDERR:2 - Order
OSTENQ:1 - Order Status Enquiry
PRILST:1 - Price List
REMADV:2 - Remittance Advice
REPDEL:2 - Reply to Delivery Instruction
REPINV:2 - Reply to Invoice
REPORT:2 - Reply to Order

Message Type
STATAC:2 - Statement of Account
STOACT:1 - Stock Activities
SYNCRO:2 - Sequenced Delivery
SYNPAC:2 - Sequenced Packaged Delivery

The Odette delivery contains 26 different message types. For most of the message types only the newest version is provided. For example, INVOIC:3 represents Version 3 of the Invoice message. For the message types CONTRL and ORDERR both versions 1 and 2 are available.

i Note

Despatch Advice message structure remains unchanged between AVIEXP Versions 3 and 4. However, several nodes' requirements were updated from mandatory to conditional. Despite these changes, Odette did not revise segment version numbers. Consequently, segments display AVIEXP:4 configurations for AVIEXP:3. You can optionally update these fields to mandatory in your Message Implementation Guideline (MIG), if necessary.

Tradacoms

GS1 TRADACOMS was the core UK message standard from the early days of EDI in the UK retail industry. It was widely used by retailers and wholesalers for EDI order and invoice processes between buyers and their mainly UK-based suppliers. However, since 1998, existing users have only received basic TRADACOMS support from GS1 UK for current applications and there has been no development of any new message types. Since 1 July 2017, GS1 UK has no longer provided any support for TRADACOMS. All users have been advised to adopt the recognized GS1 EDI standards EANCOM, GS1 XML and the GS1 UN/CEFACT XML profile.

Version

The Tradacoms Message Standard used a different versioning strategy. It did not publish complete release versions like other standards (see e.g. UN/EDIFACT). Tradacoms versioned all their messages separately.

Due to this versioning strategy and the fact that no updates to the Tradacoms messages are expected, you will find all definitions together in one version called **All**.

Message Types

Tradacoms contains **Transmission Files** which are a predefined combination of single message types. For example, the **ORDHDR_FILE** Transmission File is the combination of ORDHDR, ORDERS and ORDTLR message types. The Transmission Files are available under the **Messages** tab. The single message types (such as ORDHDR, ORDERS, ORDTLR) are not offered as they can't be used separately by themselves.

The Tradacoms delivery contains 25 different message types. For most of the message types only the newest version is available. For the message types **CORHDR_FILE**, **DLCHDR_FILE**, **INVFIL_FILE**, **ORDHDR_FILE**, **SRMHDR_FILE** and **UTLHDR_FILE** two versions are available. For example, **ORDHDR_FILE:9** represents Version 9 of the Order file and **INVFIL_FILE:8** represents Version 8 of the Invoice file.

The following conventions are used to define Identifier and Version number of the Transmission Files by:

- The Header Message of the Transmission File determines the Identifier.
 - This convention was derived from the Tradacoms Codelist 16 and the use of these code values in envelope field STX > APRF.

- Moreover, **_FILE** was added to the Identifier to distinguish between the overall Transmission File and its Header Message.
- The Version of a Transmission File is consistently derived from the version of the Main Message.

For example: The older transmission file for Order (Order File) consists of three single messages: header message **ORDHDR** (Version 6), main message **ORDERS** (Version 8) and trailer message **ORDTLR** (Version 4). Applying the rules mentioned before, this Order File will be **ORDHDR_FILE:8 – Order File (Version 8)**.

i Note

The identifiers and versions of the Transmission Files are not defined by Tradacoms standard (by GS1 organization) itself. Different users and data providers have applied different conventions due to which sometimes transmission files are named after the main message and sometimes versions of the transmission files are derived from the version of their respective header message. For example, the older Order File (see before) might also be called **ORDERS (Version 8)** or **ORDHDR (Version 6)**.

Message Types (31)

Transmission Files	Included Single Messages	Remark
ACKHDR_FILE: 4 - Acknowledgement File (Version 4)	• ACKHDR (Version 4) • ACKMNT (Version 4) • ACKTLR (Version 4) • RSGRSG (Version 2)	
AVLHDR_FILE: 4 - Availability File (Version 4)	• AVLHDR (Version 4) • AVLDET (Version 4) • AVLTLR (Version 4) • RSGRSG (Version 2)	
CORHDR_FILE: 4 - Complex Order File (Version 4)	• CORHDR (Version 4) • CORDER (Version 4) • CORTLR (Version 4) • RSGRSG (Version 2)	
CORHDR_FILE: 6 - Complex Order File (Version 6)	• CORHDR (Version 6) • CORDER (Version 6) • CORTLR (Version 6) • RSGRSG (Version 2)	

Transmission Files	Included Single Messages	Remark
CRAHDR_FILE: 3 - Credit Advice File (Version 3)	• CRAHDR (Version 3) • CRAINP (Version 3) • CRATLR (Version 3) • RSGRSG (Version 2)	
CREHDR_FILE: 9 - Credit Note File (Version 9)	• CREHDR (Version 9) • CREDIT (Version 9) • VATTLR (Version 9) • CRETLR (Version 9) • RSGRSG (Version 2)	
CUSHDR_FILE: 8 - Customer Information File (Version 8)	• CUSHDR (Version 8) • CUSINF (Version 8) • CUSTLR (Version 8) • RSGRSG (Version 2)	
DELHDR_FILE: 9 - Delivery File (Version 9)	• DELHDR (Version 9) • DELIVR (Version 9) • DELTLR (Version 9) • RSGRSG (Version 2)	
DLCHDR_FILE: 4 - Delivery Confirmation File (Version 4)	• DLCHDR (Version 4) • DLCDET (Version 4) • DLCTLR (Version 4) • RSGRSG (Version 2)	
DLCHDR_FILE: 5 - Delivery Confirmation File (Version 5)	• DLCHDR (Version 5) • DLCDET (Version 5) • DLCTLR (Version 5) • RSGRSG (Version 2)	

Transmission Files	Included Single Messages	Remark
DRAHDR_FILE: 3 - Debit Advice File (Version 3)	• DRAHDR (Version 3) • DRAINF (Version 3) • DRATLR (Version 3) • RSGRSG (Version 2)	
EXCHDR_FILE: 3 - Exception Condition File (Version 3)	• EXCHDR (Version 3) • EXCINF (Version 3) • EXCTLR (Version 3) • RSGRSG (Version 2)	
GENHDR_FILE: 3 - General Communications File (Version 3)	• GENHDR (Version 3) • GENRAL (Version 3) • GENTLR (Version 3) • RSGRSG (Version 2)	
INVFIL_FILE: 8 - Invoice File (Version 8)	• INVFIL (Version 6) • INVOIC (Version 8) • VATTLR (Version 6) • INVTLR (Version 5) • RSGRSG (Version 2)	Also known as Version 6 based on version of header message INVFIL.
INVFIL_FILE: 9 - Invoice File (Version 9)	• INVFIL (Version 9) • INVOIC (Version 9) • VATTLR (Version 9) • INVTLR (Version 9) • RSGRSG (Version 2)	
LPRHDR_FILE: 2 - Location Planning Report File (Version 2)	• LPRHDR (Version 2) • LPRDET (Version 2) • LPRTLR (Version 2) • RSGRSG (Version 2)	
ORDHDR_FILE: 8 - Order File (Version 8)	• ORDHDR (Version 6) • ORDERS (Version 8) • ORDTLR (Version 4)	Also known as Version 6 based on version of header message ORDHDR.

Transmission Files	Included Single Messages	Remark
	• RSGRSG (Version 2)	
ORDHDR_FILE: 9 - Order File (Version 9)	• ORDHDR (Version 9) • ORDERS (Version 9) • ORDTLR (Version 9) • RSGRSG (Version 2)	
PAYHDR_FILE: 3 - Payment Order File (Version 3)	• PAYHDR (Version 3) • PAYINF (Version 3) • PAYTLR (Version 3) • RSGRSG (Version 2)	
PICHDR_FILE: 4 - Picking Instructions File (Version 4)	• PICHDR (Version 4) • PICKER (Version 4) • PICTLR (Version 4) • RSGRSG (Version 2)	
PPRHDR_FILE: 2 - Product Planning Report File (Version 2)	• PPRHDR (Version 2) • PPRDET (Version 2) • PPRTLRL (Version 2) • RSGRSG (Version 2)	
PRIHDR_FILE: 8 - Price File (Version 8)	• PRIHDR (Version 8) • PRIINF (Version 8) • PRITLR (Version 8) • RSGRSG (Version 2)	
PROHDR_FILE: 8 - Product File (Version 8)	• PROHDR (Version 8) • PROINF (Version 8) • PROTLR (Version 8) • RSGRSG (Version 2)	

Transmission Files	Included Single Messages	Remark
SADHDR_FILE: 3 - Stock Adjustment File (Version 3)	• SADHDR (Version 3) • SADDET (Version 3) • SADTLR (Version 3) • RSGRSG (Version 2)	
SNPHDR_FILE: 3 - Stock Snapshot File (Version 3)	• SNPHDR (Version 3) • SNPSTS (Version 3) • SNPTLR (Version 3) • RSGRSG (Version 2)	
SRMHDR_FILE: 7 - Statement/Remittance File (Version 7)	• SRMHDR (Version 6) • SRMINF (Version 7) • SRMTLR (Version 4) • RSGRSG (Version 2)	Also known as Version 6 based on version of header message SRMHDR.
SRMHDR_FILE: 9 - Statement/Remittance File (Version 9)	• SRMHDR (Version 9) • SRMINF (Version 9) • SRMTLR (Version 9) • RSGRSG (Version 2)	
UCNHDR_FILE: 3 - Uplift Confirmation File (Version 3)	• UCNHDR (Version 3) • UCNDDET (Version 3) • UCNTLR (Version 3) • RSGRSG (Version 2)	
UPLHDR_FILE: 4 - Uplift Instruction File (Version 4)	• UPLHDR (Version 4) • UPLIFT (Version 4) • UPLTLR (Version 4) • RSGRSG (Version 2)	Currently dispalyed as Uplift Confirmation File (Version 4) in the system.

Transmission Files	Included Single Messages	Remark
UTLHDR_FILE: 2 - Utility Bill File (Version 2)	• UTLHDR (Version 2) • UTLBIL (Version 2) • UVATLR (Version 2) • UTLTLR (Version 2) • RSGRSG (Version 2)	
UTLHDR_FILE: 3 - Utility Bill File (Version 3)	• UTLHDR (Version 3) • UTLBIL (Version 3) • UVATLR (Version 3) • UTLTLR (Version 3) • RSGRSG (Version 2)	

UN/EDIFACT

SAP Integration Advisor supports the libraries for syntax version 2 (S2) and syntax version 3 (S3).

Versions

Versions List (63)

Version
90.2 S2
91.1 S2
91.2 S2
92.1 S3
93.2 S3
D.93A S3
D.94A S3
D.94B S3
D.95A S3
D.95B S3
D.96A S3
D.96B S3
D.97A S3
D.97B S3
D.98A S3

Version
D.98B S3
D.99A S3
D.99B S3
D.00A S3
D.00B S3
D.01A S3
D.01B S3
D.01C S3
D.02A S3
D.02B S3
D.03A S3
D.03B S3
D.04A S3
D.04B S3
D.05A S3
D.05B S3
D.06A S3
D.06B S3
D.07A S3
D.07B S3
D.08A S3
D.08B S3
D.09A S3
D.09B S3
D.10A S3
D.10B S3
D.11A S3
D.11B S3
D.12A S3
D.12B S3
D.13A S3

Version
D.13B S3
D.14A S3
D.14B S3
D.15A S3
D.15B S3
D.16A S3
D.16B S3
D.17A S3
D.17B S3
D.18A S3
D.18B S3
D.19A S3
D.19B S3
D.20A S3
D.20B S3
D.21A S3
D.21B S3

Message Types

Message Types Since Version D.93A S3 (200)

Message	Available from Version	Last Version (if discontinued)
APERAK - Application error and acknowledgment message	D.95A S3	
AUTHOR - Authorization message	D.94A S3	
BALANC - Balance message	D.98A S3	
BANSTA - Banking status message	D.93A S3	
BAPLIE - Bayplan/stowage plan occupied and empty locations message	D.93A S3	
BAPLTE - Bayplan/stowage plan total numbers message	D.93A S3	D.02B S3
BERMAN - Berth management message	D.00A S3	
BMISRM - Bulk marine inspection summary report message	D.98B S3	

Message	Available from Version	Last Version (if discontinued)
BOPBNK - Bank transactions and portfolio transactions report message	D.94B S3	
BOPCUS - Balance of payment customer transaction report message	D.94B S3	
BOPDIR - Direct balance of payment declaration message	D.94B S3	
BOPINF - Balance of payment information from customer message	D.94B S3	
BUSCRD - Business credit report message	D.99B S3	
CALINF - Vessel call information message	D.95A S3	
CASINT - Request for legal administration action in civil proceedings message	D.96B S3	
CASRES - Legal administration response in civil proceedings message	D.96B S3	
CHACCO - Chart of accounts message	D.98A S3	
CLASET - Classification information set message	D.99A S3	
CNTCND - Contractual conditions message	D.99A S3	
COACSU - Commercial account summary message	D.99B S3	
COARRI - Container discharge/loading report message	D.95A S3	
CODECO - Container gate-in/gate-out report message	D.95A S3	
CODENO - Permit expiration/clearance ready notice message	D.95A S3	
COEDOR - Transport equipment stock and profile report message	D.95A S3	
COHAOR - Container special handling order message	D.95A S3	
COLREQ - Request for a documentary collection message	D.97B S3	
COMDIS - Commercial dispute message	D.94A S3	
CONAPW - Advice on pending works message	D.94A S3	
CONDPV - Direct payment valuation message	D.93A S3	
CONDRA - Drawing administration message	D.95A S3	

Message	Available from Version	Last Version (if discontinued)
CONDRO - Drawing organisation message	D.95A S3	
CONEST - Establishment of contract message	D.93A S3	
CONITT - Invitation to tender message	D.93A S3	
CONPVA - Payment valuation message	D.93A S3	
CONQVA - Quantity valuation message	D.93A S3	
CONRPW - Response of pending works message	D.94A S3	
CONTEN - Tender message	D.93A S3	
CONTRL - Syntax and service report message	D.99B S3	
CONWQD - Work item quantity determination message	D.94A S3	
COPARN - Container announcement message	D.95A S3	
COPAYM - Contributions for payment	D.99B S3	
COPINO - Container pre-notification message	D.95A S3	
COPRAR - Container discharge/loading order message	D.95A S3	
COREOR - Container release order message	D.95A S3	
COSTCO - Container stuffing/stripping confirmation message	D.95A S3	
COSTOR - Container stuffing/stripping order message	D.95A S3	
CREADV - Credit advice message	D.93A S3	
CREEXT - Extended credit advice message	D.93A S3	
CREMUL - Multiple credit advice message	D.95A S3	
CUSCAR - Customs cargo report message	D.93A S3	
CUSDEC - Customs declaration message	D.93A S3	
CUSEXP - Customs express consignment declaration message	D.94A S3	
CUSPED - Periodic customs declaration message	D.97B S3	
CUSREP - Customs conveyance report message	D.93A S3	

Message	Available from Version	Last Version (if discontinued)
CUSRES - Customs response message	D.93A S3	
DAPLOS - Data Plot Sheet	D.05A S3	
DEBADV - Debit advice message	D.93A S3	
DEBMUL - Multiple debit advice message	D.95A S3	
DEBREC - Debts recovery message	D.99B S3	
DELFOR - Delivery schedule message	D.93A S3	
DELJIT - Delivery just in time message	D.93A S3	
DESADV - Despatch advice message	D.93A S3	
DESTIM - Equipment damage and repair estimate message	D.96B S3	
DGRECA - Dangerous goods recapitulation message	D.97A S3	
DIRDEB - Direct debit message	D.93A S3	
DIRDEF - Directory definition message	D.95B S3	
DMRDEF - Data maintenance request definition message	D.98B S3	
DMSTAT - Data maintenance status report/query message	D.98B S3	
DOCADV - Documentary credit advice message	D.93A S3	
DOCAMA - Advice of an amendment of a documentary credit message	D.95A S3	
DOCAMI - Documentary credit amendment information message	D.95A S3	
DOCAMR - Request for an amendment of a documentary credit message	D.95A S3	
DOCAPP - Documentary credit application message	D.93A S3	
DOCARE - Response to an amendment of a documentary credit message	D.95A S3	
DOCINF - Documentary credit issuance information message	D.93A S3	
ENTREC - Accounting entries message	D.98A S3	
EnvelopeForMessagesS3 – Envelope for Messages (Syntax Versions 1/2/3)	D.93A S3	
i Note		

Message	Available from Version	Last Version (if discontinued)
This message is specifically designed to capture and define the envelope structure.		
EnvelopeWithFunctionalGroupsS3 – Envelope with Functional Groups of Messages (Syntax Versions 1/2/3) i Note This message is specifically designed to capture and define the envelope structure.	D.93A S3	
FINCAN - Financial cancellation message	D.95A S3	
FINPAY - Multiple interbank funds transfer message	D.98A S3	
FINSTA - Financial statement of an account message	D.95A S3	
GENRAL - General purpose message	D.96A S3	
GESMES - Generic statistical message	D.95A S3	
GOVCBR - Government Cross Border Regulatory message	D.09A S3	
HANMOV - Cargo/goods handling and movement message	D.94A S3	
ICASRP - Insurance claim assessment and reporting message	D.01A S3	
ICSOLI - Insurance claim solicitor's instruction message	D.00B S3	
IFCSUM - Forwarding and consolidation summary message	D.93A S3	
IFTCCA - Forwarding and transport shipment charge calculation message	D.93A S3	
IFTDGN - Dangerous goods notification message	D.94A S3	
IFTFCC - International transport freight costs and other charges message	D.96A S3	
IFTIAG - Dangerous cargo list message	D.95A S3	D.02B S3
IFTICL - Cargo insurance claims message	D.00B S3	
IFTMAN - Arrival notice message	D.93A S3	
IFTMBC - Booking confirmation message	D.93A S3	
IFTMBF - Firm booking message	D.93A S3	

Message	Available from Version	Last Version (if discontinued)
IFTMBP - Provisional booking message	D.93A S3	
IFTMCA - Consignment advice message	D.99B S3	
IFTMCS - Instruction contract status message	D.93A S3	
IFTMIN - Instruction message	D.93A S3	
IFTRIN - Forwarding and transport rate information message	D.93A S3	
IFTSAI - Forwarding and transport schedule and availability information message	D.93A S3	
IFTSTA - International multimodal status report message	D.93A S3	
IFTSTQ - International multimodal status request message	D.94A S3	
IMPDEF - EDI implementation guide definition message	D.98B S3	
INFCON - Infrastructure condition message	D.00B S3	
INFENT - Enterprise accounting information message	D.97B S3	
INSDDES - Instruction to despatch message	D.98A S3	
INSPRE - Insurance premium message	D.94A S3	
INSREQ - Inspection request message	D.99A S3	
INSRPT - Inspection report message	D.99A S3	
INVOIC - Invoice message	D.93A S3	
INVRPT - Inventory report message	D.93A S3	
IPPOAD - Insurance policy administration message	D.00A S3	
IPPOMO - Motor insurance policy message	D.98B S3	
ISENDS - Intermediary system enablement or disablement message	D.00B S3	
ITRRPT - In transit report detail message	D.96B S3	
JAPRES - Job application result message	D.95B S3	
JINFDE - Job information demand message	D.95B S3	
JOBAPP - Job application proposal message	D.95B S3	
JOBCON - Job order confirmation message	D.95B S3	
JOBMOD - Job order modification message	D.95B S3	

Message	Available from Version	Last Version (if discontinued)
JOBOFF - Job order message	D.95B S3	
JUPREQ - Justified payment request message	D.99B S3	D.02B S3
LEDGER - Ledger message	D.99B S3	
LREACT - Life reinsurance activity message	D.97B S3	
LRECLM - Life reinsurance claims message	D.98B S3	
MEDPID - Person identification message	D.95B S3	
MEDPRE - Medical prescription message	D.99B S3	
MEDREQ - Medical service request message	D.96B S3	
MEDRPT - Medical service report message	D.96B S3	
MEDRUC - Medical resource usage and cost message	D.97B S3	
MEQPOS - Means of transport and equipment position message	D.97A S3	
MOVINS - Stowage instruction message	D.94A S3	
MSCONS - Metered services consumption report message	D.97A S3	
ORDCHG - Purchase order change request message	D.93A S3	
ORDERS - Purchase order message	D.93A S3	
ORDRSP - Purchase order response message	D.93A S3	
OSTENQ - Order status enquiry message	D.97A S3	
OSTRPT - Order status report message	D.97A S3	
PARTIN - Party information message	D.93A S3	
PAXLST - Passenger list message	D.93A S3	
PAYDUC - Payroll deductions advice message	D.93A S3	D.02B S3
PAYEXT - Extended payment order message	D.93A S3	
PAYMUL - Multiple payment order message	D.93A S3	
PAYORD - Payment order message	D.93A S3	
PRICAT - Price/sales catalogue message	D.93A S3	
PRIHIS - Pricing history message	D.97B S3	

Message	Available from Version	Last Version (if discontinued)
PROCST - Project cost reporting message	D.99B S3	
PRODAT - Product data message	D.96B S3	
PRODEX - Product exchange reconciliation message	D.94A S3	
PROINQ - Product inquiry message	D.97A S3	
PROSRV - Product service message	D.00A S3	
PROTAP - Project tasks planning message	D.98B S3	
PRPAID - Insurance premium payment message	D.94A S3	
QUALITY - Quality data message	D.93A S3	
QUOTES - Quote message	D.93A S3	
RDRMES - Raw data reporting message	D.96A S3	
REBORD - Reinsurance bordereau message	D.96B S3	
RECADV - Receiving advice message	D.96A S3	
RECALC - Reinsurance calculation message	D.96B S3	
RECECO - Credit risk cover message	D.94B S3	
RECLAM - Reinsurance claims message	D.96A S3	
RECORD - Reinsurance core data message	D.00A S3	
REGENT - Registration of enterprise message	D.99B S3	
RELIST - Reinsured objects list message	D.00A S3	
REMADV - Remittance advice message	D.93A S3	
REPREM - Reinsurance premium message	D.96B S3	
REQDOC - Request for document message	D.94B S3	
REQOTE - Request for quote message	D.93A S3	
RESETT - Reinsurance settlement message	D.96A S3	
RESMSG - Reservation message	D.96A S3	
RETACC - Reinsurance technical account message	D.96A S3	
RETANN - Announcement for returns message	D.97B S3	
RETINS - Instruction for returns message	D.97B S3	
RPCALL - Repair call message	D.00A S3	

Message	Available from Version	Last Version (if discontinued)
SAFHAZ - Safety and hazard data message	D.95B S3	
SANCRT - International movement of goods governmental regulatory message	D.93A S3	
SLSFCT - Sales forecast message	D.94B S3	
SLSRPT - Sales data report message	D.93A S3	
SOCADE - Social administration message	D.98A S3	
SSIMOD - Modification of identity details message	D.95B S3	
SSRECH - Worker's insurance history message	D.95B S3	
SSREGW - Notification of registration of a worker message	D.95B S3	
STATAC - Statement of account message	D.93A S3	
STLRPT - Settlement transaction reporting message	D.98B S3	
SUPCOT - Superannuation contributions advice message	D.93A S3	
SUPMAN - Superannuation maintenance message	D.93A S3	
SUPRES - Supplier response message	D.96A S3	
TANSTA - Tank status report message	D.96A S3	
TAXCON - Tax control message	D.99B S3	
TPFREP - Terminal performance message	D.00A S3	
UTILMD - Utilities master data message	D.01A S3	
UTILTS - Utilities time series message	D.01A S3	
VATDEC - Value added tax message	D.96B S3	
VERMAS - Verified gross mass message	D.16A S3	
VESDEP - Vessel departure message	D.95A S3	
WASDIS - Waste disposal information message	D.97A S3	
WKGRDC - Work grant decision message	D.95B S3	
WKGRRE - Work grant request message	D.95B S3	

Message Types Before Version D.93A S3 (54)

Message	90.2 S2	91.1 S2	91.2 S2	92.1 S3	93.2 S3
BANSTA - Banking status message				X	
BAPLIE - Bayplan/stowage plan occupied and empty locations message		X		X	X
BAPLTE - Bayplan/stowage plan total numbers message		X		X	X
CONDPV - Construction direct payment valuation				X	
CONEST - Construction establishment of contract				X	
CONITT - Construction invitation to tender				X	
CONPVA - Construction payment valuation				X	
CONQVA - Construction quantity valuation				X	
CONTEN - Construction tender				X	
CREADV - Credit advice	X	X	X	X	X
CREEXT - Extended credit advice	X	X	X	X	X
CUSCAR - Customs cargo report message		X	X	X	X
CUSDEC - Customs declaration message	X	X	X	X	X
CUSREP - Customs conveyance report message		X	X	X	X
CUSRES - Customs response message	X	X	X	X	X
DEBADV - Debit advice	X	X	X	X	X
DELFOR - Delivery schedule message		X		X	
DELJIT - Delivery just-in-time message		X		X	X
DESADV - Despatch advice message		X		X	
DIRDEB - Direct debit message				X	
DOCADV - Documentary credit advice message				X	

Message	90.2 S2	91.1 S2	91.2 S2	92.1 S3	93.2 S3
DOCAPP - Application for a documentary credit		X		X	
DOCINF - Documentary credit issuance information message				X	
EnvelopeForMessagesS3 - Envelope for Messages (Syntax Versions 1/2/3) i Note This message is specifically designed to capture and define the envelope structure.	X	X	X	X	X
EnvelopeWithFunctionalGroupsS3 - Envelope with Functional Groups of Messages (Syntax Versions 1/2/3) i Note This message is specifically designed to capture and define the envelope structure.	X	X	X	X	X
IFCSUM - International forwarding and consolidation summary message		X		X	X
IFTMAN - Arrival notice message	X	X	X	X	X
IFTMBC - Booking confirmation message	X	X	X	X	X
IFTMBF - Firm booking message	X	X	X	X	X
IFTMBP - Provisional booking message	X	X	X	X	X
IFTMCS - Instruction contract status message	X	X	X	X	X
IFTMFR - International forwarding and transport message framework	X	X	X	X	
IFTMIN - Instruction message	X	X	X	X	X
IFTSTA - International multimodal status report message				X	
INVOIC - Invoice message		X	X	X	X
INVRPT - Inventory report message				X	

Message	90.2 S2	91.1 S2	91.2 S2	92.1 S3	93.2 S3
ORDCHG - Purchase order change message		X		X	
ORDERS - Purchase order message		X	X	X	X
ORDRSP - Purchase order response message		X		X	
PARTIN - Party information message		X		X	X
PAXLST - Passenger list message				X	
PAYDUC - Payroll Deductions Advice				X	
PAYEXT - Extended payment order	X	X	X	X	X
PAYMUL - Multiple payment order				X	
PAYORD - Payment order	X	X	X	X	X
PRICAT - Price sales catalogue message		X		X	
QALITY - Quality data message		X		X	X
QUOTES - Quote message		X		X	X
REMADV - Remittance advice message	X	X	X	X	X
REQOTE - Request for quote message		X		X	X
SLSRPT - Sales data report message				X	
STATAC - Statement of account message		X		X	X
SUPCOT - Superannuation Contribution Advice				X	
SUPMAN - Superannuation Maintenance Message				X	

i Note

Edifact versions are always shown together with the syntax version that was used at the time the Edifact version was published (for example, **91.1 S2** and **D.01B S3**). However, syntax versions 2 and 3 are almost identical and only differ in the allowed code pages (UNOA-UNOB and UNOA-UNOF). Therefore, you can also use later Edifact versions together with syntax version 2 or use earlier Edifact versions together with syntax version 3.

The VDA message standard consists of the early VDA messages and codelists which were defined by VDA in the 1980s and 1990s and were used in the German automotive industry.

Newer VDA offerings are based directly on Edifact and form an Edifact subset. To view these VDA Edifact messages, see [Automotive EDIFACT](#).

Version

The VDA message standard used a different versioning strategy than other standards (for example, [UN/EDIFACT](#)), as it didn't publish full release versions. VDA versioned all their messages and segments separately.

Due to this versioning strategy and the fact that no updates to the VDA messages are expected, you can find all definitions in a version called [All](#).

Message Types (20)

Message Types

Message Type	Corresponding Message Type in SAP Process Integration & SAP Process Orchestration
VDA4905:3 - Delivery Schedule	
VDA4905:4 - Delivery Schedule	VDA4905
VDA4906:2 - Invoice	VDA4906
VDA4907:2 - Payment Advice	VDA4907
VDA4908:3 - Credit Note and Self-Billing Invoice	VDA4908
VDA4911:1 - Price Data	
VDA4913:3 - Despatch Note	
VDA4913:4 - Despatch Note	
VDA4913:5 - Despatch Note	VDA4913
VDA4915:2 - JIT Delivery Schedule	VDA4915
VDA4916:1 - Call-Offs in Sync with Production	VDA4916
VDA4918:1 - Vehicle Identification and Shipment Data	
VDA4919:1 - Vehicle Entry and Exit Notification	
VDA4920:1 - Shipping Order Data (Supplier - Forwarding Agent)	VDA4920
VDA4921:1 - Delivery Data (Freight Forwarder - Customer)	VDA4921
VDA4927_1:3 - Load Carrier Account Statements	VDA4927
VDA4927_2:3 - Load Carrier Movements (Empty Containers)	
VDA4927_3:3 - Load Carrier Inventory (Inventory Requests)	
VDA4927_4:3 - Load Carrier Inventory (Inventory Confirmation)	
VDA4927_5:3 - Load Carrier Requirements	

The VDA delivery contains 13 different message types. For many of the message types only the newest version is provided. For example, VDA4906:2 represents version 2 of the invoice message. For the message types VDA4905 and VDA4913, 2 and 3 versions are available. And for the message type VDA4927, 5 different variants (all in version 3) are defined.

cXML

cXML is a streamlined protocol intended for consistent communication of business documents between procurement applications, e-commerce hubs and suppliers.

Versions List (30)

Version
1.2.037
1.2.038
1.2.039
1.2.040
1.2.041
1.2.042
1.2.043
1.2.044
1.2.045
1.2.046
1.2.047
1.2.048
1.2.049
1.2.050
1.2.051
1.2.052
1.2.053
1.2.054
1.2.055
1.2.056
1.2.057
1.2.058
1.2.059
1.2.060
1.2.061

Version
1.2.062
1.2.063
1.2.064
1.2.065
1.2.066

Message Types (105)

Message Type	Available from Version
ApprovalRequest	1.2.037
AuthRequest	1.2.037
AuthResponse	1.2.043
BlanketOrderStatusUpdateRequest	1.2.037
CatalogChangeRequest	1.2.037
CatalogUploadRequest	1.2.037
ChargeFileRequest	1.2.045
CollaborationRequest	1.2.037
ComponentConsumptionRequest	1.2.037
ConfirmationRequest	1.2.037
ContentRequest	1.2.037
ContentResponse	1.2.043
ContractRequest	1.2.037
ContractStatusUpdateRequest	1.2.037
CopyRequest	1.2.037
DataAvailableMessage	1.2.037
DataRequest	1.2.037
DataResponse	1.2.043
DataSyncChangeListRequest	1.2.037
DataSyncChangeListResponse	1.2.043
DataSyncDataRequest	1.2.037
DataSyncDataResponse	1.2.043
EnvelopeForMessages	1.2.044
EnvelopeForRequests	1.2.044

Message Type	Available from Version
EnvelopeForResponses	1.2.044
GetPendingRequest	1.2.037
GetPendingResponse	1.2.043
InvoiceDetailRequest	1.2.037
InvoiceRequest	1.2.037
L3ChargeFileRequest	1.2.037
MapSyncRequest	1.2.037
MasterAgreementRequest	1.2.037
NotificationRequest	1.2.037
OrderRequest	1.2.037
OrderStatusDataRequest	1.2.037
OrderStatusDoneMessage	1.2.037
OrderStatusRequest	1.2.037
OrderStatusSetupRequest	1.2.037
OrderStatusSetupResponse	1.2.043
OrganizationChangeMessage	1.2.037
OrganizationDataRequest	1.2.037
OrganizationDataResponse	1.2.043
PaymentBatchRequest	1.2.037
PaymentProposalRequest	1.2.037
PaymentReceiptConfirmationRequest	1.2.045
PaymentRemittanceRequest	1.2.037
PaymentRemittanceStatusUpdateRequest	1.2.037
PriceAvailabilityRequest	1.2.037
PriceAvailabilityResponse	1.2.043
PrivateOrganizationRequest	1.2.037
PrivateOrganizationResponse	1.2.043
ProductActivityMessage	1.2.037
ProductReplenishmentMessage	1.2.037
ProfileRequest	1.2.037
ProfileResponse	1.2.043

Message Type	Available from Version
ProofOfServiceRequest	1.2.042
ProviderDataRequest	1.2.037
ProviderDataResponse	1.2.043
ProviderDoneMessage	1.2.037
ProviderSetupRequest	1.2.037
ProviderSetupResponse	1.2.043
PunchOutOrderMessage	1.2.037
PunchOutSetupRequest	1.2.037
PunchOutSetupResponse	1.2.043
PurchaseRequisitionRequest	1.2.037
QualityInspectionDecisionRequest	1.2.037
QualityInspectionRequest	1.2.037
QualityInspectionResultRequest	1.2.037
QualityNotificationRequest	1.2.037
QuoteDataMessage	1.2.037
QuoteMessage	1.2.037
QuoteRequest	1.2.037
ReceiptRequest	1.2.037
RelationshipRequest	1.2.037
RelationshipResponse	1.2.043
RFxEventRequest	1.2.037
RFxPublishRequest	1.2.037
S4DataPushRequest	1.2.037
S4DataPushResponse	1.2.043
SalesOrderRequest	1.2.037
SearchRequest	1.2.037
SearchResponse	1.2.043
ServiceEntryRequest	1.2.037
SessionStatusRequest	1.2.037
SessionStatusResponse	1.2.043
ShipNoticeRequest	1.2.037

Message Type	Available from Version
SourcingUpdateRequest	1.2.037
StatusUpdateRequest	1.2.037
SubscriptionChangeMessage	1.2.037
SubscriptionContentRequest	1.2.037
SubscriptionContentResponse	1.2.043
SubscriptionListRequest	1.2.037
SubscriptionListResponse	1.2.043
SubscriptionStatusUpdateRequest	1.2.037
SupplierChangeMessage	1.2.037
SupplierDataRequest	1.2.037
SupplierDataResponse	1.2.043
SupplierListRequest	1.2.037
SupplierListResponse	1.2.043
TimeCardInfoRequest	1.2.037
TimeCardRequest	1.2.037
TradeRequest	1.2.037
TransportConfirmation	1.2.037
TransportRequest	1.2.037
UpdateMigratedSupplierRequest	1.2.037

GS1 XML

GS1 uses XML to create a set of standard messages for the GS1 EDI (Electronic Data Interchange). GS1 XML messages can be exchanged using any technical solution or internet transport protocol.

Versions (18)

Version
2.0.2
2.1
2.1.1
2.2
2.3
2.4
2.5

Version
2.6
2.7
2.8
3.0
3.1
3.2
3.3
3.4
3.4.1
3.5.1
3.6

Message Types for Versions 3.x (59)

Message Type	Available from Version	Last Version (if discontinued)
advancedRemittanceNotificationMessage - Advanced Remittance Notification Message	3.0	
applicationReceiptAcknowledgementMessage - Application Receipt Acknowledgement Message	3.0	
artworkContentMessage - Artwork Content Message	3.1	
artworkContentResponseMessage - Artwork Content Response Message	3.1	
buyerReconciliationOfRequestForPaymentMessage - Buyer Reconciliation Of Request For Payment Message	3.1	
cashCountNotificationMessage - Cash Count Notification Message	3.4	
cashCountReportMessage - Cash Count Report Message	3.4	
cashCountStatisticReportMessage - Cash Count Statistic Report Message	3.4	
cashPackagingReportMessage - Cash Packaging Report Message	3.4	
claimsNotificationMessage - Claims Notification Message	3.0	
clinicalTrialsDespatchAdviceMessage - Clinical Trials Despatch Advice Message	3.6	

Message Type	Available from Version	Last Version (if discontinued)
clinicalTrialsInventoryReportMessage - Clinical Trials Inventory Report Message	3.6	
clinicalTrialsKitStatusChangeMessage - Clinical Trials Kit Status Change Message	3.6	
clinicalTrialsReceivingAdviceMessage - Clinical Trials Receiving Advice Message	3.6	
clinicalTrialsRequestForInventoryReportMessage - Clinical Trials Request For Inventory Report Message	3.6	
configureToOrderMessage - Configure To Order Message	3.0	
consumptionReportMessage - Consumption Report Message	3.0	
debitCreditAdviceMessage - Debit Credit Advice Message	3.0	
despatchAdviceMessage - Despatch Advice Message	3.0	
dispensingAdviceMessage - Dispensing Advice Message	3.6	
goodsRequirementsMessage - Goods Requirements Message	3.0	
goodsRequirementsResponseMessage - Goods Requirements Response Message	3.0	
inventoryReleaseFileMessage - Inventory Release File Message	3.6	
inventoryReportMessage - Inventory Report Message	3.0	
invoiceMessage - Invoice Message	3.0	
itemDataNotificationMessage - Item Data Notification Message	3.0	
logisticsInventoryReportMessage - Logistics Inventory Report Message	3.1	
logisticsInventoryReportRequestMessage - Logistics Inventory Report Request Message	3.1	
orderMessage - Order Message	3.0	
orderResponseMessage - Order Response Message	3.0	
performanceMeasurementMessage - Performance Measurement Message	3.0	
productRecallCloseoutNotificationMessage - Product Recall Closeout Notification Message	3.0	

Message Type	Available from Version	Last Version (if discontinued)
productRecallNotificationMessage - Product Recall Notification Message	3.0	
productRemovalConfirmationMessage - Product Removal Confirmation Message	3.0	
purchaseConditionsMessage - Purchase Conditions Message	3.0	
receivingAdviceMessage - Receiving Advice Message	3.0	
replenishmentProposalMessage - Replenishment Proposal Message	3.0	
replenishmentRequestMessage - Replenishment Request Message	3.0	
requestForPaymentMessage - Request For Payment Message	3.0	
settlementMessage - Settlement Message	3.0	
shipmentConfirmationMessage - Shipment Confirmation Message	3.6	
shipmentNotificationMessage - Shipment Notification Message	3.6	
shipmentRequestMessage - Shipment Request Message	3.6	
transportCapacityBookingMessage - Transport Capacity Booking Message	3.0	
transportCapacityBookingResponseMessage - Transport Capacity Booking Response Message	3.0	
transportCapacityPlanMessage - Transport Capacity Plan Message	3.0	
transportCapacityRequirementsMessage - Transport Capacity Requirements Message	3.0	
transportInstructionMessage - Transport Instruction Message	3.0	
transportInstructionResponseMessage - Transport Instruction Response Message	3.0	
transportPickUpDropOffConfirmationMessage - Transport Pick Up Drop Off Confirmation Message	3.0	
transportPickUpDropOffRequestMessage - Transport Pick Up Drop Off Request Message	3.0	
transportStatusNotificationMessage - Transport Status Notification Message	3.0	

Message Type	Available from Version	Last Version (if discontinued)
transportStatusRequestMessage - Transport Status Request Message	3.0	
warehousingInboundInstructionMessage - Warehousing Inbound Instruction Message	3.1	
warehousingInboundNotificationMessage - Warehousing Inbound Notification Message	3.1	
warehousingOperationsInstructionMessage - Warehousing Operations Instruction Message	3.1	
warehousingOperationsNotificationMessage - Warehousing Operations Notification Message	3.1	
warehousingOutboundInstructionMessage - Warehousing Outbound Instruction Message	3.1	
warehousingOutboundNotificationMessage - Warehousing Outbound Notification Message	3.1	

Message Types for Versions 2.x (90)

Message	Namespace	Message Name
applicationReceiptAcknowledgement	urn:ean.ucc:2	Application Receipt Acknowledgement
basicPartyRegistration	urn:ean.ucc:gdsn:2	Basic Party Registration
buyerReconciliationOfRequestForPayment	urn:ean.ucc:pay:2	Buyer Reconciliation Request For Payment
carrierRoutingInstruction	urn:ean.ucc:deliver:2	Carrier Routing Instruction
catalogueItemAuthorisation	urn:ean.ucc:gdsn:2	Catalogue Authorisation
catalogueItemAuthorisationResponse	urn:ean.ucc:gdsn:2	Catalogue Authorisation Response
catalogueItemConfirmation	urn:ean.ucc:gdsn:2	Catalogue Confirmation
catalogueItemLink	urn:ean.ucc:gdsn:2	Catalogue Link
catalogueItemNotification	urn:ean.ucc:gdsn:2	Catalogue Notification
catalogueItemPublication	urn:ean.ucc:gdsn:2	Catalogue Publication

Message	Namespace	Message Name
catalogueItemRegistrationResponse	urn:ean.ucc:gdsn:2	Catalogue Registratic Response
catalogueItemSubscription	urn:ean.ucc:gdsn:2	Catalogue Subscripti
conditionDocument	urn:ean.ucc:align:2	Condition Document
configureToOrder	urn:ean.ucc:order:2	Configure Order
confirmationOfSettlementInstructions	urn:ean.ucc:pay:2	Confirmati Settlement Instructior
consumptionReport	urn:ean.ucc:deliver:2	Consumpt Report
controlTotal	urn:ean.ucc:pay:2	Control To
dataSynchronisationDataPoolProfile	urn:ean.ucc:gdsn:2	Data Synchroni Data Pool I
debitCreditAdvice	urn:ean.ucc:pay:2	Debit Cred Advice
despatchAdvice	urn:ean.ucc:deliver:2	Despatch /
eANUCCResponse	urn:ean.ucc:gdsn:2	EANUCC Response
Envelope1	http://www.unece.org/cefact/namespaces/StandardBusinessDocumentHeader	Envelope f level hiera
Envelope2	http://www.unece.org/cefact/namespaces/StandardBusinessDocumentHeader	Envelope f level hiera
Envelope3	http://www.unece.org/cefact/namespaces/StandardBusinessDocumentHeader	Envelope f level hiera
Envelope4	http://www.unece.org/cefact/namespaces/StandardBusinessDocumentHeader	Envelope f level hiera
event	urn:ean.ucc:plan:2	Event
exceptionCriteria	urn:ean.ucc:plan:2	Exception
exceptionNotification	urn:ean.ucc:plan:2	Exception Notificatio
financialInstitutionControlTotal	urn:ean.ucc:pay:2	Financial Institution Total
forecast	urn:ean.ucc:plan:2	Forecast

Message	Namespace	Message Name
forecastBulkData	urn:ean.ucc:plan:2	Forecast Bulk Data
forecastRevision	urn:ean.ucc:plan:2	Forecast Revision
fTLLTLShipmentStatus	urn:ean.ucc:deliver:2	FTLLTL Shipment Status
fTLLTLShipmentStatusInquiry	urn:ean.ucc:deliver:2	FTLLTL Shipment Status Inquiry
fullTruckloadFreightInvoice	urn:ean.ucc:deliver:2	Full Truckload Freight Invoice
fullTruckloadLoadTender	urn:ean.ucc:deliver:2	Full Truckload Load Tender
gDSNException	urn:ean.ucc:gdsn:2	GDSN Exception
goodsRequirements	urn:ean.ucc:plan:2	Goods Requirements
goodsRequirementsResponse	urn:ean.ucc:plan:2	Goods Requirements Response
inventoryActivityOrInventoryStatus	urn:ean.ucc:deliver:2	Inventory Activity Or Inventory Status
invoice	urn:ean.ucc:pay:2	Invoice
invoiceResponse	urn:ean.ucc:pay:2	Invoice Response
itemDataNotification	urn:ean.ucc:align:cpg:cpg_direct_materials:2	Item Data Notification
lessThanTruckloadFreightInvoice	urn:ean.ucc:deliver:2	Less Than Truckload Freight Invoice
ITLCarrierBillOfLading	urn:ean.ucc:deliver:2	LTL Carrier Bill of Lading
monetaryDocument	urn:ean.ucc:align:2	Monetary Document
motorFuelCarrierBillOfLading	urn:ean.ucc:deliver:2	Motor Fuel Carrier Bill of Lading
multiShipmentOrder	urn:ean.ucc:order:2	Multi Shipment Order
multiShipToOrder	urn:ean.ucc:order:2	Multi Ship To Order
order	urn:ean.ucc:order:2	Order
orderResponse	urn:ean.ucc:order:2	Order Response

Message	Namespace	Message Name
partyConfirmation	urn:ean.ucc:gdsn:2	Party Confirmation
partyDocument	urn:ean.ucc:align:2	Party Document
partyNotification	urn:ean.ucc:gdsn:2	Party Notification
partyPublication	urn:ean.ucc:gdsn:2	Party Publication
partyRegistration	urn:ean.ucc:gdsn:2	Party Registration
partyRegistrationResponse	urn:ean.ucc:gdsn:2	Party Registration Response
partySubscription	urn:ean.ucc:gdsn:2	Party Subscription
performanceHistory	urn:ean.ucc:plan:2	Performance History
pickupNotificationLTL	urn:ean.ucc:deliver:2	Pickup Notification
priceBracketDocument	urn:ean.ucc:align:2	Price Bracket Document
priceDocument	urn:ean.ucc:align:2	Price Document
priceSynchronisationConfirmation	urn:ean.ucc:gdsn:2	Price Synchronisation Confirmation
priceSynchronisationDocument	urn:ean.ucc:gdsn:2	Price Synchronisation Document
productActivity	urn:ean.ucc:plan:2	Product Activity
productActivityBulkData	urn:ean.ucc:plan:2	Product Activity Bulk Data
purchaseConditions	urn:ean.ucc:plan:2	Purchase Conditions
receivingAdvice	urn:ean.ucc:deliver:2	Receiving Advice
registryCatalogueItem	urn:ean.ucc:gdsn:2	Registry Catalogue Item
registryPartyDataDump	urn:ean.ucc:gdsn:2	Registry Party Data Dump
replenishmentProposal	urn:ean.ucc:plan:2	Replenishment Proposal
replenishmentRequest	urn:ean.ucc:plan:2	Replenishment Request
requestForCatalogueItemNotification	urn:ean.ucc:gdsn:2	Request For Catalogue Item Notification

Message	Namespace	Message Name
		Notification
requestForPartyNotification	urn:ean.ucc:gdsn:2	Request For Notification
requestForPayment	urn:ean.ucc:pay:2	Request For Payment
retailEvent	urn:ean.ucc:plan:2	Retail Event
searchRequest	urn:ean.ucc:gdsn:2	Search Request
searchResult	urn:ean.ucc:gdsn:2	Search Result
settlement	urn:ean.ucc:pay:2	Settlement
simpleInvoice	urn:ean.ucc:pay:2	Simple Invoice
smallPackageCarrierFreightInvoice	urn:ean.ucc:deliver:2	Small Package Carrier Freight Invoice
smallPackagePickupManifest	urn:ean.ucc:deliver:2	Small Package Pickup Manifest
sPCShipmentStatus	urn:ean.ucc:deliver:2	SPC Shipment Status
sPCShipmentStatusInquiry	urn:ean.ucc:deliver:2	SPC Shipment Status Inquiry
tradeItemDocument	urn:ean.ucc:align:2	Trade Item Document
tradeItemDocument	urn:ean.ucc:gdsn:2	Trade Item Document
tradeItemInformationRequest	urn:ean.ucc:plan:2	Trade Item Information Request
tradeItemLocationProfile	urn:ean.ucc:plan:2	Trade Item Location Profile
tradingPartnerProfileDocument	urn:ean.ucc:align:2	Trading Partner Profile Document
warehouseShippingOrder	urn:ean.ucc:deliver:2	Warehouse Shipping Order

xCBL

Versions (3)

Version
3.0
3.5

Version
4.0

Message Types (135)

Message Type	Namespace	V
AccountCheckRequest - Account Check Request	rrn:org.xcbl:schemas/xcbl/v4_0/applicationintegration/v1_0/applicationintegration.xsd	4
AdvanceShipmentNotice - Advance Shipment Notice	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AdvanceShipmentNotice - Advance Shipment Notice	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AdvanceShipmentNotice - Advance Shipment Notice	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
ApplicationResponse - Application Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
ApplicationResponse - Application Response	rrn:org.xcbl:schemas/xcbl/v4_0/messagemanagement/v1_0/messagemanagement.xsd	4
AuctionCreate - Auction Create	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AuctionCreate - Auction Create	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AuctionCreateResponse - Auction Create Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AuctionCreateResponse - Auction Create Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AuctionResult - Auction Result	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AuctionResult - Auction Result	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AuctionResultResponse - Auction Result Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AuctionResultResponse - Auction Result Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AvailabilityCheckRequest - Availability Check Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AvailabilityCheckRequest - Availability Check Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AvailabilityCheckRequest - Availability Check Request	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
AvailabilityCheckResult - Availability Check Result	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AvailabilityCheckResult - Availability Check Result	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3

Message Type	Namespace	V
AvailabilityCheckResult - Availability Check Result	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
AvailabilityToPromise - Availability To Promise	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AvailabilityToPromise - Availability To Promise	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AvailabilityToPromise - Availability To Promise	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
AvailabilityToPromiseResponse - Availability To Promise Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
AvailabilityToPromiseResponse - Availability To Promise Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
AvailabilityToPromiseResponse - Availability To Promise Response	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
ChangeOrder - Change Order	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
ChangeOrder - Change Order	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
ChangeOrder - Change Order	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
ErrorResponse - Error Response	rrn:org.xcbl:schemas/xcbl/v4_0/messagemanagement/v1_0/messagemanagement.xsd	4
FXRateRequest - FX Rate Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
FXRateRequest - FX Rate Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
FXRateRequest - FX Rate Request	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
FXRateResponse - FX Rate Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
FXRateResponse - FX Rate Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
FXRateResponse - FX Rate Response	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
GetERPData - Get ERP Data	rrn:org.xcbl:schemas/xcbl/v4_0/applicationintegration/v1_0/applicationintegration.xsd	4
GetERPDataResponse - Get ERP Data Response	rrn:org.xcbl:schemas/xcbl/v4_0/applicationintegration/v1_0/applicationintegration.xsd	4
GetOrder - Get Order	rrn:org.xcbl:schemas/xcbl/v4_0/applicationintegration/v1_0/applicationintegration.xsd	4
GoodsReceipt - Goods Receipt	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
GoodsReceipt - Goods Receipt	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
InventoryReport - Inventory Report	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
Invoice	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
Invoice	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
Invoice	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
InvoiceResponse - Invoice Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3

Message Type	Namespace	V
InvoiceResponse - Invoice Response	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
MessageAcknowledgement - Message Acknowledgement	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
MessageAcknowledgement - Message Acknowledgement	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
MessageAcknowledgement - Message Acknowledgement	rrn:org.xcbl:schemas/xcbl/v4_0/messagemanagement/v1_0/messagemanagement.xsd	4
Order	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
Order	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
Order	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
OrderConfirmation - Order Confirmation	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
OrderConfirmation - Order Confirmation	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
OrderConfirmationResponse - Order Confirmation Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
OrderConfirmationResponse - Order Confirmation Response	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
OrderRequest - Order Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
OrderRequest - Order Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
OrderRequest - Order Request	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
OrderResponse - Order Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
OrderResponse - Order Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
OrderResponse - Order Response	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
OrderStatusRequest - Order Status Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
OrderStatusRequest - Order Status Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
OrderStatusRequest - Order Status Request	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
OrderStatusResult - Order Status Result	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
OrderStatusResult - Order Status Result	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
OrderStatusResult - Order Status Result	rrn:org.xcbl:schemas/xcbl/v4_0/ordermanagement/v1_0/ordermanagement.xsd	4
PaymentRequest - Payment Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3

Message Type	Namespace	V
PaymentRequest - Payment Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PaymentRequest - Payment Request	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
PaymentRequestAcknowledgment - Payment Request Acknowledgment	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PaymentRequestAcknowledgment - Payment Request Acknowledgment	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PaymentRequestAcknowledgment - Payment Request Acknowledgment	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
PaymentStatusRequest - Payment Status Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PaymentStatusRequest - Payment Status Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PaymentStatusRequest - Payment Status Request	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
PaymentStatusResponse - Payment Status Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PaymentStatusResponse - Payment Status Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PaymentStatusResponse - Payment Status Response	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
PlanningSchedule - Planning Schedule	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PlanningSchedule - Planning Schedule	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PlanningSchedule - Planning Schedule	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
PlanningScheduleResponse - Planning Schedule Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PlanningScheduleResponse - Planning Schedule Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PlanningScheduleResponse - Planning Schedule Response	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
PriceCheckRequest - Price Check Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PriceCheckRequest - Price Check Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PriceCheckRequest - Price Check Request	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
PriceCheckResult - Price Check Result	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
PriceCheckResult - Price Check Result	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
PriceCheckResult - Price Check Result	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4

Message Type	Namespace	V
ProductCatalog - Product Catalog	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
ProductCatalog - Product Catalog	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
ProductCatalog - Product Catalog	rrn:org.xcbl:schemas/xcbl/v4_0/catalog/v1_0/catalog.xsd	4
Quote	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
Quote	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
Quote	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
RemittanceAdvice - Remittance Advice	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
RemittanceAdvice - Remittance Advice	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
RemittanceAdvice - Remittance Advice	rrn:org.xcbl:schemas/xcbl/v4_0/financial/v1_0/financial.xsd	4
RequestForQuotation - Request For Quotation	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
RequestForQuotation - Request For Quotation	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
RequestForQuotation - Request For Quotation	rrn:org.xcbl:schemas/xcbl/v4_0/preordermanagement/v1_0/preordermanagement.xsd	4
Requisition	rrn:org.xcbl:schemas/xcbl/v4_0/applicationintegration/v1_0/applicationintegration.xsd	4
ShippingSchedule - Shipping Schedule	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
ShippingSchedule - Shipping Schedule	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
ShippingSchedule - Shipping Schedule	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
ShippingScheduleResponse - Shipping Schedule Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
ShippingScheduleResponse - Shipping Schedule Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
ShippingScheduleResponse - Shipping Schedule Response	rrn:org.xcbl:schemas/xcbl/v4_0/materialsmanagement/v1_0/materialsmanagement.xsd	4
SourcingCreate - Sourcing Create	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
SourcingCreateResponse - Sourcing Create Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
SourcingResult - Sourcing Result	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
SourcingResultResponse - Sourcing Result Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TimeSeries - Time Series	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TimeSeries - Time Series	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TimeSeries - Time Series	rrn:org.xcbl:schemas/xcbl/v4_0/statisticsandforecasting/v1_0/statisticsandforecasting.xsd	4

Message Type	Namespace	V
TimeSeriesRequest - Time Series Request	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TimeSeriesRequest - Time Series Request	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TimeSeriesRequest - Time Series Request	rrn:org.xcbl:schemas/xcbl/v4_0/statisticsandforecasting/v1_0/statisticsandforecasting.xsd	4
TimeSeriesResponse - Time Series Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TimeSeriesResponse - Time Series Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TimeSeriesResponse - Time Series Response	rrn:org.xcbl:schemas/xcbl/v4_0/statisticsandforecasting/v1_0/statisticsandforecasting.xsd	4
TradingPartnerOrganizationDelete - Trading Partner Organization Delete	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TradingPartnerOrganizationDelete - Trading Partner Organization Delete	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TradingPartnerOrganizationInformation - Trading Partner Organization Information	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TradingPartnerOrganizationInformation - Trading Partner Organization Information	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TradingPartnerResponse - Trading Partner Response	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TradingPartnerResponse - Trading Partner Response	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TradingPartnerUserDelete - Trading Partner User Delete	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TradingPartnerUserDelete - Trading Partner User Delete	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3
TradingPartnerUserInformation - Trading Partner User Information	rrn:org.xcbl:schemas/xcbl/v3_0/xcbl30.xsd	3
TradingPartnerUserInformation - Trading Partner User Information	rrn:org.xcbl:schemas/xcbl/v3_5/xcbl35.xsd	3

The standard organization **xCBL.org** has used a design where the namespace is specific to a version. As a result, each Message Type is applicable to 1 version only (because a Message Type is always identified by Identifier + Namespace).

i Note

Message Types with same Identifier (like Order) are nevertheless considered as logical predecessors.

Automotive Codelists

JAIF Codelists

The JAIF Codelists has only one version which is **2020-12** and consists of codelists all pertaining to the same version. These codelists are to be used in Message Guidelines based on JAIF EDIFACT, ODETTE EDIFACT and VDA EDIFACT.

Codelists (47)

Codelist
JAI001 - Document Type
JAI002 - Mode of Transport
JAI003 - Tax Exemption Reason Code
JAI004 - Article Status Code
JAI005 - Delivery Instruction Reason Code
JAI006 - Payment Means (Instrument of Payment)
JAI007 - Payment conditions (Payment Method) Odette
JAI008 - Reason for Adjustment
JAI009 - Transaction Status Code
JAI010 - Customs Preference Origin Code
JAI011 - Quality Control Status Code
JAI012 - Reply Status Code
JAI013 - Material Category Code
JAI014 - Invoice Correction Action Code
JAI015 - Terms of Transport (Odette)
JAI016 - JAI Subset Identification
JAI017 - Minority Owned Business Identification Code
JAI018 - Party ID Type Code
JAI019 - Label Type Code
JAI020 - Price Specification Code (STAR)
JAI021 - Order Status Code (STAR)
JAI022 - Order Type Code (STAR)
JAI023 - Order Line Status Code (STAR)
JAI024 - Invoice corrective action code
JAI025 - Business legal structure type
JAI026 - Type Of Label Marking

Codelist
JAI030 - Quantity qualifier
JAI031 - Configuration level (Despatched article)
JAI032 - Allowance Charge Type
JAI033 - Delivery Instruction Type
JAI034 - Quality Conformance (Odette)
JAI035 - Safety Regulations Indicator Code (AIAG)
JAI036 - Booking key (self billed invoice line item)
JAI037 - Transport service description code
JAI038 - Package ownership
JAI039 - JIS Information Status code
JAI040 - Vehicle reference data (JIS Process)
JAI041 - Supplier organises customs clearance indicator
JAI042 - Properties of steel
JAI043 - Date qualifier code
JAI044 - Object identification qualifier code
JAI045 - Stock regime status code
JAI046 - Stock standard ID
JAI047 - Date/Time Format Qualifier
JAI048 - Dutiable goods indicator
JAI049 - Agency responsible for a package type code list
JAI050 - MAT label - package level code

ODETTE Codelists

The ODETTE Codelists has only one version which is **2020-12** and consists of codelists all pertaining to the same version. These codelists are to be used in Message Guidelines based on JAIF EDIFACT, ODETTE EDIFACT and VDA EDIFACT.

Codelists (179)

Codelist
3055 - Code list responsible agency code
ODDC001(1) - Document Type based on EDIFACT 1001
ODDC001(2) - Document Type
ODDC001(A) - Document Type (Odette)
ODDC001(C) - Document Type based on 1153 EDIFACT Internal

Codelist
ODDC003(1) - Mode of Transport
ODDC004(1) - Incoterms
ODDC006(1) - COUNTRIES GEONOMENCLATURE
ODDC007(1) - Currencies
ODDC008(1) - Prepaid/Collect Indicator
ODDC009(1) - Charge Type (based on EDIFACT 7161)
ODDC009(A) - Allowance Charge Type
ODDC009(B) - Allowance Charge Type
ODDC010(1) - VAT Category
ODDC013(1) - Nature of Transaction based on EDIFACT 4183
ODDC014(1) - Transport Ownership based on EDIFACT 8281
ODDC015(1) - Article status
ODDC017(1) - Schedule Status Indicator (based on EDIFACT 4017)
ODDC018(1) - Delivery instruction reason based on EDIFACT 6063
ODDC018(2) - Delivery Instruction Reason (Odette)
ODDC021(1) - Schedule Release Category based on EDIFACT 1229
ODDC021(2) - Schedule Release Category (Odette)
ODDC022(1) - Type of Period based on EDIFACT 2379
ODDC022(2) - Type of Period (Odette)
ODDC022(B) - Type of Period based On EDIFACT 2151
ODDC023(1) - Payment Means based on EDIFACT 4461
ODDC023(2) - Payment Means (Instrument of Payment)
ODDC024(1) - Terms Time Information based on EDIFACT 2009
ODDC025(1) - Unit of Measurement (based on UN recommendation 20)
ODDC026(1) - Payment Conditions based on EDIFACT 4277
ODDC026(2) - Payment conditions (Payment Method) Odette
ODDC027(1) - Reason for Adjustment based on 4465 EDIFACT
ODDC027(2) - Reason for Adjustment (Odette)
ODDC031(1) - Message Status based on EDIFACT 1225
ODDC031(2) - Status based on EDIFACT 1225
ODDC031(3) - Status based on EDIFACT 1229

Codelist
ODDC031(4) - Status based on EDIFACT 1245
ODDC031(5) - Status (Odette)
ODDC032(1) - Message Indicator based on EDIFACT 1225
ODDC032(2) - Reply Indicator based on EDIFACT 1229
ODDC032(3) - Reply Indicator (Odette)
ODDC033(1) - Release Type based on EDIFACT 2005
ODDC034(1) - Preference Origin based on EDIFACT 9213
ODDC034(2) - Preference Origin (Odette)
ODDC037(1) - Quality Control (Odette)
ODDC038(1) - Quantity Descriptor based on EDIFACT 6063
ODDC038(2) - Quantity Descriptor based on EDIFACT 6167
ODDC040(2) - Qualifier for Document Balance (Odette)
ODDC041(1) - General Conditions (based on EDIFACT 4279)
ODDC041(2) - General Conditons (Odette)
ODDC042(1) - Ownership/Durability (based on EDIFACT 7073)
ODDC042(2) - Ownership/Durability (Odette)
ODDC043(1) - Product Usage Type (Odette)
ODDC045(1) - Product/Service Type (Odette)
ODDC046(1) - Packaging level (based on EDIFACT 7075)
ODDC047(1) - Material Category (Odette)
ODDC048(1) - Price Type (based on EDIFACT 5125)
ODDC048(2) - Price Type (based on EDIFACT 5387)
ODDC048(3) - Price Type (Odette)
ODDC050(1) - Packaging Charge (based on EDIFACT 7073)
ODDC050(2) - Packaging Charge (Odette)
ODDC052(1) - Drawing Type (Odette)
ODDC054(1) - Domestic/Export Indicator (based on 7073 EDIFACT)
ODDC055(1) - Industry Identification
ODDC056(1) - Reference Table Identification
ODDC057(1) - Product Specification (based on 6313 EDIFACT)
ODDC057(2) - Product Specification (Odette)

Codelist
ODDC058(1) - Action (Odette)
ODDC059(1) - Information Requested (based on EDIFACT 4511)
ODDC059(2) - Information Requested (Odette)
ODDC062(1) - Package Code Identifier (based on 3055 EDIFACT)
ODDC062(2) - Package Code Identifier (Odette)
ODDC063(1) - Flow Dest. Indicator (based on EDIFACT 4183)
ODDC065(1) - Schedule Release Frequency (EDIFACT 2013)
ODDC065(2) - Schedule Release Frequency (EDIFACT 2379)
ODDC065(3) - Schedule Release Frequency (Odette)
ODDC066(1) - Terms of Transport (Odette)
ODDC067(1) - Supply Condition (Odette)
ODDC068(1) - Supplier's Stock Turnover (Odette)
ODDC069(1) - Tooling Type (Odette)
ODDC070(1) - Tooling Specifier (Odette)
ODDC071(1) - Reply Reason (based on EDIFACT 4465)
ODDC071(2) - Reply Reason (Odette)
ODDC072(1) - Price List Type (Odette)
ODDC075(1) - Anomalies (Odette)
ODDC076(1) - Split Consignments (Odette)
ODDC079(1) - File Status (Odette)
ODDC080(1) - Link Purpose (Odette)
ODDC081(1) - Quality Conformance (Odette)
ODDC082(1) - Safety Regulation Specifier (Odette)
ODDC083(1) - Stock Balance Details (based on 4503 EDIFACT)
ODDC083(2) - Stock Balance Details (Odette)
ODDC084(1) - Stock Movement Details
ODDC084(A) - Nature of Stock (based on EDIFACT 7491)
ODDC084(B) - Stock Movement Direction (EDIFACT 4501)
ODDC085(1) - Issue Types (Odette)
ODDC087(1) - Type of Means of Transport (EDIFACT 8179)
ODDC087(2) - Delivery Instruction Type

Codelist
ODDC088(1) - Dangerous Goods Regulations (EDIFACT 8273)
ODDC090(1) - Type of Equipment (based on EDIFACT 8053)
ODDC091(1) - Equipment Size and Type (based on 8155 EDIFACT)
ODDC092(1) - Transport Priority (based on 4219 EDIFACT)
ODDC093(1) - Nature of Cargo (based on EDIFACT 7085)
ODDC093(A) - Nature of Cargo (Odette)
ODDC094(1) - Government Agency (based on EDIFACT 9415)
ODDC095(1) - Government Involvement (based on EDIFACT 9411)
ODDC096(1) - Government Procedure (based on EDIFACT 9353)
ODDC097(1) - Identity Number (based on EDIFACT 7405)
ODDC098(1) - Transport Movement (based on EDIFACT 8323)
ODDC100(1) - Allowance/Charge Qualifier (EDIFACT 5463)
ODDC101(1) - Article number source (based on EDIFACT 7143)
ODDC102(1) - Currency Type (based on EDIFACT 6343)
ODDC103(1) - Recalculation (Odette)
ODDC104(1) - Monetary Amount Type (based on EDIFACT 5025)
ODDC105(1) - Duty Tax Fee Type (based on EDIFACT 5153)
ODDC106(1) - Summary Total Type (based on EDIFACT 6069)
ODDC107(1) - Contract and Carriage Condition (EDIFACT 4065)
ODDC108(1) - Handling Unit category
ODDC109(1) - Free Text Qualifier (based on EDIFACT 4451)
ODDC110(1) - Opening/Closing Balance (based on EDIFACT 6063)
ODDC111(1) - Stock Movement Reason (based on EDIFACT 4499)
ODDC112(1) - Constants identifying volume measurement
ODDC113(1) - Constants identifying dimension
ODDC114(1) - Constants identifying weight
ODDC115(1) - Standard date format (based on EDIFACT 2379)
ODDC116(1) - Payment Time Reference (based on EDIFACT 2475)
ODDC117(1) - Configuration level (material)
ODDC117(2) - Configuration level (Despatched article)
ODDC118(1) - Payment Variation (based on EDIFACT 5245)

Codelist
ODDC119(1) - Document Response Type (EDIFACT 4343)
ODDC121(1) - Document Function Coded (Based on EDIFACT 1225)
ODDC122(1) - Odette Subset Identification
ODDC123(1) - Product Group Type
ODDC124(1) - Delivery Instruction Type
ODDC126(1) - Reported Discrepancy
ODDC128(1) - Drawing Reference Qualifier based on EDIFACT 1153
ODDC128(2) - Status of Despatch Advice
ODDC129(1) - TYPE OF LABEL MARKING
ODDC130(1) - Type of Label Marking based on EDIFACT 4233
ODDC131(1) - PARTY TYPE (BASED ON EDIFACT 3035)
ODDC132(1) - ODDC132 NATURE OF PACKAGING SERVICES
ODDC133(1) - Nature of Packaging
ODDC134(1) - Message Function (Based on EDIFACT 1225)
ODDC135(1) - Additional Article Information
ODDC137(1) - Administration action code
ODDC139(1) - Error code
ODDC140(1) - Functional Entity Type
ODDC141(1) - Functional Reference Type
ODDC142(1) - Certificate name code
ODDC143(1) - Package Management Action Code
ODDC144(1) - Transport Status Code
ODDC145(1) - Complaint reason code
ODDC146(1) - Delivery status code
ODDC147(1) - Package booking key
ODDC148(1) - Package Order Reason Code
ODDC149(1) - Package marking type code
ODDC150(1) - Exception reason code
ODDC151(1) - Order Information Source Qualifier
ODDC152(1) - Article number source (Odette)
ODDC153(1) - Delivery priority code

Codelist
ODDC154(1) - Item characteristic code based on EDIFACT 7081
ODDC155(1) - RTI Quantities (based on EDIFACT 6063)
ODDC156(1) - Transport stage based on EDIFACT 8051
ODDC157(1) - Status category code based on EDIFACT 9015
ODDC158(1) - Handling instruction description code based on EDIFACT 4079
ODDC159(1) - Dutiable goods indicator
ODDC160(1) - Status attribute, coded
ODDC161(1) - Damage remark code
ODDC162(1) - Transport mode information
ODDC163(1) - Reason for ETA change
ODDC164(1) - Vehicle request purpose
ODDC165(1) - Geofence zone level
ODDC166(1) - Movement type
ODDC167(1) - Power type code (based on EDIFACT 7041)
ODDC168(1) - Vehicle request confirmation code
ODDC169(1) - Vehicle order type code
ODDC170(1) - Vehicle service order code
ODDC171(1) - Reason of vehicle service order rejection code
ODDC172(1) - Reason for estimated time of completion change code

VDA Codelists

These automotive codelists are defined by VDA - to be used in Message Guidelines based on VDA EDIFACT. It has only one version which is **2020-12** and consists of codelists all pertaining to the same version.

Codelists (39)

Codelist
0057 - Subset identification
01 - Freight prepayment
02 - Means of transportation
03 - Delivery medium
04 - Country of origin
05 - Quantity unit
06 - Call-off key

Codelist
07 - Use
08 - Preference status
09 - Transaction key
10 - Truck type key
11 - Package type code
1101 - Packaging coding
12 - Warehousekeeper key
13 - Delivery code
14 - Key to SA 712 Pos. 16
15 - Bonded goods
16 - Stock status
17 - Label identification
18 - Packaging identification
19 - Property identification
3055 - Code list responsible agency code
4183 - Reason for the exception
4401 - Action code
4405 - Status description code
4405(2) - Reason for complaint, code
4441(1) - Delivery status
4441(2) - Author order reference
4441(3) - Premium services for transports
4499 - Booking key
6063 - Quantity, qualifier
6411 - Measure unit, code
7064 - Packaging type, code (also valid as CL 11 for VDA 4913)
7295 - Requirement reason for packaging material
7405 - Object identification code qualifier
7511 - Marking type
8393 - Reusable packaging, cargo content, code
9015 - Status category, code

Codelist
9321 - APERAK Error Codes - VDA

UNC Codelists

The UNC Codelists have only one version which is [2020-12](#) and consists of codelists all pertaining to the same version.

Codelists (304)

Codelist
0001 - Syntax identifier
0002 - Syntax version number
0007 - Partner identification code qualifier
0013 - Service segment tag, coded
0025 - Recipient's reference/password qualifier
0029 - Processing priority code
0031 - Acknowledgement request
0035 - Test indicator
0051 - Controlling agency
0052 - Message version number
0054 - Message release number
0065 - Message type identifier
0073 - First/last sequence message transfer indication
0081 - Section identification
0083 - Action, coded
0085 - Syntax error, coded
1001 - Document name code
1049 - Message section code
1073 - Document line action code
1153 - Reference code qualifier
1159 - Sequence identifier source code
1225 - Message function code
1227 - Calculation sequence code
1229 - Action code
1373 - Document status code
1501 - Computer environment details code qualifier

Codelist
1503 - Data format description code
1505 - Value list type code
1507 - Designated class code
2005 - Date or time or period function code qualifier
2009 - Terms time relation code
2013 - Frequency code
2015 - Despatch pattern code
2017 - Despatch pattern timing code
2023 - Period type code qualifier
2151 - Period type code
2155 - Charge period type code
2379 - Date or time or period format code
2475 - Event time reference code
3035 - Party function code qualifier
3045 - Party name format code
3055 - Code list responsible agency code
3077 - Test medium code
3079 - Organisation classification code
3083 - Organisational class name code
3131 - Address type code
3139 - Contact function code
3153 - Communication medium type code
3155 - Communication means type code
3207 - Country name code
3227 - Location function code qualifier
3237 - Sample location description code
3239 - Country of origin name code
3279 - Geographic area code
3285 - Instruction receiving party identifier
3289 - Person characteristic code qualifier
3295 - Name original alphabet code

Codelist
3299 - Address purpose code
3301 - Enacting party identifier
3397 - Name status code
3401 - Name component usage code
3403 - Name type code
3405 - Name component type code qualifier
3453 - Language name code
3455 - Language code qualifier
3457 - Originator type code
3475 - Address status code
3477 - Address format code
3479 - Marital status description code
3493 - Nationality code qualifier
3499 - Gender code
4017 - Delivery plan commitment level code
4025 - Business function code
4027 - Business function type code qualifier
4035 - Priority type code qualifier
4037 - Priority description code
4043 - Trade class code
4049 - Certainty description code
4051 - Characteristic relevance code
4053 - Delivery or transport terms description code
4055 - Delivery or transport terms function code
4059 - Clause code qualifier
4065 - Contract and carriage condition code
4071 - Proviso code qualifier
4079 - Handling instruction description code
4153 - Information details code qualifier
4183 - Special condition code
4215 - Transport charges payment method code

Codelist
4219 - Transport service priority code
4221 - Discrepancy nature identification code
4233 - Marking instructions code
4237 - Payment arrangement code
4277 - Payment terms description identifier
4279 - Payment terms type code qualifier
4295 - Change reason description code
4343 - Response type code
4347 - Product identifier code qualifier
4383 - Bank operation code
4401 - Instruction description code
4403 - Instruction type code qualifier
4405 - Status description code
4407 - Sample process step code
4419 - Test administration method code
4431 - Payment guarantee means code
4435 - Payment channel code
4437 - Account type code qualifier
4439 - Payment conditions code
4441 - Tunnel restriction code
4447 - Free text format code
4451 - Text subject code qualifier
4453 - Free text function code
4455 - Back order arrangement type code
4457 - Substitution condition code
4461 - Payment means code
4463 - Intra-company payment indicator code
4465 - Adjustment reason description code
4471 - Settlement means code
4475 - Accounting entry type name code
4487 - Financial transaction type code

Codelist
4493 - Delivery instruction code
4499 - Inventory movement reason code
4501 - Inventory movement direction code
4503 - Inventory balance method code
4505 - Credit cover request type code
4507 - Credit cover response type code
4509 - Credit cover response reason code
4511 - Requested information description code
4513 - Maintenance operation code
4517 - Seal condition code
4525 - Seal type code
5007 - Monetary amount function description code
5013 - Index code qualifier
5025 - Monetary amount type code qualifier
5027 - Index type identifier
5039 - Index representation code
5047 - Contribution code qualifier
5049 - Contribution type description code
5125 - Price code qualifier
5153 - Duty or tax or fee type name code
5189 - Allowance or charge identification code
5213 - Sub-line item price change operation code
5237 - Charge category code
5243 - Rate or tariff class description code
5245 - Percentage type code qualifier
5249 - Percentage basis identification code
5261 - Charge unit code
5267 - Service type code
5273 - Duty or tax or fee rate basis code
5283 - Duty or tax or fee function code qualifier
5305 - Duty or tax or fee category code

Codelist
5307 - Tax or duty or fee payment due date code
5315 - Remuneration type name code
5375 - Price type code
5379 - Product group type code
5387 - Price specification code
5393 - Price multiplier type code qualifier
5419 - Rate type code qualifier
5463 - Allowance or charge code qualifier
5495 - Sub-line indicator code
5501 - Rate plan code
6029 - Geographical position code qualifier
6063 - Quantity type code qualifier
6069 - Control total type code qualifier
6071 - Frequency code qualifier
6077 - Result representation code
6079 - Result normalcy code
6085 - Dosage administration code qualifier
6087 - Result value type code qualifier
6113 - Length type code
6145 - Dimension type code qualifier
6155 - Non-discrete measurement name code
6167 - Range type code qualifier
6173 - Size type code qualifier
6245 - Temperature type code qualifier
6311 - Measurement purpose code qualifier
6313 - Measured attribute code
6321 - Measurement significance code
6331 - Statistic type code qualifier
6341 - Exchange rate currency market identifier
6343 - Currency type code qualifier
6345 - Currency identification code

Codelist
6347 - Currency usage code qualifier
6353 - Unit type code qualifier
6411 - Measurement unit code
6415 - Clinical information type code qualifier
6417 - Measurement unit name code
7001 - Physical or logical state type code qualifier
7007 - Physical or logical state description code
7009 - Item description code
7011 - Item availability code
7039 - Sample selection method code
7041 - Power type code
7045 - Sample state code
7047 - Sample direction code
7059 - Class type code
7065 - Package type description code
7073 - Packaging terms and conditions code
7075 - Packaging level code
7077 - Description format code
7081 - Item characteristic code
7083 - Configuration operation code
7085 - Cargo type classification code
7111 - Characteristic value description code
7133 - Product details type code qualifier
7139 - Product characteristic identification code
7143 - Item type identification code
7161 - Special service description code
7171 - Hierarchical structure relationship code
7173 - Hierarchy object code qualifier
7187 - Process type description code
7233 - Packaging related description code
7273 - Service requirement code

Codelist
7293 - Sector area identification code qualifier
7295 - Requirement or condition description identifier
7297 - Set type code qualifier
7299 - Requirement designator code
7365 - Processing indicator description code
7383 - Surface or layer code
7405 - Object identification code qualifier
7429 - Indexing structure code qualifier
7431 - Agreement type code qualifier
7433 - Agreement type description code
7449 - Membership type code qualifier
7451 - Membership category description code
7455 - Membership level code qualifier
7459 - Attendee category description code
7491 - Inventory type code
7493 - Damage details code qualifier
7495 - Object type code qualifier
7497 - Structure component function code qualifier
7511 - Marking type code
7515 - Structure type code
8015 - Traffic restriction code
8023 - Freight and other charges description identifier
8025 - Conveyance call purpose description code
8035 - Traffic restriction type code qualifier
8051 - Transport stage code qualifier
8053 - Equipment type code qualifier
8067 - Transport mode name code
8077 - Equipment supplier code
8101 - Transit direction indicator code
8155 - Equipment size and type description code
8169 - Full or empty indicator code

Codelist
8179 - Transport means description code
8249 - Equipment status code
8273 - Dangerous goods regulations code
8275 - Container or package contents indicator code
8281 - Transport means ownership indicator code
8323 - Transport movement code
8325 - Hazardous means of transport category code
8335 - Movement type description code
8339 - Packaging danger level code
8341 - Haulage arrangements code
8393 - Returnable package load contents code
8395 - Returnable package freight payment responsibility code
8453 - Transport means nationality code
8457 - Excess transportation reason code
8459 - Excess transportation responsibility code
9003 - Employment details code qualifier
9013 - Status reason description code
9015 - Status category code
9017 - Attribute function code qualifier
9023 - Definition function code
9025 - Definition extent code
9029 - Value definition code qualifier
9031 - Edit mask representation code
9035 - Qualification application area code
9037 - Qualification type code qualifier
9039 - Facility type description code
9043 - Reservation identifier code qualifier
9045 - Basis code qualifier
9051 - Applicability code qualifier
9141 - Relationship type code qualifier
9143 - Relationship description code

Codelist
9153 - Simple data element character representation code
9155 - Length type code
9161 - Code set indicator code
9169 - Data representation type code
9175 - Data element usage type code
9213 - Duty regime type code
9285 - Validation criteria code
9303 - Sealing party name code
9353 - Government procedure code
9411 - Government involvement code
9415 - Government agency identification code
9417 - Government action code
9421 - Process stage code qualifier
9437 - Clinical intervention description code
9441 - Clinical intervention type code qualifier
9443 - Attendance type code qualifier
9447 - Discharge type description code
9453 - Code value source code
9501 - Formula type code qualifier
9507 - Formula sequence code qualifier
9509 - Formula sequence operand code
9601 - Information category code
9623 - Diagnosis type code
9625 - Related cause code
9633 - Information details code qualifier
9635 - Event details code qualifier
9641 - Service basis code qualifier
9643 - Supporting evidence type code qualifier
9645 - Payer responsibility level code
9649 - Processing information code qualifier

Error Codelists

The Error Codelists has only one version which is **2020-12** and consists of codelists all pertaining to the same version.

Codelists (2)

Codelist
4441 - Application error severity code
9321 - Application error code

GS1 Codelists

The GS1 Codelists has only one version which is **2021-05** and consists of 211 codelists all pertaining to the same version. This is the collection of all code lists defined and maintained by GS1.

Codelists (211)

Codelist
AdditionalConsignmentIdentificationTypeCode
AdditionalIndividualAssetIdentificationTypeCode
AdditionalLogisticUnitIdentificationTypeCode
AdditionalPartyIdentificationTypeCode
AdditionalReturnableAssetIdentificationTypeCode
AdditionalServiceRelationIdentificationTypeCode
AdditionalShipmentIdentificationTypeCode
AdditionalTradeItemClassificationCodeListCode
AdditionalTradeItemIdentificationTypeCode
AdministrativeUnitTypeCode
AllowanceChargeTypeCode
AllowanceOrChargeEnumeration
AnimalIdentificationTypeCode
ApplicationResponseStatusEnumeration
AppointmentTimeMeasurementBasisTypeCode
BarCodeTypeCode
BusinessDocumentProcessingTypeCode
CapacityPlanStatusEnumeration
CargoTypeCode
CashCountStatisticsTypeCode
CashCountTypeCode
CashProcessingFlagCode

Codelist
ClaimsNotificationTypeEnumeration
ColourCodeListCode
ColourTintCode
CommunicationChannelCode
ContactTypeCode
CopyElementTypeCode
CountryCode
CreditReasonCode
CustomsWarehouseStatusCode
DangerousGoodsAttributeTypeCode
DangerousGoodsRegulationCode
DataCarrierTypeCode
DateFormatCode
DayOfTheWeekEnumeration
DebitCreditEnumeration
DeliverDateTypeCode
DeliveryMethodCode
DeliveryTimeMeasurementBasisTypeCode
DeliveryTypeCode
DemandEstimationTypeCode
DespatchAdviceTypeCode
DiscountAgreementCode
DMEOrderPriorityCode
DockTypeCode
DocumentActionEnumeration
DocumentStatusEnumeration
DutyFeeTaxExemptionReasonCode
DutyFeeTaxLiabilityCode
DutyFeeTaxRegistrationTypeCode
DutyFeeTaxTypeCode
EffectiveDateTypeCode

Codelist
EntityTypeCode
EquipmentTypeCode
ErrorOrWarningCode
EuUniqueIDTypeCode
FinancialAccountNumberTypeCode
FinancialAdjustmentReasonCode
FinancialRoutingNumberTypeCode
ForecastPurposeCode
ForecastTypeCode
GenderEnumeration
GoodsReceiptReportingCode
GoodsRequirementsTypeEnumeration
GraphicElementTypeCode
GS1TradeItemIdentificationKeyCode
HandlingInstructionCode
IdentityDocumentTypeCode
IncidentTypeCode
InstructionOrResponseEnumeration
InventoryActivityTypeCode
InventoryBusinessStepCode
InventoryDispositionCode
InventoryEventReasonCode
InventoryMeasurementBasisTypeCode
InventoryMovementTypeCode
InventoryReportTypeEnumeration
InventoryStatusCode
InventorySubLocationFunctionCode
InventorySubLocationTypeCode
InvoiceTypeCode
ItemScopeTypeCode
ItemSourceCode

Codelist
ItemTypeCode
LegalRegistrationCode
LineItemActionCode
LocationScopeParameterTypeCode
LocationScopeTypeCode
LogisticEventTypeCode
LogisticServiceReferenceTypeCode
LogisticServiceRequirementCode
LogisticsInventoryReportTypeEnumeration
MarginSchemeCode
MaterialTypeCode
MeasurementTypeCode
MeatProcessingActivityTypeCode
MeatProcessingPartyIdentificationTypeCode
MeatProcessingPartyTypeCode
MeatWorkItemTypeCode
NonBinaryLogicEnumeration
NutrientTypeCode
ObservationTypeCode
OrderEntryTypeCode
OrderInstructionCode
OrderRelationshipTypeCode
OrderTypeCode
OutOfStockMeasurementTypeCode
OwnershipTransferConditionCode
PackageLevelCode
PackageTypeCode
PackagingConditionCode
PackagingMarkingTypeCode
PackagingTermsAndConditionsCode
PartyRoleCode

Codelist
PassengerCategoryCode
PatternCode
PaymentFormatCode
PaymentMethodCode
PaymentTermsEventCode
PaymentTermsTypeCode
PerformanceMeasureTypeCode
PeriodicityTypeCode
PhysicalOrLogicalStateDescriptionCode
PlanBucketSizeCode
PlanCommitmentLevelCode
PrecautionQualifierCode
PrintingInstructionCode
ProductionMethodForFishAndSeafoodCode
ProductRecallIncidentRiskLevelCode
ProductRecallNotificationStatusEnumeration
ProductRecallNotificationVersionStatusEnumeration
ProductRecallReasonCode
ProductRecallReimbursementReasonCode
ProductRecallTrackingTypeCode
PurchaseConditionsCommitmentTypeCode
QualityControlCode
QuantitySpecificationTypeCode
ReadingMethodCode
ReceivingConditionCode
ReconciliationStatusEnumeration
RemainingQuantityStatusCode
ReplenishmentProposalTypeEnumeration
ReplenishmentRequestStatusEnumeration
ReplenishmentRequestTypeEnumeration
ResponseStatusCode

Codelist
ResponseStatusEnumeration
ResponsibleAgencyCode
RevisionStatusCode
SalesMeasurementTypeCode
SealConditionCode
SealTypeCode
ServiceLevelBasisTypeCode
ServiceLevelMeasurementBasisTypeCode
SettlementAdjustmentReferenceTypeCode
SettlementHandlingTypeCode
SettlementTypeCode
ShipmentRequestTypeCode
ShipmentSplitMethodCode
SizeCodeListCode
SizeTypeCode
SpecialServiceTypeCode
StatusCode
StockRequirementTypeCode
StorageStateCode
StructuredCopyElementTypeCode
StructureTypeCode
SymbolComponentCode
SynchronisationCalculationTypeCode
TaxCategoryCode
TemperatureMeasurementUnitCode
TemperatureQualifierCode_EDI
TimeMeasurementUnitCode
TimePeriodScopeTypeCode
TradeItemDataOwnerCode
TradeItemUnitDescriptorCode
TransactionalReferenceTypeCode

Codelist
TransportAppointmentStatusEnumeration
TransportAppointmentTypeEnumeration
TransportCapacityBookingStatusEnumeration
TransportChargesPaymentMethodCode
TransportInstructionFunctionEnumeration
TransportInstructionResponseTypeEnumeration
TransportInstructionStatusEnumeration
TransportMeansTypeCode
TransportModeCode
TransportPartyRoleCode
TransportPaymentMethodCode
TransportPlanningDateTypeEnumeration
TransportReferenceTypeCode
TransportServiceCategoryCode
TransportServiceConditionTypeCode
TransportServiceLevelCode
TransportStatusConditionCode
TransportStatusInformationEnumeration
TransportStatusObjectEnumeration
TransportStatusReasonCode
TypeOfServiceTransactionCode
VarianceReasonCode
WarehouseABCClassificationCode
WarehousingDeliveryInstructionCode
WarehousingDespatchTypeCode
WarehousingInboundExceptionCode
WarehousingInboundStatusCode
WarehousingOutboundExceptionCode
WarehousingOutboundStatusCode
WarehousingQuantityVarianceCode
WarehousingReceiptTypeCode

Codelist
WarehousingServiceAttributeTypeCode
WarehousingServiceConditionCode
WeekendDayEnumeration

ISO Codelists

Versions (4)

Version
2002
2004
2013
2015

Codelists (5)

Codelist	Version
ISO 3166-1 - Country codes	2013
ISO 3166-2 - Country subdivision code	2013
ISO_4217 - Currency codes	2015
ISO 5218 - Gender codes	2004
ISO 639-1 - Language code (Alpha-2 code)	2002

UN/CEFACT Recommendations

Versions (7)

Version
1982
2001
2011
2012
2014
2016
2017

Codelists (8)

Codelist	Version
Rec 17 - PAYTERMS	1982
Rec 19 - Code for Modes of Transport	2001
Rec_20 - Measurement unit code	2016
Rec 21 - Package type code	2012
Rec 23 - Freight Cost Code - FCC	2012
Rec_24 - Trade and Transport Status Code	2017
Rec 28 - Codes for Types of Means of Transport	2014
Rec 5 - Abbreviations of Incoterms	2011

Further Reading

- [GS1 EANCOM Library in Integration Advisor](#) 
- [Automotive Edifact Subsets](#) 
- [Odette Message Standard](#) 
- [GS1 XML 3.x Messages](#) 
- [GS1 Codelists](#) 
- [Tradacoms Message Standard](#) 
- [EDI Envelopes in Integration Advisor](#) 
- [xCBL Messages](#) 
- [Integration Advisor: Early UN/EDIFACT Versions with Syntax Version 2 Now Available](#) 
- [Integration Advisor: GS1 XML Messages of the Version 2.X Series \(EAN.UCC\)](#) 
- [Using cXML Envelopes in Integration Advisor](#) 
- [Integration Advisor - VDA Message Standard](#) 

Custom Type Systems

The library of custom type systems allows you to upload custom messages, IDocs, SOA messages and custom codelists – this enables you to create MIGs and MAGs based on your own message structures. Use cases for this could be proprietary messages, standard messages extended by you and messages from B2B standards not available in the system. Within each custom type system, you can:

- Create custom messages based on xsd and wsdl files
- Create custom codelists
- Create MIG and MAG
- Customize your MIG
- View the list of your custom messages

To upload and use custom messages/codelists, you need to have a custom type system. Follow the procedure given below to do the same.

Procedure

1. Navigate to **Custom Type Systems** (🔗) from the left pane of your application.
2. Choose **Add** to create a custom type system.

i Note

Currently the application supports only one custom type system for each kind (XML/IDoc/SOA/Codelists).

3. Choose and open the type system. The **Overview** tab provides you with the following details :
 - General information
 - Documentation
 - Administrative Data

To know how to add a message to the custom type system, see: [Adding a Custom Message](#).

To know how to add a codelist to the custom type system, see: [Adding a Custom Codelist](#).

Adding a Custom Message

Context

You can add a custom message to a custom type system library.

Procedure

1. Go to **Custom Type Systems**.
2. Choose and open the custom type system and navigate to the **Messages** tab. This tab displays the list of uploaded custom messages.
3. To upload a custom message, choose **Create** or ► **Create** ► **Custom Message** ►. A wizard opens.
4. In the **XSD File/ZIP/WSDL File** step, choose **Browse** and upload the XSD/ZIP/WSDL file.

i Note

The following applies:

- XSD files are intended for XML and IDoc custom type systems and WSDL files for SOA type system.
- For **custom messages**, use either a single **XSD** file or a **ZIP of multiple XSD files**.

For **custom IDocs**, use a single **XSD** file.

For **custom SOA messages**, use a **WSDL** file.

- The maximum file size that you can upload is 10 MB, with a maximum limit of 20000 nodes.
- Uploading custom IDocs requires an XSD in the format as downloaded from the Enterprise Service Repository (ESR) or from the transaction **WE60** of your SAP S/4HANA backend system.

- If you use a ZIP file, it can contain multiple related XSD files (for different `targetNamespaces`) that refer to each other using the `xsd:import` feature.

Please note the existing [Limitations](#).

5. The next step shows you all the messages present in your XSD/ZIP/WSDL. Choose the required message from the **Messages** step.
6. The details of the message are auto-filled under the **Message Creation** step. You can verify and modify the details.

i Note

- The fields **Identifier**, **Namespace** and **Version** uniquely identify your message.
- You can create multiple versions for a message with the same **Identifier** and **Namespace**.
- If you upload a message with already existing **Identifier**, **Namespace** and **Version**, a new **Revision** of the message gets created. And, only the latest revision of the message gets displayed in the **Messages** list.

7. For custom IDoc creation, the **Message Creation** step displays the following additional fields:

- a. **Message Type**: This is a mandatory field which is auto-filled by the proposals picked from the xsd file. If not, you need to enter a meaningful text.
- b. **IDoc Type** : This is a mandatory field which is auto-filled by the proposals picked from the xsd file. If not, you need to enter a meaningful text.
- c. **Extension Type** : This is an optional field which is auto-filled by the proposals picked from the xsd file.

The **Identifier** field is read-only and auto-populated using the values provided in the previous fields. The format is as follows: **Message Type.IDoc Type.Extension Type**.

❖ Example

For the following values, the **Identifier** field is populated as INVOIC.INVOIC02.ZINVOIC02:

- **Message Type**: INVOIC
- **IDoc Type** : INVOIC02
- **Extension Type** : ZINVOIC02

i Note

Newly created custom IDoc message types (or message type versions) automatically include an additional hierarchy level called **IDOC** below the root node. You can set this node to repeating in your MIG and model bulk IDocs in Integration Advisor.

8. Choose **Create**.

Results

The custom message is imported into the system.

Extend Standard Message for Custom IDoc

For custom IDocs type systems, the **Create** button also has the option **Extended Standard Message**. With this option you can upload an IDoc message using an XSD file and then reference that message to a standard IDoc. Standard IDocs contain an enriched structural definition such as codelists, qualifiers, long-text documentation and DateTime information. The custom IDoc

xsd files (as downloaded from transaction WE60 of your SAP S/4HANA backend system) that you upload don't carry this additional information attached to these segment definitions. Now, with this option, these details (codelists, qualifiers, long-text documentation and DateTime information) are copied from the referenced standard IDoc to the custom IDoc.

i Note

This additional information can only be copied for the standard segments (from the standard IDoc). Custom segments cannot be enriched using this option. For custom segments, you can manually enrich the MIG.

To add an extended standard message, complete the following steps:

1. Choose **Create** > **Extended Standard Message**.
2. Choose **Browse** and upload the XSD file under the **XSD File** step.
3. The next step shows you all the messages present in your XSD. Choose the required message from the **Messages** step.
4. In the **Standard Type System** step, select the type system from which you want to choose the standard message.
5. The **Standard Message** step displays the list of IDoc messages available in the selected type system. Select the required message from the list.
6. The next step, **Standard Versions**, shows the list of versions available for the message that you chose. Select the version that you need to refer to your custom IDoc.
7. The next step gives an overview of your selection. You can view a consolidated list of the standard message that you chose. Select **Create**.
8. Select and open the newly created custom IDoc. You can now see the standard segments enriched with additional information.

Adding a Custom Codelist

Create and add a codelist to the custom type system **Custom Codelist**.

Context

Follow the procedure to add a codelist to a custom codelist.

Procedure

1. Navigate to the **Custom Type Systems** from the left pane of your application.
2. Choose the **Custom Codelists** and navigate to the **Codelists** tab.
3. Choose **Add** and maintain the following parameters:

Field	Value
ID	Enter an ID for the codelist
Name	Enter a meaningful name for your codelist
Definition	Provide a decription for the codelist. You can select the Formatted Text checkbox if you want the description to be formatted.

4. Select the checkbox for **Copy code values from Codelist** if you want to copy existing code values to your custom codelist.
 - a. If selected, you have to maintain the following fields that appear:
 - i. **Type System**: Select the value help  button to select the type system. You can switch between **Standard** and **Custom** type systems in the resulting dialog and select a type system from the list.
 - ii. **Codelist**: Select a codelist from the list using the drop-down  button. The drop-down list displays the codelists pertaining to the type system you selected.
 - iii. **Version**: Select a codelist version from the list using the drop-down  button.
 - b. The available code values of the selected codelist gets displayed in the table below these fields. Use the checkbox to select the required code values.
 - c. You can also use the **Search** option to search for specific code values. If you want to select all the values, use the **Select All** checkbox.
5. Choose **Create**. This will create and open the codelist.
6. In the **Code Values** tab of the codelist, you have the following options for adding code values:
 - o Choose **Add an Entry** to add an entry in the table with no values. You can then maintain the necessary details.
 - o Choose  **Add an Entry**  **Add from CSV file**  to upload code values using a CSV file.
 In the resulting dialog, browse for and upload your CSV file by choosing **Select**.
 If you don't have a CSV file yet, see the section *Creating Codelists Using a CSV File*.
 - o Choose **Add from other codelists** to add code values from existing codelists in the system.
 In the resulting dialog, maintain the following:
 - a. **Type System**: Select a type system by choosing the  *value help*. In the resulting dialog, you can switch between **Standard** and **Custom** type systems, and select a type system from the list.
 - b. **Codelist**: Select a codelist from the list using the  *value help*.
 - c. **Version**: Select a codelist version from the list using the drop-down list.
 - d. Choose **Next**. The next step displays all the code values pertaining to the codelist you selected. All the values are selected by default.
 - e. Select the required code values from the list and choose **Add Code Values**.
 - o Choose  **Add an entry**  **Add empty code value**  to add a code value with the value (empty).
 You can't edit the value of this code value, but you can maintain a name and definition.
7. Choose **Save** to save your changes.
8. Choose **Activate** to activate your codelist. Once activated, the codelist becomes read only. To make new edits, create a new draft version of the codelist.
9. To delete a codelist, see [Deleting a Custom Codelist](#).

Creating Codelists Using a CSV File

Create a codelist using a csv file.

Procedure

1. In the **Code Values** tab of the codelist, choose the  next to **Add an Entry** and choose **Add from CSV file** to upload a CSV file.

2. In the **Code Value Upload** dialog, choose **How to... > Download the template(*.csv)** option. This downloads the CSV file template locally with the name `CodeValueDownload.csv`.
3. Edit the downloaded template file `CodeValueDownload.csv` to enter the required values and save.
Ensure you don't change or remove the header of the template.
4. Once done, you can choose **Select** to upload the updated file containing code values in the **Code Value Upload** dialog.

i Note

You can also download a standard codelist from a type system in CSV format and upload it here. To do so,

- Go to **Type Systems** and open the type system that you require.
 - Navigate to the **Codelists** tab and select a version of the codelist that you want to download.
 - In the **Code Values** tab, choose **Download**. This downloads the standard codelist in CSV format.
5. If the csv file contains any code value with an escape character, on uploading the file the list of those code values are displayed. Select the rows for which you want the escape character to be removed and choose **Remove**.
 6. If you want to skip this step, choose **Skip All**.
 7. If you want to add more code values to the uploaded values, choose **Add** and enter the required fields in the new entry.

i Note

You can use the upload option even if code values exist already. The code values in the file get appended to the existing list. However, if there are conflicting values between the list and the csv file, during upload the dialog box displays the list of already existing values. You can select the checkbox of those values that you wish to overwrite and choose **Overwrite**.

Deleting a Custom Message

Delete a custom message from the custom type system.

Context

You can also delete an uploaded custom message from the library.

Procedure

1. Navigate to the **Custom Type Systems**  from the left pane.
2. Choose and open the custom type system and navigate to the **Messages** tab.
3. This tab lists the number of custom messages uploaded for the type system. Choose **Delete**  for the specific custom message that you want to delete.

i Note

You can either delete the complete message, one complete version or one specific revision of the message.

4. In the resulting dialog box, select the desired checkbox of the message/version/revision and choose **Delete**.

i Note

- Only one revision of a message can be selected for deletion.
- If a version of the message is selected, then all the respective revision under that version gets auto-selected.

- Deletion of a revision or a version of a message that is referenced in an MIG is not possible.

Deleting a Custom Codelist

Delete a codelist from the custom type system **Custom Codelists**.

Context

Follow the procedure below to delete a codelist from **Custom Codelists**.

Procedure

1. Navigate to the **Custom Type Systems**  from the left pane.
2. Open the **Custom Codelists**. The **Codelists** tab is displayed by default with all the codelists created.
3. Choose **Delete**  for the specific custom codelist that you want to delete.
4. In the resulting dialog box, the following options are displayed. Select the desired option and choose **Delete**.
 - a. **Delete this (latest) version**: This will delete only the latest version of the codelist.
 - b. **Delete complete custom codelist (including history)**: This will delete the entire codelist including its previous versions.
5. Alternately, if you want to delete a specific version of a codelist, you can do so by following the steps below:
 - a. Select the expand  button next to the codelist to view all the versions.
 - b. Select the more options  button for the version you want to delete and choose **Delete**.

i Note

You cannot delete codelists that are in **Active** state.

- c. In the resulting dialog, choose **Delete** again to confirm deletion.

Limitations

Consider the following limitations before uploading a Custom Message/IDoc/SOA.

The following limitation applies to all custom messages:

- The maximum file size of custom message that you can upload is 10MB, with a maximum limit of 20000 nodes.

Custom Type System - Custom Messages

XSD Features with Limited Support

The following XSD features are currently only supported in a limited way. You can upload your XSD as custom message, but you might not be able to use the full scope of the XSD feature in your MIGs.

Custom Messages - Limitations		
Limitation	Detail	Resolution / Workaround

Limitation	Detail	Resolution / Workaround
Only native-XML messages supported	Full support is only given to messages that are natively XML (all messages are imported with SyntaxType = XML .) You can still import XSDs representing non-XML messages and SAP Integration Advisor will support standard XML-handling. But extended special features such as XSD creation for EDI Flow Steps are not supported.	
Recursion (limited support)	Self-recursion occurs when a child node refers to its parent complexType. In such cases, the message will be imported as Custom Message. But this recursive node cannot be used and will be disabled for selection in the MIG. Chain recursion occurs when a child node refers to an ancestor complexType. This scenario is not supported and such elements will be removed automatically when imported into the tenant .	
Global <code>xml:lang</code> attribute must be declared explicitly	If your message definition includes <code>xml:lang</code> attributes, you must explicitly declare the XML namespace.	Before the upload, add an appropriate <code>xml.xsd</code> file to your ZIP file. This <code>xml.xsd</code> file should only declare the required global <code>xml:lang</code> attribute without using more complex, unsupported XSD features. This workaround requires intermediate XSD knowledge.

Unsupported XSD Features

The following XSD features are not supported. If they are present in your XSD, XSD validation will fail and the custom message will not be uploaded. In certain cases you can resolve the situation by manually modifying the XSD before upload.

Custom Messages - Limitations

Limitation	Detail	Resolution / Workaround
Multiple XSD files for same target namespace not supported	Your message definition can't be based on multiple XSD files for the same target namespace.	Merge all XSD files of the same namespace into one XSD file. This workaround requires basic XSD knowledge.
Multiple namespace prefixes for the same namespace in the same XSD file		Use only one namespace prefix for every namespace within the same XSD file. This workaround requires intermediate XSD knowledge.
<code>xsd:include</code> not supported	Include of another XSD (same namespace) is not supported.	Merge all XSD files of the same namespace into one XSD file. (Basic XSD knowledge required.)

Limitation	Detail	Resolution / Workaround
No support for schema extension via: • <code>xsd:any</code> • <code>xsd:anyType</code>	SAP Integration Advisor always works on specific message structures and not on generic undeclared nodes. Therefore any generic schema extensions must be replaced by the specifically required elements and attributes before the upload.	Firstly, ensure whether you need any extension of your custom message structure. If yes, extend your XSD file before the upload: replace the <code>xsd:any</code> with element(s) you want to use and replace the <code>xsd:anyType</code> with the specific type you want to use. If not, restrict your XSD file before the upload: simply remove the respective elements or attributes from the xsd file. (Intermediate XSD knowledge required.)
No support for advanced XSD features like: • <code>xsd:complexContent</code> • mixed content • <code>xsd:extension</code> (of complex or simple types) • <code>xsd:substitutionGroup</code>		Some of these missing advanced XSD features can be replaced by semantically equivalent standard XSD features which are supported. It depends a lot on the specific use case to decide if this is possible or not. (Advanced XSD knowledge required.)

Custom Type System - SOA Messages

Unsupported WSDL Features

The following WSDL features are not supported. If they are present in your WSDL, it cannot be uploaded. In some cases you can resolve the situation by manually modifying the WSDL before upload.

Custom SOA - Limitations

Limitation	Detail
WSDL version 2.0 not supported	SAP Integration Advisor works on the predominantly used WSDL version 1.1 (= WSDL file starting with tag <code><definitions></code>). WSDL version 2.0 is not supported (= WSDL file starting with tag <code><description></code>).
Only one wsdl:part allowed	A wsdl:message can only refer to one wsdl:part declaration. A declaration of multiple wsdl:part for one wsdl:message is not supported.
Same global element cannot be referenced by multiple messages	All messages within one WSDL file must refer to different global elements and it is not allowed to reuse a global element in multiple messages.

XSD Features Unsupported or with Limited Support

For all the XSD Schemas declared within the **wsdl:types** section of your WSDL file, the same restrictions and limitations apply as described in section for **Custom Messages**.

Custom Type System - Custom IDocs

For custom IDocs, an IDoc-XSD file is expected as it can be downloaded from the SAP S/4HANA system (transaction WE60).

In principle the same restrictions and limitations apply as described in section for [Custom Messages](#), however, none of the advanced XSD features are used in the IDoc-XSDs.

Integration Advisor message types don't support the slash character "/" as part of their identifier. This means that IDoc types with a slash "/" in its identifier can't be uploaded.

Message Implementation Guidelines (MIGs)

Message implementation guideline, also referred to as MIG, is used as the source or target in a mapping guideline (MAG) of SAP Integration Advisor. This is created using one of the messages in the type system that is relevant to your scenario as the template.

The MIG contains all the information for implementing a customized message interface. SAP Integration Advisor uses the message in a type system as a starting point to ensure that you don't have to refer to any additional documents to implement the interface. By providing a MIG, you ensure that all users who are involved in the process of implementing the interface have a clear understanding of the guidelines.

Filtering MIGs

The [Message Implementation Guidelines](#) screen displays an overview of all the MIGs that you've created. Filters are used to search and display MIGs based on various criteria. You can use the [Free text filter](#) search or the  (*Extended filter*) at the top of the screen to specify your search criteria. The [Free text filter](#) search allows you to filter and display MIGs that match your search text. It can be used as a quick filter to match your search text against multiple attributes of your MIGs.

You can also choose  (*Extended filter*) to filter MIGs based on specific criteria. You can use the required filter fields to list MIGs that match your query. You can also customize the filter bar by choosing [Filters](#). You can expand or collapse the filter area using [Show Filter Bar](#) or [Hide Filter Bar](#).

Filter Criteria

Filter	Description
MIG Name	The Name of the MIG provided in the General Information section of the Overview tab.
Summary	The Summary of the MIG provided in the Documentation section of the Overview tab.
MIG Version	The Version of the MIG provided in the General Information section of the Overview tab.
Status	The Status of the MIG provided in the General Information section of the Overview tab.
Type System	The Type System of the MIG provided in the General Information section of the Overview tab.
Type System Version	The Type System Version of the MIG provided in the General Information section of the Overview tab.
Message Type	The Message Type of the MIG provided in the General Information section of the Overview tab.

Filter	Description
Created By	The unique identifier of the user who created the MIG. It is available in the Administrative Data section of the Overview tab.
Created Between	The date range during which the MIG was created.
Modified By	The unique identifier of the user who modified the MIG. It is available in the Administrative Data section of the Overview tab.
Modified Between	The date range during which the MIG was modified.
Last Imported By	The unique identifier of the user who last imported the MIG. It is available in the Administrative Data section of the Overview tab.
Last Imported Between	The date range during which the MIG was imported.

You can also view the version history of a specific message implementation guideline. Choose the **History** button provided below the version details of a MIG in the **Message Implementation Guidelines** page. The resulting screen would display the following version history details of the MIG:

- **Version:** This column displays the version numbers of the MIG
- **Status:** This column displays the status of each version of the MIG. There is always only one active version of a MIG. The previous version of the active MIG will be in draft status and all other MIGs versions before that will be deprecated.
- **Name:** Displays the name of the MIG in each version.
- **Modified On:** Displays the last modified timestamp.
- **Modified By:** Displays the unique identifier of the user who modified the MIG.
- **Last Imported On:** Displays the last imported date and time.
- **Last Imported By:** Displays the unique identifier of the user who imported the MIG.
- **Imported:** This column lets you know if the MIG is imported or not.
- Under ... *More Options*, you can copy, migrate, or compare your MIG.

Here are the next steps for you:

- [Creating Message Implementation Guidelines](#)
- [Working with a Message Implementation Guideline](#)

Creating Message Implementation Guidelines

Learn how to create new message implementation guidelines.

Prerequisites

You've logged into SAP Integration Advisor.

Context

A message implementation guideline (MIG) is the source/target that you use in a mapping guideline. You use a template from one of the type systems available in the library as the starting point and build your MIG based on your requirement. The application also provides you with proposals on which nodes are most appropriate for you using the existing MIGs as the reference.

Procedure

1. Go to  *MIGs*.
2. Choose **Create** in the resulting screen to create a MIG.
3. The **Create Message Implementation Guideline** wizard opens. Choose a type system from the list displayed under **Type System** step.

By default, all standard type systems are listed. If you want to select a custom type system, select the **Custom** button.
4. The list of messages under the selected type system is displayed under the **Messages** step. Choose a message from the list.
5. The **Versions** step displays the available versions of the message. Choose the desired version from the list.
6. In the **Envelopes** step, select an envelope for the type system from the list.

For **cXML**, the envelope must fit the message type as follows:

- For cXML requests, like `InvoiceRequest`, choose **EnvelopeForRequests**.
- For cXML responses, like `SearchResponse`, choose **EnvelopeForResponses**.
- For cXML messages, like `QuoteMessage`, choose **EnvelopeForMessages**.

For **ASC X12** and **UN/EDIFACT**, no limitations apply.

i Note

To see the messages allowed for an envelope, go to **Type Systems** and open the relevant type system. In the tab **Messages**, search for the envelope and open the relevant version. In the **Structure** tab, find the message placeholder and check theodelist assigned to it. The list of code values represents the allowed messages.

To continue **without an envelope**, choose **None**.

7. In the optional **Sample Payload** step, you can provide sample payload files that automatically enrich your newly created MIG. Any elements or attributes in the message structure are then auto-selected if they're present in a sample payload file.

To skip the whole step, choose **Skip**.

- a. To select and upload either a single payload file or a ZIP file with multiple payload files, choose **Browse**.
- b. Set the format of the payload in the **Data Format** field. The following formats are supported:
 - SAP Cloud Integration (XML)
 - SAP Process Integration (XML)
 - EDI Flat File: You can use this format only for type systems ASC X12, Odette, TRADACOMS, UN/EDIFACT and its subsets, and VDA. For more information, refer to the blog [Direct Support of EDI Payloads](#) .

i Note

The flat files might contain envelope segments outside the message structure. Integration Advisor will automatically consider the messages inside this envelope.

- Multiple Payloads SAP Cloud Integration (ZIP)
- Multiple Payloads SAP Process Integration (ZIP)

- Multiple Payloads EDI Flat File (ZIP)

i Note

For multiple payload files (ZIP), the upload is limited to 25 payload files and an overall size of 10MB for the zip archive.

To learn more, see the blog [Integration Advisor - multi-payload support for MIG creation](#) .

- For single payloads files, you can additionally select **Use file content as example values** to use payload values as example values.
- Choose **Next**.

8. In the **Node Selection** step, you can select nodes that are automatically qualified in your MIG based on the qualifier values present in your payload.

i Note

This step works only if required qualifying values are available in the payload. If available, the nodes selected for qualification are automatically qualified with the qualifier values contained in the payload thereby simplifying the qualification process.

Nodes without a unique qualification marker are disabled for selection.

See also the blog [Integration Advisor – payload-based qualification at time of MIG creation](#) .

9. In the **MIG Creation** step, maintain the following fields:

- Name:** Name of the message implementation guideline
- Direction:** Direction is used together with your **Own Business Context** and your **Partners Business Context** to make the **Proposal Service** more precise. The values given describe the direction from a B2B interaction point of view.
 - **In:** The MIG describes a message that you receive from a business partner.
 - **Out:** The MIG describes a message that you send to a business partner.
 - **Both:** The MIG describes a message that can both be received from a partner or sent to a partner. Alternatively, the MIG describes an A2A communication where only the **Own Business Context** is set (and where **Partner Business Context** isn't relevant).

Example: Say you receive an EDIFACT message from your Business Partner and map it to an IDoc message that is sent to your backend system. In this case, you should classify both your Source EDIFACT MIG and your Target IDoc MIG as **"In"**, because they both represent a message where your Business Partner is the logical sender and your company is the logical receiver.

- Summary:** You can provide a text description of the Message Implementation Guideline (MIG). This description is visible as short text documentation in the MIG overview list.
- Business Context :** Select **+**, and choose the business context that you want to add. Based on the business context that you add, you're provided with further options in dropdown list. To understand, **Business Context** refer to [Terminology & Glossary for SAP Integration Advisor](#).

10. Choose **Create**.

i Note

You can also create a MIG from the Library of Type Systems. Navigate to  *Type Systems*, choose the type systems based on your requirement and navigate to the **Messages** tab.

- Select your message and the respective version and choose  to create the MIG.
- Alternately, open the message in the required version. The application provides a **Create MIG** button in the message type page that allows you to directly create a message implementation guideline.

Creating Message Implementation Guidelines by Payload

Use payload-based creation to automatically generate a message implementation guideline from an existing EDI flat file, with the wizard suggesting matching type systems, message types, versions, and envelopes based on your payload.

Prerequisites

- You've logged into SAP Integration Advisor.
- Automatic recommendations are possible for the following files:
 - **EDI files:** all EDI type systems (ASC X12, UN/EDIFACT and its subsets, Odette, TRADACOMS, VDA)
 - **XML files:** only SAP IDoc and Custom IDocs

For other XML-based standards, you can still use the wizard, but it doesn't give recommendations based on the payload.

Context

A message implementation guideline (MIG) is the source/target that you use in a mapping guideline. You use a template from one of the type systems available in the library as the starting point and build your MIG based on your requirement. The application also provides you with proposals on which nodes are most appropriate for you using the existing MIGs as the reference.

In this process, you use a payload as the base for your new message implementation guideline. To create one without a payload, see [Creating Message Implementation Guidelines](#).

Based on the payload you provide, the wizard suggests options for type systems, message types, versions, and envelopes according to how closely they match this payload file. The objects and matches are grouped as follows:

- **Best match based on payload:** The one object that fits the payload the best
- **Possible matches based on payload:** Other possible matches that fit less than the best match, but are still suitable
- **Other options:** All remaining objects that don't match the payload criteria of the previous categories, but you can still select them manually

If there's no match, you choose from the full list of available options.

Procedure

1. Go to [MIGs](#).
2. Choose  **Create**  **Create by Payload** .
3. In the first step of the wizard, browse for or upload a payload file, either an EDI payload or a ZIP file containing EDI payloads, or an XML payload or a ZIP file containing XML payloads. If you select a ZIP file, you can then select the exact file before continuing.

In the next steps, the wizard then suggests available options for type systems, message types, versions, and envelopes by how closely they match this payload file. For XML payloads, suggestions are only available for SAP IDocs and Custom IDocs.
4. In the next step, choose a type system from the list of matches and other available options.

By default, all standard type systems are listed. If you want to select a custom type system, select the **Custom** button. If custom IDocs are the best match, the wizard automatically selects **Custom**.
5. Next, the list of messages under the selected type system is displayed under the [Messages](#) step. Choose a message from the list of matches and other available options.
6. The [Versions](#) step displays the available versions of the message. Choose the desired version from the list.

7. In the **Envelopes** step, select an envelope for the type system from the list.

i Note

To see the messages allowed for an envelope, go to **Type Systems** and open the relevant type system. In the tab **Messages**, search for the envelope and open the relevant version. In the **Structure** tab, find the message placeholder and check theodelist assigned to it. The list of code values represents the allowed messages.

To continue **without an envelope**, choose **None**.

8. The **Sample Payload** step shows the payload that you selected in the first step of the wizard. Decide whether to use the sample payload for **pre-configuration** by selecting **Use Payload to Pre-Configure Message Implementation Guideline**. If you do, the following sub features appear:

- o **Select Nodes** (pre-selected): All nodes that are part of the sample payload are included in the new message implementation guideline.
- o **Perform Qualification** (pre-selected): In an additional step, you can select relevant nodes for automatic qualification. If there are qualifier values in the sample payload, the nodes selected for qualification are automatically qualified with those values.
- o **Set Example Values**: Choose if the values present in the sample payload should be transferred to the new message implementation guideline. The values then show up as example values when you select a node and go to **Details Properties**, or when you run a simulation for the MIG.

To skip the whole step, choose **Skip**.

9. In the **Qualification** step, select nodes that the system then automatically qualifies in your MIG based on the qualifier values present in your payload.

This step only comes up when you chose to use the payload for pre-configuration in the previous step.

i Note

This step works only if required qualifying values are available in the payload. If available, the nodes selected for qualification are automatically qualified with the qualifier values contained in the payload thereby simplifying the qualification process.

Nodes without a unique qualification marker are disabled for selection.

See also the blog [Integration Advisor – payload-based qualification at time of MIG creation](#).

10. Finally, in the **MIG Creation** step, maintain the following fields:

- a. **Name**: Name of the message implementation guideline
- b. **Direction**: Direction is used together with your **Own Business Context** and your **Partners Business Context** to make the **Proposal Service** more precise. The values given describe the direction from a B2B interaction point of view.
 - **In**: The MIG describes a message that you receive from a business partner.
 - **Out**: The MIG describes a message that you send to a business partner.
 - **Both**: The MIG describes a message that can both be received from a partner or sent to a partner. Alternatively, the MIG describes an A2A communication where only the **Own Business Context** is set (and where **Partner Business Context** isn't relevant).

❖ Example

You receive an EDIFACT message from your business partner and map it to an IDoc message that's sent to your backend system. In this case, you should classify both your source EDIFACT MIG and your target IDoc MIG as **In** because they both represent a message in which your business partner is the logical sender and your company is the logical receiver.

- c. **Summary**: You can provide a text description of the message implementation guideline. This description is visible as short text documentation in the MIG overview list.
- d. **Business Context** : Select **+ Add Own Business Context**, and choose the business context that you want to add. Based on the business context that you add, you're provided with further options in drop-down list. To understand, **business context**, see [Terminology & Glossary for SAP Integration Advisor](#).

11. Choose **Create**.

Related Information

[Integration Advisor - Simplified MIG Creation based on EDI Payload](#) 

Working with a Message Implementation Guideline

Prerequisites

You have created a message implementation guideline (MIG). For more information on creating a new MIG, see [Creating Message Implementation Guidelines](#). You have accessed the MIG that you want to work with by creating a new MIG or choosing one from **Message Implementation Guidelines (MIGs)** section of SAP Integration Advisor home page.

Context

Once you have created a message implementation guideline, you need to select the fields that are relevant to your scenario and build the MIG accordingly. You can also get proposals from the application on which fields might be most relevant for you based on existing MIGs.

Here are a few considerations to be mindful of before you work with MIG:

- In the MIG structure, if there is one or more nodes with the checkbox selected, that node is a mandatory element as defined by the original message template.
- If you want to edit details associated with the nodes like **Name**, **Simple Type Properties**, **Description**, etc., you must select the checkbox associated with the node.
- You can maintain the qualifier markers if it was defined by the source template. You can also maintain additional markers by selecting an element that can be qualified, editing the property sheet, and maintain the instance by selecting the qualifier values.

Procedure

1. Choose **Edit**.

i Note

Ensure that the MIG you are trying to edit is not locked by other users. The **Locked By** field displays the ID of the user who's currently editing the MIG. You need to have admin rights to unlock it. To know more about the role, see [Assigning Users for SAP Integration Advisor](#)

2. Use **>** and **▼** to expand and collapse nodes.

→ Tip

- If you want to search for specific elements in the MIG, select the column header. This activates a **Filter** where you can manually enter your filter criteria.

- You can right-click on the expand/collapse icon to perform a bulk expand/collapse or select/deselect operation.
3. Choose **Get Proposals** to activate a proposal indicator that displays which fields might be most relevant for you. This is calculated based on the other available MIGs.
 4. If you want to edit the properties of a node, select the checkbox associated with the node and choose the element. This will open the **Details** pane in which you can view and change the parameters.

i Note

You can also add or modify additional notes details in the **Notes** tab.

If the data element refers to a codelist, maintain those details in the **Codelist** tab.

To know more on editing these properties, see [Working with a Node](#)

Refer to the [Terminology & Glossary for SAP Integration Advisor](#) to get an understanding of Qualifier, Notes and Codelist.

5. Review the **Message Implementation Guideline** by setting a review status, or comparing the MIG
 - a. **Set Review Status** to describe the editing status of a node regarding its properties
 - b. **Compare the structure**, documentation, properties, code values, status and example values of multiple MIGs

Related Information

- [Codelists](#)
- [Qualifying Nodes](#)
- [Primitive Data Types](#)
- [Simulating Message Implementation Guidelines](#)

Working with a Node

You can edit the properties of a node

Once you have created a message implementation guideline, you can edit and modify the properties of the nodes in the structure.

1. Choose **Edit** to edit your message implementation guideline.
2. Navigate to the **Structure** tab and choose a node that you want to edit. The node details are displayed to the right side of the structure.
3. Each node has the following details. Certain fields and section pertain only to a leaf node or a group node which is also mentioned below.

Details Tab			
Section Name	Field Name	Description	Editable
Identifier Information	Identifier	Displays the name of the node	No
	Namespace	Displays the namespace of the node	No
	Namespace Prefix	Namespace Prefix is only shown if the node comes with a namespace prefix (XML namespace qualified). If a	No

Section Name	Field Name	Description	Editable
		default namespace is used, this is indicated through <code><default></code> .	
	XML Node Name	Displays the corresponding XML tag name of the node. This might include a namespace prefix, if required. The name need not necessarily be identical to the node name.	No
	ComplexType (Group node)	Displays the complex data type of the group node	No
	Complex Type Namespace (Group node)	Displays the complex type namespace of the group node	No
	SimpleType (Leaf node)	Displays the data type of the leaf node	No
	Simple Type Namespace (Leaf node)	Displays the type namespace of the leaf node	No
Properties	Identifier	Displays the name of the node	No
	Cardinality	Displays the cardinality of the node with <code>min</code> and <code>max</code> values	Yes
	Nillable	Displays if the XML-node is nillable	No
	Qualifier (Leaf node)	This field appears only when the leaf node is a qualifying node.	You can add multiple qualifiers using the add + icon. If you want to delete a qualifier from the list, choose the delete ⊗ icon. To know about qualifying nodes and how to qualify an instance, see Qualifying Nodes
	Fixed Value (Leaf node)	You can provide a fixed value to a particular node. You can use this field for a node that can have only one fixed value. This value will be included during XML validation of any payload data.	Yes
	Example Values (Leaf node)	You can provide example values to a particular leaf node.	Choose + to add an example value.

Section Name	Field Name	Description	Editable
		You can set these example values to provide the business user a better understanding of the semantics and expected syntax of the field. These example values will be also be used during simulation when you choose Simulate with Example Data.	Choose  to edit an example value. Choose  to delete an example value. Choose  to move a value up the list. Choose  to move a value down the list.
Documentation		You can enter a detailed description of the purpose of the node.	Yes If you want to format the entered text, select the checkbox next to Formatted Text.
Complex Type Properties (Group node)	XSD Assertions	Displays the XSD assertions of the group node. i Note These assertions will be added to the MIG XSD runtime artefact and will be used to improve the payload validation. Each entry should be a valid boolean XPath expression. If the XPath value you add references to a node in your MIG message structure, you need to enter the value of the node name used in the XML payload. This information can be found in the XML Node Name field. If you specify more than one XSD Assertion, then all the assertions must evaluate to true.	Yes. Choose the add button  to add a XSD assertion. Select a XSD assertion and choose the edit button  to edit the XSD assertion. Choose delete button  to delete a XSD assertion Choose  to move an assertion up the list. Choose  to move an assertion down the list.
Simple Type Properties This section is displayed only for leaf nodes	Primitive Type i Note Depending on the data type, other relevant fields may appear. Maintain the values for these fields if necessary. For example, for the data type Date , Time or	You can select a primitive data type for the node	Yes Choose the right data type for the node from the drop-down list.

Section Name	Field Name	Description	Editable
	DateTime , you can maintain the field Date Time Format .		
	Length	You can specify the minimum and maximum length of the value for the node	Enter the minimum and maximum length of the value in the min and max fields respectively.
	Format	You can specify a value pattern for the data type. For example, nnnn for a four-digit number. i Note This is free-style text and is used only for documentation purposes. For Date or Time patterns preferably use Date Time Format.	Yes
	Date Time Format	You can choose one of the predefined Date, Time or DateTime patterns. This will be used when defining a Date Time Conversion in the mapping.	Yes
	XSD-Patterns	Enter the XSD patterns for the leaf node. i Note These patterns will be added to the MIG XSD runtime artefact and will be used to improve the payload validation. Each entry should be a valid boolean XSD Pattern expression. If the XPath value you add references to a node in your MIG message structure, you need to enter the value of the node name used in the XML payload. This information can be found in the XML Node Name field. If you specify more than one XSD Pattern, then the	Yes Choose the add button + to add a XSD pattern. Choose the edit button ✎ to edit the XSD pattern. Choose delete button ⊗ to delete a XSD pattern. Choose ^ to move a pattern up the list. Choose ▼ to move a pattern down the list.

Section Name	Field Name	Description	Editable
		field value must match to one of the given patterns.	
Syntax Type Related	External Category	Displays the external category detail of the node.	No
	Syntax Data Type	Displays the data type of the syntax of a leaf node as it is originally defined by the B2B standard.	No
	Position	Displays the position of the syntax type of the node. i Note This field data is relevant for EDI standards such as, ASC X12 and UN/EDIFACT.	No
	Level	Displays the level of the syntax type. i Note This field data is relevant for EDI standards such as, ASC X12 and UN/EDIFACT.	No
XPath Expressions	CI Format	Displays the XPath expression for the node which can be used in Cloud Integration version of the message before pre-processing.	No You can copy this XPath expression using the copy  button provided next to it.
	IA Format	Displays the XPath expression for the node which can be used in Integration Advisor version of the message after pre-processing.	No You can copy this XPath expression using the copy  button provided next to it.

Notes Tab

You can use this tab to enter notes specific to that node.

1. To add a note, choose **Add**. The **Create Note** dialog box appears.
2. Choose a **Category** for the note. The different types of notes that can be added to a node are:
 - Comment
 - Constraint
 - Example
 - Usage
 - Technical Information

3. Provide a number for the note type that you are creating under the **Number** field.
4. Enter the text under the **Text** field. If you want to format the text, select the checkbox next to **Formatted Text**.
5. Choose **Apply**.
6. You can also sort the notes under each category based on the number using the ↑↓ button.

Status Tab

You can use the **Status** tab to set the review entry for a node. This helps to identify nodes that you want to work on later.

This tab can also be used for multi-user review.

Codelist Tab

This tab is available only for leaf nodes.

To know how to create and assign codelist to a leaf node, see [Codelists](#)

Read this [blog](#)  to know more about the extended semantic validation options for MIG structures.

Codelists

A codelist is a collection of acronyms and their meaningful descriptions that provide a common understanding of the code values between business partners, reducing the risks associated with business collaboration.

Codelists are developed and maintained by international Standards Developing Organizations (SDO). For example, the codelist ISO 3166-1 that is published by the International Organization for Standardization (ISO) includes values for the identification of the name of the country or other geographical entity.

Code values provide a more precise meaning to semantically generic elements in a Message Implementation Guideline (MIG). For example, code values can be used to qualify a generic element such as "Party" to mean either a "seller party" or a "buyer party".

You can see all the available codelists provided by a type system in the **Library of Type Systems**. The total number of code values appears in brackets beside any selected codelist. The complete list of code values and their descriptions are available in the Code Values tab when you select a specific codelist.

SAP Integration Advisor currently supports four different types of codelists:

- **Type-system based codelists** are provided by the type system itself. They can be used only in MIGs that are based on the same type system.
- **Reusable codelists** are provided and maintained by organizations such as ISO or United Nations - Economic Commission for Europe (UN/ECE). They can be used in MIGs that are based on any type system. For example, the ISO-based codelists such as country code, language code, currency code, or the UN/ECE based codelist such as measure unit code. They're represented as separate type systems that only contain codelists.
- **MIG codelists** are locally defined in a MIG by the users.
- **Local codelists** are only defined in the context of a specific element and can't be used at any other place. They're predominantly used in the context of custom messages.

You can identify the MIG elements that have a reference to a codelist by the corresponding checkmarks in the Codelist column. When you select an element that refers to a codelist, the Details panel opens for the selected element. The codelist tab of the Details panel shows the code values, and additional details about the codelist. You can select or deselect any code value.

Related Information

[Creating MIG Codelists](#)

[Working with Codelists](#)

Working with Codelists

Learn how to work with codelists in message implementation guidelines, edit code values, and assign codelists to leaf nodes.

Viewing Codelist Details

To view the codelist details, do the following:

1. In **MIGs**, open a message implementation guideline.
2. Go to the **Structure** tab.
3. In the structure, select a node. If there's a check in the column **Codelist** for that node, a codelist is currently assigned to it.
4. In the panel next to the structure, go to the **Codelist** tab. Here, you can review the following information:

- o **Selected Codelist:** View the currently active codelist. There can only be **one active codelist** for a node at a given time, but you can also select none.

To learn more about changing the codelist or adding more candidates, see [Assigning Codelists to Leaf Nodes](#).

- o **General Information:** Review details like ID, name, type system, or version.
- o **Documentation:** View the definition of the codelist.
- o **Code Values:** View a list of code values of the codelist that are currently used in the MIG.

To learn more about selecting code values, see [Working with Code Values](#).

Working with Code Values

In the subsection **Code Values**, you can see which code values of the codelist are currently used in the message implementation guideline.

i Note

To work with code values, you must be in edit mode.

To use additional code values in the message implementation guideline, simply select the relevant code value from the list. When searching for a specific code value, narrow down your search by filtering for **Value** (ID), **Name**, and/or **Definition**.

The following additional functions are available to you:

- **Use all code values of the codelist:** To use the codelist as it is, with all values selected, select the checkbox **Use all code values of the codelist**. Alternatively, you can manually select or deselect code values as needed.
- You can also manually select all code values one by one **without selecting the checkbox Use all code values of the codelist**. In that case, you select all the code values without the decision to use the complete codelist as it is.

Following this behavior has the following consequences, for example, if you migrate your MIG to a new message version and the new version provides more code values:

- If you selected **Use all code values of the codelist** in your old MIG, the checkbox is also selected in your migrated MIG, and you benefit from the extended code values.
 - If you didn't select the checkbox, each code value is migrated by itself, and any new code values stay unselected.
- **No validation of code values:** If you don't want the extended semantic validation of your MIG to check the value of the field against the code values, select the checkbox **No validation of code values**.

Use this checkbox if the payload can contain other values beside the code values. If it's selected, the MIG RD XSD won't contain an enumeration for the code values, which means that the semantic validation doesn't check against the code values.

- To use a code value as an **example value**, choose **+ Add the Code Value as an Example Value**.

Assigning Codelists to Leaf Nodes

To assign a codelist to a leaf node, or to replace or remove the currently assigned codelist, perform the following steps.

Prerequisites

- You've opened the message implementation guideline (MIG) in edit mode.
- You've included the leaf node that you want to assign a codelist to in your MIG by selecting it.

Procedure

1. In the **Structure** panel, select the leaf node that you want to assign a codelist to.
2. In the panel that opens, go to the **Codelist** tab.
3. **Selected Codelist** shows the codelist that's currently used in the MIG, with additional details displayed in the subsection **General Information**. To change the codelist, do one of the following:
 - Select the desired codelist from the drop-down list, which shows all candidate codelists that you can switch to. To remove a codelist from this drop-down, choose **✕ Remove codelist from available codelists**.
 - To use a codelist that's not in the drop-down list yet, choose **+ Add Codelist to Available Codelists**. In the upcoming dialog, use one of the following options:
 - To assign a **MIG codelist** to the leaf node, choose **MIG Codelists**.
 - To assign a **type system-based codelist** to the leaf node, choose **From Type System** and select the desired type system from the drop-down list, and the desired version from the second drop-down-field, if requested. Upon selecting the type system and version, the list of available codelists is displayed. Choose the needed codelist from the list.
 - To assign a **codelist from a custom type system**, choose **From Custom Type System** and select the desired custom type system from the drop-down field. Upon selecting the custom type system, the list of available codelists is displayed. Choose the required codelist version from the list.
 - To remove the active codelist from a leaf node and not assign a new one, select **None**.

Selected Codelist now shows the codelist assigned to the leaf node, if any.

4. Save the message implementation guideline.

Creating MIG Codelists

If the reusable or type-system based codelists don't meet your requirements, you can create new message implementation guideline (MIG) codelists.

Prerequisites

You've opened the message implementation guideline in the edit mode.

Adding Code Values

Procedure

1. On the **MIG Codelists** tab, choose **Create** to create a new codelist.
2. Specify the codelist identifier, name, and definition in the corresponding fields under **MIG Codelists**.
3. Choose **Code Values**.
4. On the **Code Values** tab, add new code values by choosing one of the following options:
 - **Create**: Enter a value, name, and definition.
 -  **Create**  **Create Empty Code Value** : Enter a name and definition. The value (empty) is not editable.
 - **Upload**: Import a CSV file containing code values. See the section *Adding Code Values by Uploading a CSV File*.
5. You can download the code values in **CSV** format using the **Download** option. The downloaded file corresponds to the CSV template used for the codelist upload. You can use it, for example, to update the downloaded CSV file and re-upload it.

Adding Code Values by Uploading a CSV File

Procedure

1. On the **MIG Codelists** tab, choose **Create** to create a new codelist.
2. Specify the codelist identifier, name, and definition in the corresponding fields under **MIG Codelists**.
3. Choose **Code Values**.
4. On the **Code Values** tab, choose **Upload** to import a CSV file containing code values.
5. In the **Code Value Upload** dialog box, choose **Select**, and browse and upload the CSV file containing code values. You have the following options for uploading a file:
 - If you already have a **CSV file** containing code values, choose *Browse or drop a file* and select the file.
 - You can **download a standard codelist** from a type system in CSV format and upload it here:
 - a. Go to **Type Systems** and open the required type system.
 - b. Go to the **Codelists** tab and select the version you want to download.
 - c. In the **Code Values** tab, choose **Download**.
 - If you **don't have a CSV file** yet, in the **Code Value Upload** dialog, you can download a template by choosing  **How to...**  **Download the template (*.CSV)**. The file is saved locally with the name `CodeValueDownload.csv`. Edit the downloaded template file `CodeValueDownload.csv` to enter the required values and save, but make sure not to change or remove the template header. Then, select the file in the upload dialog.
 - You can use the upload option even if there are **already code values in the list**. In that case, the code values from the file are appended to the existing list.

However, if there are **duplicate values** between the list and the CSV file, during the upload, a dialog displays the list of already existing values. Select the checkbox of the values that you want to overwrite and choose [Overwrite](#).

6. If the selected CSV file contains any code value with an escape character, the list of those code values is displayed when on uploading the file. Select the rows for which you want the escape character to be removed and choose [Remove](#).

To skip this step, choose [Skip All](#).

7. You can add more code values manually by choosing [Create](#) and entering the required fields.

Qualifying Nodes

You can qualify nodes to provide context to the semantically generic elements of your Message Implementation Guideline (MIG).

The message structure is composed of logically related pieces of information grouped into generic data segments. You can customize and precisely specify the context of generic elements in your MIG message structure to provide business meaning to the rest of the data in a segment. For example, a generic element like Date/Time reference or Address is unclear at the semantic level as to which date or address the message interface refers to. As a result, additional information is required to remove any ambiguity related to the semantic purpose of a segment. Qualifiers can be used to specify which type of address is stored within a particular instance of a data segment.

An address segment generally consists of a similar set of fields such as - street, city, state, postal code, and country. However, there may be more than one set of addresses such as a delivery address, a contact address, or a billing address. In these situations, you could use a common address segment structure and then use a qualifier to specify which type of address is stored within a particular instance of a data segment.

Qualifiers are provided by a leaf node (qualifying node) in the form of codelists and related code values. A leaf node can qualify a group node or another leaf node (peer node) only if it has a codelist assigned to it. See [Working with Codelists](#). The link between the qualifying node (leaf node) and the node you want to qualify (qualifiable node) is expressed by a qualifier marker arrow in gray. A qualifier marker can go from a qualifying element to a parent group element or a peer element at the same level. It helps identify the elements that can be qualified. Qualifier markers are either automatically provided by message templates from type systems, or you can create your own qualifier markers to suit your business context. See [Creating a Qualifier Marker](#).

A qualified group or peer leaf node is indicated by a blue arrow marker from the qualifying leaf node to the qualified node along with the qualifier value provided by the selected code values. The name of the qualifier value is appended to the qualified instance. See [Creating a Qualified Instance](#). You can create multiple instances of a qualified element by repeating it with different qualifier values. Each qualified instance can have its own customization.

Creating a Qualifier Marker

Create qualifier markers that can be used to qualify a node.

Prerequisites

- You've opened the MIG in the edit mode.
- You've selected both the qualifying node and the node that you want to qualify to include them in your MIG.
- A qualifier marker from the qualifying node to the node you want to qualify isn't available in the **Constraint** column.
- The cardinality of the qualifying node must be either [1..1] or [0..1].
- The qualifying node has a codelist assigned to it. For more information, see [Working with Codelists](#).

Context

The qualifier feature allows you to set a qualifier marker that identifies the element that is qualified by another data element. Qualifier markers are either automatically provided by message templates from type systems, or you can create your own qualifier markers to suit your business context. The following procedure explains how to create a qualifier marker.

Procedure

1. On the **Structure** tab, select the leaf node that you want to use as the qualifying node.
2. On the **Details** tab of the resulting pane, choose **+ Add** in the **Qualifiers** field.
3. In the resulting **Qualifiable Nodes** dialog, select the node you want to qualify from the list of qualifiable nodes.

The list of qualifiable nodes contains all sibling nodes of the qualifying node and suitable ancestor nodes. The ancestor nodes are shown up to the first repeatable ancestor node (maximum cardinality > 1).

Message root nodes and the technical **IDOC** node (in MIGs for SAP IDocs) can't be qualified and are therefore not listed.

4. Choose **Create**.

Results

The selected qualifiable node appears in the list of **Qualifiers** and a new qualifier marker arrow appears in gray from the qualifying node to the qualifiable node.

Creating a Qualified Instance

Create and customize a new qualified instance.

Prerequisites

- You've opened the Message Implementation Guideline (MIG) in the edit mode.
- You've selected both the qualifying node and the node that you want to qualify to include them in your MIG.
- A qualifier marker from the qualifying node to the node that you want to qualify is available in the **Constraint** column. For more information, see the section *Creating Qualifier Markers* below.
- If you want to use multiple qualifying nodes(compound qualification), ensure that the node to be qualified contains more than one qualifier markers.

Procedure

1. On the **Structure** tab, select and right-click the node that you want to qualify.
2. Choose **Qualify Node** from the context menu.
3. This opens the **Create Qualified Node** dialog box.

The **Qualifiers** section of the dialog box displays the list of qualifiers.

4. For Qualifier that is numbered **1**, choose a qualifier marker from the drop-down list under **Selected Qualifier Marker**.
5. This will display a list of values that are available for the selected qualifier marker. Choose a value from the list by selecting the checkbox.

You can choose more than one qualifying values from the list. Choosing multiple values means that the node should qualify either of the selected values.

You can select the checkbox **No Value** if the node is not present in the payload.

6. You can also add more qualifiers to qualify the node. To do so, the node to be qualified should have more than one qualifier marker. Choose **+**. This adds one more qualifier to the **Qualifiers** list.

The **+** button will be disabled if the qualifier contains only one qualifier marker.

i Note

You can add a maximum of 3 qualifiers to qualify a node.

Adding more qualifiers means the node now has to meet the conditions mentioned in all the qualifiers.

You can disable or enable a qualifier using the toggle button provided next to each qualifier in the list.

7. Choose the newly added qualifier that is numbered as **2** and perform steps 4 and 5 to add the qualifying values.
8. Once you have entered the required qualifiers, choose **Add**.

The qualifier marker arrow turns blue and has the selected qualifier value(s). The **Qualifier** tab in the right pane displays the details of the qualification.

The readable name of the qualifier value is appended to the original name of the qualified node.

9. Optional: To customize the qualified instance, perform the following:

- a. Select or deselect the subnodes as required.
- b. Edit the properties.
- c. Customize theodelist values assigned to the selected leaf nodes.

Additional Options For Qualification

After qualifying the node, you can now perform the following functions from the context menu of the qualified node

- **Change Qualified Node** allows you to modify the qualifier values of the node.
- **Change Order of Qualified Node** allows you to change the order of the qualified nodes.
- **Duplicate Qualified Node** creates a duplicate version of the qualified node where you can set different qualifier values.
- **Duplicate Qualified Node for Remaining Values** creates a duplicate version of the qualified node that includes values that are not part of other instances of the same element group.
- **Add Qualifier** allows you to add another qualifier to the node. By using this option you can also extend a simple qualification to a compound qualification.
- **Delete Qualified Node(s)** allows you to delete one or more qualified nodes. Selecting this option provides you with two options
 - **Delete this Qualified Instance:** If the selected node has other qualified instances, then this option only removes this instance and the node stays qualified.

If the selected node has only one qualified instance, then the instance is removed and the node gets unqualified. The details of the earlier qualified node will be copied to the now unqualified node.

i Note

The name and cardinality of the qualified node as well as the selected code values of the qualifying node will not be copied and they will be returned to the standard values from the message template:

- **Delete All Qualified Instances:** On choosing this option, the list of all qualified nodes corresponding to the selected node gets displayed. If you want to retain the details of any of the qualified nodes, select the node from the list. This will copy the details of the selected qualified node to the unqualified node.

i Note

The name and cardinality of the qualified node as well as the selected code values of the qualifying node will not be copied and they will be returned to the standard values from the message template:

Primitive Data Types

Primitive data types are the basic building blocks that you use to define entities or structure types within a MIG.

You can set the data type of a node under the **Primitive Type** field in the **Simple Type Properties** section.

Date Time Type

- **Date:** The Date data type consists of year, month, and day information to represent a date value.
- **Time:** The Time data type consists of hour, minute, and second to represent a time value.
- **DateTime:** The DateTime data type consists of date and time information.

There are various supported formats for this type and you can set the required format under the **Date Time Format** field. The terms used under each type is explained here:

- Date
 - **CC** : Century
 - **YY** : Year (00-99)
 - **MM** : Month (01-12)
 - **DD** : Day (01-31)
 - Example for **CCYYMMDD**: 20210731
- Time
 - **hh** : hour (00-23)
 - **mm** = Minute (00-59)
 - **ss** = Second (00-59)
 - **f** = Fractional Second (0-9)
 - Example for **hhmmss**: 180545
- Timezone offset
 - **(+|-)** : plus or minus sign
 - **hh** : hour (00-12)
 - **mm** : Minute (00-30)
 - **Z** : short form for UTC timezone

- Example for **(+|-)hhmm**: +0100
- **'...'**: Fixed literals
 - Example for **CCYY'-'MM'-'DD**: 2021-07-31
- **[...]**: Optional value
 - Example for **hhmm[ss]**: 180545 or 1805
- **(...|...)**: Alternative values
 - Example for **(+|-)hhmm**: +0100 or -0100
- *****: Repeating parts (0 or more repetitions)
 - Example for **hhmmss.ff***: 180545.9 or 180545.99 or 180545.999 etc. (any number of fraction digits allowed)

Apart from the supported formats, the following are the list of **Date Time Format** descriptions that is used to provide additional input for a selected type system:

- **XML-based Type Systems**: Elements based on special primitive types such as **xsd:dateTime** or **xsd:date**
- **Edifact (including Edifact subsets)**: Data Elements with format defined by code values in Codelist 2379. Example values: 102 or 204
- **X12**: Data Elements with format defined by code values in Codelist 1250. Example values: D8 or DT
- **X12**: Data Elements based on special X12 data types of a certain length. For example, DT6..6 or TM4..6
- **IDoc**: Data Elements based on special ABAP built-in data types. For example, DATS or TIMS

The  button allows you to set examples using **Add Example Values** and also use user-defined format provided in the **Format** field using **Use Format** option. The example values can also be later used in getting proposals during MAG simulation.

i Note

The input field supports search by both **Format** and **Format Description**. So for example, if you type in 102, you will be directly guided to the date time format corresponding to Edifact code 102.

This type is also used in the mapping of Date Time Conversion. See: [Value Transformations](#)

Other Data Types

Numeric Data Type: The supported types are:

- Decimal
- Integer
- Float

Character String Type:

- String
- NumericChar

Boolean: Represents the concept of binary-valued logic

Enabling Arbitrary Composite Ordering

Automatic reordering of repeating composite data elements in UN/EDIFACT segments to match your message implementation guideline sequence. Use this feature to handle business partner payloads where composite instances arrive in varying orders, eliminating the need for complex mapping logic to accommodate different positioning.

Prerequisites

- This feature is only available for UN/EDIFACT type systems.
- The segment must have repeating composites, like PIA, GIN, or GIR.
- All selected composite data elements must be qualified, use exactly one qualifier, and have only one qualification variant.
- All qualifier values are unique across all composite data elements.
- The message payloads only contain qualifier values configured in the MIG. Within one segment instance, each qualifier value is only used once.

Context

In certain EDIFACT segments (for example, PIA), the same composite element can appear multiple times. Business partners can send these repeating composite instances in any order that's valid according to the EDIFACT standard. This variation in order can cause mapping errors when a mapping assumes a fixed position for each composite instance.

Integration Advisor can automatically reorder the repeating composite instances in the incoming payload to match the order defined in your message implementation guideline. No complex mapping logic is required.

The feature only works if you've applied **qualification** to the repeating composites. The qualifier helps distinguish, for example, buyer part number and supplier part number. Without qualification, the system can't determine the intended order.

Procedure

1. Go to [MIGs](#) and open the message implementation guideline you want to work with.
2. In the [Structure](#) tab, choose [Edit](#).
3. Select a segment that has repeating composites. It's marked by grayed out  *Arbitrary Ordering of Composites* icon. All repeating composites must be qualified.
4. In the [Details](#) tab, go to the [Complex Type Properties](#) section.
5. Optionally, choose [Validate](#) to check if the selected segment fulfills the recommended usage pattern. Review the validation results, but note that the validation only serves as advice and doesn't block you from enabling the feature even if not all prerequisites are fulfilled. The feature can still work if not all prerequisites are met, but it's unlikely in cases with complex payloads.
6. Select the checkbox [Enable Arbitrary Ordering of Composites](#) and save your changes.

Results

The  *Arbitrary Order of Composites* icon is now blue to show that the feature is enabled.

During preprocessing, Integration Advisor now automatically sorts the composites according to the order defined in the message implementation guideline.

Related Information

Special Cases

This chapter covers the special conditions that could occur when working with a message implementation guideline.

Recursive Nodes

A recursive node is a node in the message structure which creates an infinite recursion in the structure.

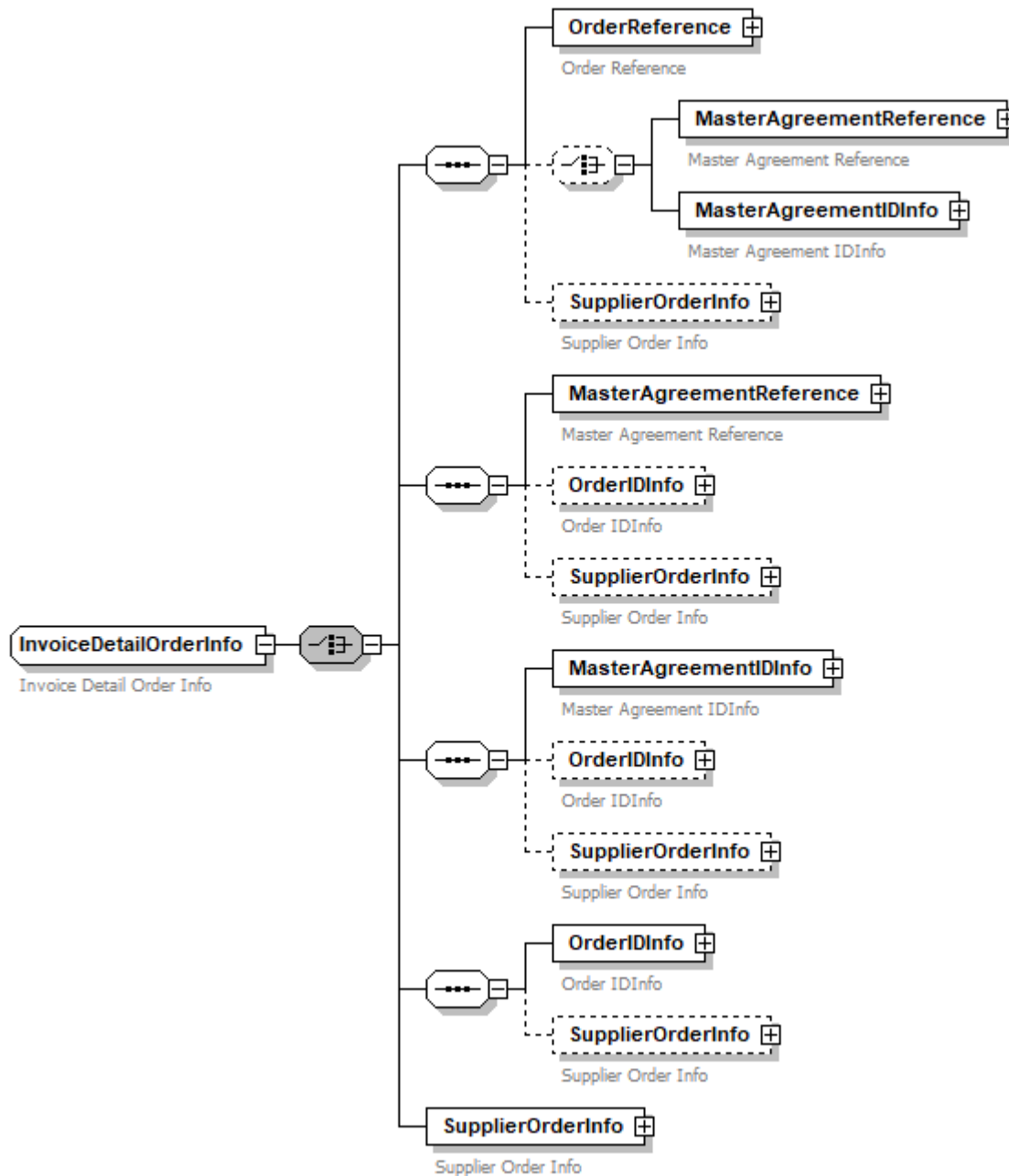
There could be two kinds of recursions:

- **Self-recursion:** The node points to its parent Complex Type.
- **Chained-recursion:** The node points to a different Complex Type, but one or more elements in the referred Complex Type again points to first Complex Type.

As the node creates an infinite substructure, it cannot be displayed in MIG Editor and also it cannot be used directly in other situations such as Mapping Guidelines. SAP Integration Advisor does not currently support recursive nodes. However, self-recursive nodes can be imported as part of Custom Messages and can be displayed in the message structure. But they still cannot be used in a MIG and hence selecting that node in the structure is disabled.

Repeating Elements

The XSD Standard allows a special situation on schema level in which a group node can contain a child node with the same identifier multiple times. This is only possible if these repeating nodes appear in different choice branches and if this can be uniquely resolved at runtime. Also, all these repeating nodes must be identical in their details and can not be customized differently.



If your MIG contains such a node, then the relevant information can be found under the [Documentation](#) section.

In such cases, the payload validation will not be possible as the XSD will be considered invalid according to the XSD specification. The validation can be achieved if your MIG uses only one node from the choice branches so that there is only one node with the same identifier. This can be done by selecting the checkbox of the respective node in your MIG structure.

Simulating Message Implementation Guidelines

Learn how to simulate a message implementation guideline.

1. Open your message implementation guideline and navigate to the [Structure](#) tab.
2. Choose [Simulate](#) to perform an embedded simulation. There are two different ways to simulate the MIG:
 - a. **Simulate with MIG Example Data:** Simulates the message implementation guideline with the existing sample MIG data.

- b. **Simulate with Payload Data...**: Prompts you to upload a sample payload file (XML or EDI Flat file) for the MIG simulation.

i Note

The EDI flat file is supported on EDI-based type systems such as ASC X12, Odette, TRADACOMS, UN/EDIFACT and its subsets, and VDA.

You can also re-run the simulation based on the option you chose using the ↺ button that appears above the MIG structure after simulation.

In case of a payload data, re-run simulation runs the simulation again with the payload file uploaded earlier. If you want to use a modified or new payload file for simulation, choose **Simulate** and upload the payload file again under **Simulate with Payload data**.

3. A new column **Simulation Data** appears in the message structure populated with the following details

- The rows with values in the structure are the input values from the payload data.
- If there are multiple occurrences of values in the payload, they are denoted by the navigation buttons <>. These buttons allow you to navigate to the other occurrences of the payload values.
- The navigation button <> is also displayed for a group node if it occurs at least once in the simulation data.
- The □ icon denotes the leaf node exists in the structure but is empty.
- The □ icon denotes that the leaf node does not exist in the structure.

How Does Simulation Work?

Simulate with Example Data

On choosing this option, a complete payload file is generated by the application that corresponds to the MIG definition and is used as input during simulation

Cardinality

The cardinality of the group node and leaf node depends on the maximum cardinality when simulating with the example data.

- If **maxOccurs**= 1, the node gets created only once (within its parent node)
- If **maxOccurs**>1, the node gets created twice independent of the **maxOccurs** value. This is done so that the repetition of the node is considered without creating very large payload files.

Values for Leaf Nodes

The data corresponding to each leaf node is fetched in the following order:

1. Example Value:

These are the values that are defined as example values for a particular node. To know how to define these values, see [Working with a Node](#). During simulation, each value from the list in the given order gets assigned to a particular occurrence of a leaf node. When the last value in the list is reached, then the first value is fetched again for the next occurrence. If there is only one example value provided, then the same value is fetched for all the occurrences of the leaf node.

2. Fixed Value:

If a leaf node has no example value but has a fixed value applied to it, the simulation then fetches the fixed value for all the occurrences of the leaf node. To know how to define a fixed value for the node, see [Working with a Node](#).

3. Code Value:

If there is neither an example value nor a fixed value for a particular leaf node, the simulation then considers the data from the code values assigned to a leaf node. To know more about creating and assigning a code list to a leaf node, see [Codelists](#).

If there are any selected code values, only those values are fetched.

If no code values are selected, then all the code values are used.

Similar to example values, each value from the code value list in the given order gets assigned to a particular occurrence of a leaf node. When the last value in the list is reached, then the first value is fetched again for the next occurrence.

4. Dummy Value

If none of the above values are defined for the node, a dummy value is generated.

For **text** nodes, the dummy value is based on the node identifier. For nodes containing **numbers**, like integers or decimals, a fitting number is produced. And for nodes containing **date or time information**, an appropriate date or time value is produced if the required format is known.

Simulate with Payload Data

Choosing this option allows you to browse and upload a payload data from your local system.

The payload data should be uploaded in the XML or a flat file (EDI text files such as EDIFACT and X12) format. The payload you are trying to import might contain additional nodes, or nodes that are not defined in the MIG. These nodes will not be considered during simulation and their details are displayed in a dialog box after importing the payload data. Choose **OK**.

i Note

- The payload file should not have any qualifiers added to it. MIG simulation converts your payload file automatically into the ICA format including those qualifiers.

Versioning a Message Implementation Guideline

This chapter covers the lifecycle of a message implementation guideline, the purpose of each status that a MIG goes through and the procedure to activate the MIG.

There are 3 different status types for an MIG:

- Draft
- Active
- Deprecated

When you create a new message implementation guideline, the newly created MIG has the **Status** field set to **Draft** and the **Version** set as **1.0**. You can now modify this new version of MIG according to your business requirements. You can also copy an existing message implementation guideline. The copied MIG will have the **Version** set as **1.0** and also the term **Copy** suffixed to its name. You can later edit the name of the MIG.

When you activate a message implementation guideline, the **Status** of the MIG is set to **Active**. Once activated, this version of the MIG will be finalised and it cannot be edited thereafter. But you can continue to export documentation and runtime artifacts from

the MIG. Activating a MIG will also anonymously push the message implementation guideline into the knowledge graph to contribute and improve the proposal service.

i Note

It is always recommended to activate a MIG to use it productively. By doing so,

- You protect the MIG against any unwanted changes
- Any new requirements for the MIG can be maintained in a new **Draft** version of the MIG without changing the active MIG version currently in use.

If you want to create the next draft version of an active MIG, open the active MIG version and choose **Edit**. You can also choose **Edit** from the action  button from the MIG overview. This will prompt you to confirm creating a draft version. Choose **OK**. Now the next version of the MIG gets created with the **Draft** status. For example, if the active version of the MIG is 2.0, then the new draft MIG will have the version set as 3.0

After the successful activation of a MIG, the previous active version of the MIG gets deprecated and its **Status** field is set to **Deprecated**.

i Note

There can be only one active and draft version of a particular MIG.

Follow the procedure below to activate a message implementation guideline.

Procedure

1. Navigate to **Message Implementation Guideline**  from the left pane.
2. Choose the MIG that you want to activate.
3. Choose **Activate** to mark a version of the MIG as a main version.

i Note

Any subsequent edit operations on the same MIG would be saved as a different version.

After successful activation of MIG, a confirmation box appears with the following information

- List of values that get published into the knowledge graph
- List of values that will not get published into the knowledge graph

If the message implementation guideline is based on a custom message, then the custom message should be activated first before activating that MIG. To do so, follow the procedure below.

Activating a Custom Message

1. In your MIG, choose ... and select **Activate Message**.
2. A message is displayed to show the status of the activation.

You can also view the status after activation under the **Message Status** field in the **Overview** tab of the MIG.

i Note

Once activated, the version of the activated custom message cannot be deleted.

3. You can now activate your MIG after activating the custom message the MIG is based on.

Migrating a Message Implementation Guideline

You can migrate your message implementation guideline (MIG) to a different version (newer or older) of the same message or even to a different message (of the same or a different type system). A new MIG is created based on the new message that you have chosen and customizations done in your previous MIG are copied over to the new MIG based on a best-effort strategy.

Context

MIGs are developed based on predefined message definitions of a specific type system version. A MIG derives its structure definition, properties, documentation, and codelists from the underlying message template. The message structure can then be customized to meet specific business requirements.

Let's consider a scenario in which you and your business partner 1 agreed on using an UN/EDIFACT D.01B S3 version ORDER message. You've created a MIG based on this specific version of the message from the type system library. You've also customized your MIG definition to meet your business needs. Later, you realize that you need to build another MIG for Business Partner 2 who is also using the same message of the same type system but a different version, say, D.16A S3. However, messages from two different versions of the same type system have similar settings except for a few variations. Earlier, you had to create a new MIG based on the new version and redo the customizations that you had done in the existing MIG. Instead, you can now migrate your existing MIG to a different version of the same type system. After migration, a new MIG is created based on the chosen version including the customizations done in the existing MIG.

i Note

Customizations or nodes that are incompatible with the new version aren't migrated. They're notified through error messages.

After migrating your MIG, you can use the status review section available in the MIG editor to view the system report, add comments, and mark review status for all the selected nodes that are part of the MIG. You can identify nodes needing further rework or discussion by reviewing the icons in the **Status** column on the **Structure** tab. You can also compare the MIGs after migration to review them. Comparisons are made across the structure definition, properties, documentation, code values, and example values of the MIGs. This allows easy identification of similarities and variations across MIGs.

Other Supported Scenarios

- When a newer revision for a specific type system version is available, you're notified of it during migration. You can choose to migrate your existing MIG, in this case, to a newer revision of the type system version so that you can use the already customized MIG in the newest revision.
- You can also choose to migrate your MIG to a different type system if the underlying message structure is similar enough, such as from EDIFACT to EANCOM or from SAP S/4HANA Cloud SOAP to SAP S/4HANA On Premise SOAP. The compatibility is determined on a case-to-case basis.
- You can also migrate your MIG to a different message such as from ORDERS05 to ORDERS03 in IDoc version 1809 FPS02 if the underlying message structure is similar enough. The compatibility is determined on a case-to-case basis.
- You can also migrate your MIG to a different message type with different root node. The migration is done on a best-effort strategy for the subnodes that are present in the old and new message structure.

Procedure

1. On the [Message Implementation Guidelines](#) screen, choose ... *More Options* on the MIG that you want to migrate and then choose [Migrate](#).

→ Tip

The [Migrate](#) option is also available on both the [Overview](#) tab and the [History](#) link of the MIG that you want to migrate.

2. In the [Migrate Message Implementation Guideline](#) dialog box, on the [Versions](#) tab, select the version that you want to migrate your MIG to.

i Note

Choose the [Messages](#) tab or the [Type Systems](#) tab to migrate your MIG to a different message or a different type system.

The resulting [Migrate Message Implementation Guideline](#) dialog box displays the details of the MIG including the new version number.

3. Optional: You can review the details and optionally choose to customize by adding more details regarding your and your partner's business context. To add your or your partner's business context, choose [+](#) *Add* to provide more details. This helps the application provide an appropriate proposal based on the defined business context. The accuracy of the proposal depends on the number of selected business context categories and business context values.
4. Choose [Create](#).

Results

A new message implementation guideline is created.

→ Tip

A few pointers to keep in mind while you review and verify your MIG after migration:

- The migration works on a best-effort strategy and considers frequently occurring situations. But there can be situations (when it comes to considering all three of previous message, previous MIG and new message) which can't be easily resolved and might need user interaction. In such cases a node or property can't be migrated and you're notified of this through the system report.
- Nodes and their properties are migrated only if they're selected in the original MIG. Unselected mandatory nodes and their properties are not migrated.
- Since mandatory nodes are preselected based on the underlying message template when a MIG is created, MIG migration does not actively unselect such mandatory nodes.
- For each selected node of the original MIG, the following applies:
 - If a matching node is available in the new message structure, details such as properties, code values, and documentation associated with this node are migrated to the matched node in the new MIG.
 - In the following situations a node can't be fully matched, so the migration finds an alternative match:
 - Example 1 (relaxed namespace handling): **Node1** in **namespace1** is not present in the new message, but there's a **Node1** in **namespace2** available. In such a case, this alternative match is migrated and you're notified through the system report to review this decision.
 - Example 2 (segment group handling for UN/EDIFACT): If **SGnn** of the old message has become **SGmm** in the new message, with the same trigger segment, then segment group **SGnn** is automatically migrated to segment group **SGmm**.

- If a matching node isn't present in the new message structure, you're notified through the system report.

→ Note

If a matching group node isn't available in the new message structure, its subnodes aren't migrated as well.

- Any modification done to an element's property, code value, or documentation in your original MIG remains intact even after MIG migration. For example, if a node's standard maximum length property of 10 was changed to 5 in the original MIG, the chosen value of 5 is retained in the new migrated MIG irrespective of the new MIG version's standard value.
- However, if you retained the standard values for an element's property, code value, or documentation in your original MIG, MIG migration keeps the standard setting in the new MIG as well. For example, if a node's standard maximum length property of 10 was unchanged in the original MIG, the matching node's standard maximum length property of 20 is retained in the new migrated MIG reflecting the new MIG version's standard value.
- In case of conflicts, an element's property, code value, or documentation might not get migrated. They are notified through the system report.

Next Steps

After the migration, the following steps are recommended for review and verification:

1. To review the system report, on the **Structure** tab, review the message structure and filter for any entry on the **Status** column to display the number of errors, warnings, informational messages, and comments.
 - a. Select the required cell in the **Status** column to view more details in the **System Report** section.
 - b. Choose **Edit** to add comments and set a status for the node in the **User Review** section. You can also choose to **Discard** any system report entries that you have reviewed and no longer need it.
 - c. Choose **Save**.

→ Tip

- On the **Structure** tab, choose  *Show Status Comments* to hide some of the columns for the prominent visibility of **Status** and **Comment**.
 - On the **Structure** tab, choose  *Settings* to customize column visibility of the properties.
2. Choose  *Show Changes Mode* to see differences between the MIG and its underlying message template.
 3. Choose **Edit** to customize your MIG as per your business needs.
 4. Choose **Save**.

Deleting Message Implementation Guidelines

This topic explains the different options available to delete a Message Implementation Guideline (MIG) that is no longer required.

You can choose to delete a MIG including its draft, active, or deprecated versions or only a specific version. You can also mass delete message implementation guidelines.

Caution

Deleting MIGs is permanent and can't be undone.

Delete MIGs Including Draft, Active, or Deprecated Versions

1. On the [Message Implementation Guidelines](#) screen, choose [⌵ More Options](#) on the MIG that you want to delete and then choose **Delete**.
2. In the [Delete Message Implementation Guideline](#) dialog box, choose **Delete complete Message Implementation Guideline (including history)**.

Delete Specific Versions of MIGs

Using the Message Implementation Guidelines Screen

1. On the [Message Implementation Guidelines](#) screen, choose [⌵ More Options](#) on the MIG that you want to delete and then choose **Delete**.
2. In the [Delete Message Implementation Guideline](#) dialog box, choose **Delete this (latest) version**.

Using the Overview Tab

1. On the [Message Implementation Guidelines](#) screen, choose the specific version of the MIG that you want to delete.
2. On the [Overview](#) tab, depending on the screen size, choose **Delete**, or choose [⋮ More](#) and then choose **Delete**.
3. In the [Confirm](#) dialog box, choose **OK**.

i Note

You can't delete active MIGs.

Using the History Link

1. On the [Message Implementation Guidelines](#) screen, choose **History** on the MIG that you want to delete.
2. From the [Message Implementation Guideline History Versions](#) table, choose [⌵ Actions](#) on the version that you want to delete.
3. In the [Confirm](#) dialog, choose **OK**.

i Note

You can't delete active MIGs.

Mass Delete Message Implementation Guidelines

To delete multiple message implementation guidelines at once, perform the following steps.

1. Go to [MIGs](#) and choose **Mass Delete MIGs**.
2. In the upcoming dialog, select the message implementation guidelines that you want to delete.

Use the search and filters to narrow down the list of artifacts and then use the checkbox in the table header to only select those artifacts. For more information on the filter, see [Filtering MIGs](#).

Choose **Next**.

i Note

Message implementation guidelines can only be deleted if they're not referenced by any other artifacts.

3. In the **Confirm Selection** step, check that you've selected the relevant message implementation guidelines and versions. If you're satisfied, delete them by choosing **Confirm Deletion**.
4. In the **Results** screen, you can see the progress of the deletion. Once the deletion is finished, you can download a results file that contains the details of the deleted message implementation guidelines by choosing [↓ Download](#).

Mapping Guidelines (MAGs)

Learn what mapping guidelines are and how to work with the mapping guideline page.

Mapping guidelines are runtime artifacts that you use in the mapping step of an integration application. They're used as reference or guidance for implementing mapping in the integration application. By providing such an artifact, you simplify the mapping task for users who use messages that adhere to the A2A/B2B type system standards.

Mapping guidelines are based on a source and a target message implementation guideline. They demonstrate how the defined nodes at each side are mapped, describing all mapping elements in detail with definitions or notes and providing further instructions for the transformation, such as functions or code value mappings.

Types of Mapping Guidelines

You can create the following types of MAGs:

- **Standard Graphical MAGs** serve as the foundation for your mapping guidelines. They contain the core mapping rules and instructions that apply to all users. Standard MAGs are used as the default or standard mapping guidelines for your organization.
- **Overlay Graphical MAGs**, on the other hand, allow you to create customized mapping guidelines that build upon the standard graphical MAGs. These overlays enable you to tailor mapping rules and instructions for specific use cases, projects, or user groups. Overlay MAGs provide flexibility and adaptability, ensuring that your mapping guidelines remain relevant and effective in diverse contexts.
- **XSLT MAGs** differ from standard graphical MAGs due to their mapping source: As opposed to graphical MAGs, in which source nodes are dragged towards the target node to create a mapping, for XSLT MAGs, you upload an XSLT file that contains the finished mapping definition.

Mapping Guidelines Tab

Global Actions

The **Mapping Guidelines** tab displays an overview of all the MAGs that you've created.

In the tab, you can perform the following actions for any selected mapping guideline:

- **Create:** Create a new mapping guideline, either **Graphical MAG: Standard**, **Graphical MAG: Overlay**, or **XSLT MAG**. See [Creating Mapping Guidelines](#).
- **Export:** Export your mapping guideline. See [Exporting MIG/MAG](#).
- **Import:** Import mapping guidelines to your tenant. See [Importing MIG/MAG](#).
- **Mass-Update MAGs:** Filter and update mapping guidelines in bulk. See [Updating Mapping Guidelines](#).

History

You can also view the version history of a specific mapping guideline. Choose **History** which you can find after the version details of a MAG in the **Mapping Guidelines** page. The resulting screen would display the following version history details of the MAG:

- **Version:** This column displays the version numbers of the MAG
- **Status:** This column displays the status of each version of the MAG. There's always only one active version of a MAG. The previous version of the active MAG is in draft status and all other MAG versions before that are deprecated.
- **Name:** Displays the name of the MAG in each version.
- **Modified On:** Displays the last modified timestamp.
- **Modified By:** Displays the unique identifier of the user who modified the MAG.
- **Last Imported On:** Displays the last imported date and time.
- **Last Imported By:** Displays the unique identifier of the user who imported the MAG.
- **Imported:** This column lets you know if the MAG is imported or not.
-  **Action** allows you to copy, migrate, or compare your MAG.

Code Value Mapping

The **Code Value Mapping** tab displays the list of global code value mappings that you can use in your mapping guideline. See [Global Code Value Mappings](#).

Filtering MAGs

Filters are used to search and display MAGs based on various criteria. You can use the **Free text filter** search or the  **Extended Filter** at the beginning of the screen to specify your search criteria. The **Free text filter** search allows you to filter and display MAGs that match your search text. It can be used as a quick filter to match your search text against multiple attributes of your MAGs.

You can also choose  **Extended Filter** to filter MAGs based on specific criteria. You can use the required filter fields to list MAGs that match your query. You can also customize the filter bar by choosing **Filters**. You can expand or collapse the filter area using **Show Filter Bar** or **Hide Filter Bar**.

Filter Criteria

Filter	Description
MAG Name	The name of the MAG provided under  Overview > General Information  .
MAG Version	The version of the MAG provided under  Overview > General Information  .
Status	The status of the MAG provided under  Overview > General Information  .
Source MIG Name	The name of the source MIG provided under  Overview > Source and Target MIGs  .
Type System of Source MIG	The type system of the source MIG provided under  Overview > Source and Target MIGs  .

Filter	Description
Type System Version of Source MIG	The type system version of the source MIG provided under Overview > Source and Target MIGs .
Message Type	The Message Type of the source MIG provided under Overview > Source and Target MIGs .
Target MIG Name	The Name of the target MIG provided under Overview > Source and Target MIGs .
Type System of Target MIG	The Type System of the target MIG provided under Overview > Source and Target MIGs .
Type System Version of Target MIG	The Type System version of the target MIG provided under Overview > Source and Target MIGs .
Message Type	The Message Type of the target MIG provided under Overview > Source and Target MIGs .
Created By	The unique identifier of the user who created the MAG. This information is available under Overview > Administrative Data .
Created Between	The date range during which the MAG was created.
Modified By	The unique identifier of the user who modified the MAG. This information is available under Overview > Administrative Data .
Modified Between	The date range during which the MAG was modified.
Last Imported By	The unique identifier of the user who last imported the MAG. This information is available under Overview > Administrative Data .
Last Imported Between	The date range during which the MAG was imported.

Related Information

[Creating Mapping Guidelines](#)

[Working with Mapping Guidelines \(MAGs\)](#)

Creating Mapping Guidelines

Before you create a new mapping guideline (MAG), you need to decide on the mapping method that you want to use.

In Integration Advisor, there are two types of mapping guidelines, based on the mapping method:

- For **graphical mapping guidelines**, you create mappings graphically, namely by dragging source nodes towards target nodes in the mapping editor.

Additionally, there are two types of graphical mapping guidelines: **standard** and **overlay**. See [Creating Standard Graphical Mapping Guidelines](#) and [Creating Overlay Graphical Mapping Guidelines](#).

- For **XSLT mapping guidelines**, you create mappings by uploading an XSLT file that defines the mapping of the source and target nodes. See [Creating XSLT Mapping Guidelines](#).

Creating Standard Graphical Mapping Guidelines

Use SAP Integration Advisor to create a standard mapping guideline for implementing mapping between systems that adhere to B2B type system standards. Choose MAGs, select source and target MIGs, and review and modify details before creating the mapping guideline.

Prerequisites

You've accessed SAP Integration Advisor.

Context

A mapping guideline can be used as a source, reference, or guidance, for implementing mapping between systems that adhere to B2B type system standards. You use this procedure to create a new standard mapping guideline.

Procedure

1. Navigate to **MAGs**.
2. In the **Mapping Guidelines** page, choose **Create** > **Graphical MAG: Standard**.
3. In the **Source MIG** step, select the MIG that you want to use as the source in your MAG.
4. In the **Target MIG** step, select the MIG that you want to use as the target in your MAG.
5. The **MAG Creation** step displays the following information:
 - o **General Information**: Contains the **Name**, **Version**, and **Status** of the MAG. You can edit the name of the MAG here.
 - o **Source and Target MIGs**: Displays the details of the source and target MIGs you chose in the previous steps.
 - o **Documentation**: Contains the **Summary** for the MAG. You can add the summary detail for this field.
 - o **Source Business Context** : Displays the business context details maintained in the source MIG. You can modify this information.
 - o **Target Business Context** : Displays the business context details maintained in the target MIG. You can modify this information.
6. Review your choices, then choose **Create**.

Once created, the newly created MAG has the label **Base** below its name to distinguish it from the overlay MAGs.

Create MAG from an existing MAG

Create a new mapping guideline by copying an existing mapping guideline.

Procedure

1. In the **Mapping Guidelines** tab, select **More Options** next to the MAG that you want to copy and choose **Copy**.
2. Then, choose one of the following copy options, based on your requirements:
 - a. **Copy** > **Copy MAG Only**: This option creates only a copy of the MAG, reusing the MIGs from the copied MAG.
 - b. **Copy** > **Copy MAG and Referenced MIGs**: This option creates a copy of both the MAG and the MIGs referenced in it.
 - c. **Copy MAG with MIGs...**: This option creates a copy of the MAG and lets you choose if source MIG, target MIG, or both should be copied as well.

A new copy of the mapping guidelines opens, and you can work with it right away.

Results

You've created a new mapping guideline as a copy of an existing mapping guideline.

Next Steps

To learn how to work mapping guidelines, see [Working with Mapping Guidelines \(MAGs\)](#).

To learn how to create overlay mapping guidelines, see [Creating Overlay Graphical Mapping Guidelines](#).

Creating Overlay Graphical Mapping Guidelines

Create an overlay mapping guideline to fulfill specific business requirements. Follow the procedure to select base MAG, source and target MIGs, and add general information, documentation, and business context details. Use it to customize standard guidelines for specific needs.

Context

Overlay mapping guidelines are extensions of standard mapping guidelines that are used to fulfill specific business requirements.

To create an overlay MAG, perform the following steps.

Procedure

1. From the left pane of the application, choose **MAGs**.
2. In the **Mapping Guidelines** page, choose **Create** > **Graphical MAG: Overlay**.
3. In the **Base MAG** step, select a standard MAG from the table that you want to use as the base.
4. The **Source MIG** step displays the MIG that is used as the source in the selected base MAG. If you want to reuse the same source MIG in your overlay MAG, simply select the MIG.

To use an alternative source MIG, refer to the **Compatible MIG** tab, which lists suitable replacement MIGs. Choose one from the list to proceed.
5. Similarly, the **Target MIG** step displays the MIG that is used as the target in the selected base MAG. If you want to reuse the same target MIG in your overlay MAG, simply select the MIG.

To use an alternative target MIG, refer to the **Use Compatible MIG** tab, which lists suitable replacement MIGs. Choose one from the list to proceed.
6. The **MAG Creation** step displays the following information, most of which is automatically populated using the data from the base MAG:
 - **General Information:** Contains the **Name**, **Version**, and **Status** of the MAG. You can edit the name of the MAG here.
 - **Base MAG:** Displays the information about the standard MAG.
 - **Source and Target MIGs:** Displays the details of the source and target MIGs you chose in the previous steps.
 - **Documentation:** Contains the **Summary** for the MAG prefilled from the standard MAG. You can add the summary detail for this field.
 - **Source Business Context :** Displays the business context details maintained in the source MIG. You can modify this information.
 - **Target Business Context :** Displays the business context details maintained in the target MIG. You can modify this information.
7. Review your choices, then choose **Create**.

Once created, the newly created MAG has the label **Overlay** below its name to distinguish it from the standard MAGs.

Next Steps

To learn how to work with an overlay MAG, see [Working with Overlay Mapping Guidelines](#).

Creating XSLT Mapping Guidelines

Learn how to create a new mapping guideline that uses an XSLT file as its source.

Context

Besides the graphical mapping guidelines, which you define by drawing lines, you can also create XSLT mapping guidelines, which use XSLT files as source.

To learn more, see the blog [Integration Advisor - Use Custom XSLT as MAG](#).

Procedure

1. Go to **MAGs**.
2. Choose **Create > XSLT MAG**.
3. In the first step of the creation wizard, select a **source MIG**. Choose **Next**.
4. Then, select a **target MIG** and choose **Next**.
5. Finally, in the tab **MAG Creation**, enter a name for the new mapping guideline and choose **Create**. The new mapping guideline opens in edit mode, with unresolved errors due to a missing XSLT file.
6. To add an XSLT file, go to **Mapping Details** and choose **Browse**, next to **New XSLT File**.
7. Select an XSLT file from your file system. To upload it to the MAG, choose **Save**.

After saving, the custom XSLT file is automatically validated. If there are any errors, you can review them in the **Mapping Details** section. Until you fix the file, the mapping guideline is considered erroneous.

If your XSLT file is valid, you can review its content by choosing *Show File Content* and you can download the file by choosing *Download XSLT File*.

To update the file, simply choose **Browse** again and upload a new file, thereby replacing and overwriting the previous file.

8. Now, you can simulate the new MAG by choosing either **Simulate > Simulate with MIG Example Data** or **Simulate > Simulate with Payload Data**. After a successful simulation, you can download the simulation results by choosing *Download Simulation Results*.
9. Finally, to activate the MAG, return to display mode by choosing **Cancel**, then choose **...** *Actions*, followed by **Activate**. The version status switches to **Active**.

Results

You've created a new XSLT mapping guideline and uploaded an XSLT file as its source.

Next Steps

After activating a MAG, you can perform the following actions:

- To edit a MAG after activating it, you have to create a new draft version of the MAG by choosing **Edit**. You can then upload a new XSLT file by choosing **Browse** in the **Mapping Details** section, and continuing as described previously.

- Copy a mapping guideline: While your mapping guideline is in draft or active status, you can create a copy of it by choosing **Copy**. You can then attach an XSLT file to the new copy.

Global Code Value Mappings

Learn more about the **Code Value Mappings** tab in the Mapping Guidelines page.

The **Code Value Mappings** tab displays the list of global code value mappings that can be used in your mapping guideline. You can open a code value mapping to view its source and target code values. You can also create a global code value mapping with both 1:1 and N:1 cardinality. To know more, follow the following procedure:

1. Login to your application and select the **MAGs** section.
2. In the **Mapping Guidelines** screen, navigate to the **Code Value Mappings** tab and choose **Create** to create a Global Code Value Mapping. This opens the **Create Code Value Mapping** dialog.
3. Maintain the following fields for **Source Codelist** :
 - **Type System**: Select the  *value help* for this field. This displays the **Standard** type systems available in the tenant. If you want to use a custom code list, select the **Custom** button and select the custom type system from the list.
 - **Codelist**: Select the codelist using the  *value help* button. The button displays the list of codelists available in the type system that you chose in the previous field.
 - **Version**: Select a version of the codelist
4. Choose **+ Add** to add after maintaining the fields.
5. If you want to add more source codelists, you can do so by changing the corresponding values of the fields using the  *value help* button and then selecting **+ Add**.
6. Select **Next** and maintain the following fields for **Target Codelist**.
 - Type System
 - Codelist
 - Version
7. Choose **Next** and in the **Mapping Creation** step, you are provided with a summary of your selection.
8. You can modify the name of this new global code value mapping and if you want to modify your selection, you can do so using the **Edit** button provided under the **Source Codelist** and **Target Codelist**.
9. At first, no values appear in the **Code Value Mapping** tab. You can populate it as follows:
 - **Manually add entries** as follows:
 - a. Select a value for the **Default Value** field.
 - b. To add an entry in the code values table, choose **►Add► One source value ►**.
 - c. Add all remaining primary source values. For details, check the **Code Value Mapping** section.
 - d. After maintaining the entries, choose  *Sort* to reorder the existing code value mappings. This ensures that more accurate mappings are applied first.
 - e. For n:1 code value mappings: Choose  *Reorder* to open a **Reorder Source Codelists** dialog, which allows you to move the order of the source codelists using **^ Up** and **▼ Down**. Choose **OK** to see your changes reflecting in the code values table.

- o You can also **upload a local CSV file** with predefined code value mappings as follows:

- Choose **Import**. A wizard opens.

- In the first step, **Download** the .CSV template and add your data as new rows. Then choose **Select** to browse for and upload the file. Choose **Next**.

- If there are any code issues, they're raised in the second step. Correct the issues by selecting valid codes from the allowed list. Once you're finished, choose **Next**.

- Finally, review the imported code value mappings and choose **Import**. Your new entries appear in the table of code value mappings.

If needed, use the **Export** function to export the current entries in the code value mapping table to a .CSV file.

10. After making the necessary changes, choose **Save**.

11. You can use the proposal service to get proposals on code value mappings that might be most relevant for you.

- Choose **Get Proposals**. A list of proposals is loaded, showing proposed code value mapping entries, which are calculated based on other available global code value mappings.

- After reviewing the proposals, you can choose **+ Add this proposal as value mapping element** or **↺ Overwrite existing value mapping element with this proposal** from the **Action** column.

- Alternatively, you can also choose **Select Best Proposal** to automatically highlight the proposed mappings with the highest confidence for you.

- Review the list of proposals anytime by choosing **Proposals**. To remove all proposals, choose **Clear Proposal**. You can always retrieve the latest proposals by choosing **Get Proposals** again.

12. The **Overview** tab of the code value mapping provides an overview of the source and target codelists. The **References** section displays the list of MAGs wherever the codelist is used.

13. You can also edit the **Name** of the code value mapping.

Copying a Global Code Value Mapping

If you want to create a code value mapping from an existing code value mapping:

- You can open the existing code value mapping and choose **Copy**.
- In the main **Code Value Mapping** tab, you can select the more options **...** button and choose **Copy**.

i Note

Copying a global code value mapping creates a new **Draft** global code value mapping with version set to 1.0.

Versioning a Global Code Value Mapping

Version and activate your global code value mapping.

Context

Once you've created and saved your global code value mapping, you can activate the mapping to use it productively. You can set the following different types of status for a code value mapping:

- **Draft:** When you create a new global code value mapping, the newly created code value mapping has the **Status** field set to **Draft** and the **Version** set as **1.0**. These details can be viewed in the **Overview** tab of your code value mapping. You can modify this new version according to your business requirements.

i Note

You can also copy an existing code value mapping. The copied code value mapping has the **Version** set as **1.0** and also the term **Copy** suffixed to its name. See [Global Code Value Mappings](#).

- **Active:** After activation, the global code value mapping's **Status** is set to **Active**. Once activated, this version of the global code value mapping is finalized and it can't be edited anymore.

i Note

It's always recommended to activate a global code value mapping to use it productively. By doing so,

- You protect the global code value mapping against any unwanted changes.
 - Any new requirements for the global code value mapping can be maintained in a new **Draft** version without changing the active global code value mapping version currently in use.
- **Deprecated:** After activating a global code value mapping, its previous active version gets deprecated and its **Status** field is set to **Deprecated**.

i Note

There can be only one active and draft version of a particular global code value mapping.

To activate a global code value mapping, perform the following steps.

Procedure

1. Open the global code value mapping.
2. Choose **Activate**.

i Note

Activating a global code value mapping renders it read-only. To make further edits, you need to create a new draft version of the mapping.

3. In the confirmation dialog, choose **Activate**.

Working with Mapping Guidelines (MAGs)

Learn how to work with mapping guidelines, also known as MAGs.

Prerequisites

- You've created a mapping guideline (MAG). For more information on creating a new MAG, see [Creating Standard Graphical Mapping Guidelines](#).
- You've opened the MAG that you want to work with, either after creating a new MAG or by choosing one from the **Mapping Guidelines (MAGs)** section of the Integration Advisor home page.

Context

Once you've created a mapping guideline (MAG), you can edit it to change the mapping details to suit your scenario. Additionally, you can also add documentation that provides the users of the MAG with more information on its relevance.

In some scenarios, significant differences between the message implementation guidelines, such as different node hierarchies, difference in the message structure, and so on, can complicate the mapping. To overcome this issue, the application provides you with an approach that helps transform the structure of a source MIG to ensure that its content can be mapped easily with the target MIG structure. To learn more, see [Pretransforming a Message Implementation Guideline](#).

Procedure

1. Choose **Edit**.

i Note

Ensure that the MAG you want to edit isn't **locked** by other users. The **Locked By** field displays the ID of the user who's currently editing the MAG. To unlock a MAG, you need admin rights. To learn more about the role, see [Assigning Users for SAP Integration Advisor](#).

2. The **Overview** tab displays the general information that you provided at the time of creation, such as the name, version, source, and target MIGs used in the MAG.

For standard MAGs, the **Overview** tab includes a **References** section, which lists all overlay MAGs that reference this standard MAG.

3. In the **Mapping** tab, you can edit the mapping details of the MAG. For more information on mapping nodes, see [Mapping the Source and Target Nodes](#).

- To create a **mapping**, drag the field from **Source** to the field on **Target** that you want to map to.
- You can also use the **Proposal Service** to get a proposal on fields that might be most relevant for you.
 - Choose **Proposals: Get Proposals** to activate a proposal indicator that displays which fields might be most relevant for you. This proposal is calculated based on the other available MAGs.
 - Choose **Proposals: Select Best Proposal** to automatically select all proposed mapping entities or select single proposed mapping entities by clicking on the checkboxes within the mapping list.
- The **mapping list** displays all the relevant information of the source and target nodes such as source nodes, target nodes, their cardinality and so on. The **Type** column displays the type of each entry in the mapping list. You can filter and view the mappings that are erroneous and mappings that have proposals using the drop-down filter available above the mapping list table.

To **delete** one or more mappings from the list, select them and choose **Delete**.

- To **search** for a particular value in the **Mapping List** tab, select one of the following column headers and enter the search value in the upcoming filter .
- Source
- Source Name
- Target
- Target Node Name

This filter option is also available for the **Code Value Mapping** tab.

- To **search** for all XSLT snippets and XPath expressions for a certain text pattern and filter the list of mapping elements by those elements that contain the search pattern, use the **Search** field provided above the **Mapping List**

table. Enter the function pattern and choose . This displays the list of mapping elements pertaining to that function pattern.

- o To **mass validate** all mapping elements that include an XSLT snippet, choose **Validate XSLT Snippets** in the header of the mapping list table. Any mapping element with invalid XSLT snippets is then marked as erroneous in the mapping list. The validation applies to the whole mapping list, independent of any active selection. For more details on the error, check the tool tip in the **Status** column or navigate to the details of the erroneous mapping element.
- o You can automatically **create mappings between identical nodes** using the **Create 1:1 Mappings** function:

- a. Choose **Edit**.

- b. In the **Mapping** tab, select a group node in the source MIG and then select a group node in the target MIG, either by clicking or drawing a mapping line between them.

- c. In the header of the **Mapping List** section, choose **Create 1:1 Mappings** and confirm the upcoming dialog with **Create**.

For all identical nodes in the selected subtrees, 1:1 mapping elements are created and added to the mapping list. Existing mapping elements are not modified or removed.

- o You can add or manage **additional information or functions** about each element by selecting the mapping entity line.

The mapping also supports various value transformations. To know about these features, see [Value Transformations](#).

4. In the **Global Parameters** tab, you can declare external parameters that you want to use in your mapping guideline. Before you can consume them during the execution of the mapping guideline, you must fill the parameters outside the mapping, for example, as headers or properties in Cloud Integration.

i Note

When you create a new global parameter, the value help provides you with the most frequently used parameters from the EDI flow steps of Cloud Integration. These parameters are automatically filled if you either use the EDI Splitter or the EDI Extractor in your integration flow.

5. Once you've finished editing the mapping guideline, save your changes.

Related Information

[Mapping the Source and Target Nodes](#)

[Functions](#)

[Value Transformations](#)

[Simulating Mapping Guidelines](#)

[Integration Advisor - Create 1:1 Mapping Elements in MAG Editor](#) 

Pretransforming a Message Implementation Guideline

Learn how to transform the structure of a message implementation guideline to better use it in a mapping guideline.

Context

When working with a mapping guideline, in some scenarios the source MIG and the target MIG can't be mapped due to significant structural differences between them, for example, node hierarchies. The goal of **pretransformation** is to allow you to define, simulate, and pretransform a source MIG including the source message instance so that it fits better to the structure of the target MIG. Once pretransformed, you can use the MIG in the normal mapping of a MAG.

Pretransformation is a way of modifying the source structure of a MIG with the purpose of allowing later mappings that wouldn't be possible in the application without such a modified source structure being applied. The creation of mappings, that is, the drawing of mapping lines in a mapping guideline, always requires a MIG on either side of the mapping. Therefore, pretransformation applies a certain structural transformation to the hierarchy of the incoming payload and its corresponding MIG defined by a pretransformation script (PTS). As a next step for the payload to be executable at runtime in an integration flow, PTS needs to provide a corresponding runtime artifact. So, pretransformation consists of two major operations:

- The **MIG transformation**, which is performed at design time. Its goal is to provide the corresponding transformed message structure for the MIG that replaces the MIG structure on the left side of a MAG.
- The **payload transformation**, which is performed in an integration flow. To achieve the transformation, the application exports an additional XSLT script which is, however, contained in the same XSL file as the main XSLT script.

Procedure

1. Let's assume that you've created a MAG with two MIGs and you want to transform a MIG to continue with the mapping.

Open your MAG and choose **Edit**.

2. Navigate to the **Overview** tab. The **Pretransformation** field under **Source and Target MIGs** displays the name of the pretransformation. If no transformation is done, the value is **None**.
3. Next to the field, choose **+ Add** to open the **Mapping** tab with the pretransformation function. By default, the source MIG of the mapping structure is considered for pretransformation.

The page consists of the following sections:

- **MIG Structure**: The structure of the MIG chosen for pretransformation. It always shows the unmodified source MIG structure.
 - **Pretransformation/Main Transformation**: The button that helps you toggle between pretransformation and the main mapping guideline. Below **Pretransformation**, you have the tables to define your transformation steps and the corresponding parameters.
 - **Pretransformed Structure**: The pretransformation that you define is applied on this structure and this modified MIG becomes the pretransformed MIG. You can check the results and move to **Main Transformation** for mapping.
4. The field **New Pretransformation** shows the name of the transformation. Edit it to give it a meaningful name according to your business purpose.
 5. Next to **Transformation Steps**, choose **Add**. Choose one of the following options:
 - **Group by Key**: Group the elements based on the values of specified node and puts them under a new wrapper node. If you choose this step, continue with step 6.
 - **Copy referenced node**: Copy an element referenced by a key or ID into a specified node. If you chose this step, continue with step 7.
 6. If you chose **Group by Key**, maintain the following parameters in the **Operation Parameters** table displayed below the **Transformation Steps** table:

Operation Parameters

Parameter	Value
Grouped node	The group node whose instances are grouped.
Key node	The leaf node that contains the key for grouping. This node must be a child node of the grouped node.
Wrapper name	The name of the group node that contains the grouped nodes. This node becomes the parent node of the grouped node.

Parameter	Value
Group key node	Child node of the wrapper node that contains the key of the group for each instance.

7. If you chose **Copy referenced node**, maintain the following parameters in the **Operation Parameters** table displayed below the **Transformation Steps** table.

Operation Parameters

Parameter	Value
Receiving node	The group node to which the referenced node is copied into.
Node with reference ID	Leaf node that contains the reference ID of the node to be copied. This node must be a child node of the receiving node.
Referenced Node	Group node which is referenced and copied.
Referenced ID Node	Leaf node that contains the ID which is referenced. This node must be the child node of the referenced node.

8. After maintaining all the entries, choose **Execute** to apply the operation to the MIG structure. The pretransformed structure is displayed under **Pretransformed Structure**.
9. Review the results and modify your operation accordingly. You can also delete your transformation step by choosing  **Delete**.
10. If you're satisfied with the outcome of the transformation, choose **Save**.
11. Continue to the mapping by choosing **Main Transformation**.

You can now see that source MIG is the latest pretransformed MIG.

To simulate and see the effects of your pretransformation on a payload, choose **Simulate**.

Example

To learn more about pretransformation and look at an example, see the blog [New Feature in Integration Advisor: Reordering of Source Structure](#) .

Next Steps

You can start working on mapping using this MIG. See [Mapping the Source and Target Nodes](#).

Mapping the Source and Target Nodes

This topic helps you understand the concepts involved in mapping source and target nodes.

You can define a mapping between a source and a target node using the Mapping Guidelines (MAGs) editor. The source and target Message Implementation Guidelines (MIGs) are displayed in a hierarchical structure in the **Mapping** tab. You can drag a source node to a target node to specify a mapping, and apply functions based on your requirements to define a data transformation. For more information, see [Functions](#) and [Value Transformations](#).

The resulting target document structure varies based on:

- The type of mapping defined between the source and target nodes.

- The cardinality of the target node.

The following table explains the basic terms involved in mapping nodes:

Term	Description
Group Node	A node that has subnodes and doesn't contain data. It can also be a subnode to another group node.
Leaf Node	A subnode that contains data.
Cardinality	Specifies the number of times an element can occur in a document. There are two values: minimum and maximum. ❖ Example A cardinality of 1..n where n=5, means that a minimum of one element is instantiated and a maximum of five elements are instantiated at runtime.
Mapping Line	Mapping associates elements of a source structure with one or more elements of a target structure and determines target node construction. A mapping line is drawn to connect nodes and subnodes from source to target.
Mapping Entity	A mapping entity represents a mapping from one or more source elements to a single target element.

Mapping Cardinality

Mapping cardinality represents the number of nodes associated in the source and target document structure.

❖ Example

You can map one source node to one target node or one or more source nodes to one target node.

The root node is the highest node in the document structure. As you traverse down the structure from the root node, each mapped element sets the context for interpreting the cardinalities. The target document structure differs when you map either only the leaf nodes or the leaf nodes along with their group nodes.

When you map a source group node to a target group node whose target maximum occurrence is greater than one, an instance of the target group node is created for each instance of the source group node. The target group node serves as the context for realizing the cardinalities and the mappings of its leaf nodes.

If you map leaf nodes without including their group nodes, an instance of the target group node is created based on the instantiation of its leaf nodes.

Group to Leaf Mapping

Mapping a source group node to a target leaf node creates a target node, where the cardinality of the target node is determined by the actual cardinality of the source group node. The newly created target node will be empty and you need to map the relevant source leaf nodes with this target node to provide data for the target instances.

For this mapping

- If the target maximum occurrence equals 1 and

- If the actual occurrence of the source group node is at least one, the target leaf node gets created
 - If the actual occurrence of the source group node is zero and the minimum occurrence of the target leaf node is zero, the target leaf node does not get created.
 - If the actual occurrence of the source group node is zero and the minimum occurrence of the target leaf node is one, the target leaf node gets created.
- If the target maximum occurrence is greater than 1, then the cardinality of the target leaf node corresponds to that of the mapped source group node. Each instance of the source group node creates exactly one instance of the group node in the target independent of the target cardinality, even if the actual occurrence of the source node is greater than the maximal occurrence of the target node.

i Note

String Processing is not supported in group to leaf mapping.

If there is only one leaf node mapped to the target node and

- If only one instance of this single node is relevant for the respective target node, then the content of each instance of the target node is filled with the content of the corresponding instance of the source node.
- If more than one instances of this single leaf node are relevant, the values are concatenated for each occurrence of the source group node separately and assigned to the corresponding instance of the target leaf node.

If more than one leaf node is mapped, the values of all the nodes are concatenated for each occurrence of the source group node and the concatenated value is used for the target leaf node.

The following options are supported when mapping nodes from source to target.

Mapping Cardinalities	Mapping Type
1:1	Leaf to Leaf
1:1	Group to Group
1:1	Leaf to Group
N:1	Leaf to Leaf
0:1	None to Leaf (Constants)
1:1	Group to Leaf
N:1	Group to Leaf

1:1 Mapping

- It represents an association in which one element of the source is mapped to only one element of the target.
- If the maximum occurrence value of the target node is greater than one, then a target node is created for each instance of the source node.
- If the maximum occurrence value of the target node is equal to one, then the contents of the source nodes are concatenated and written onto the target node.

N:1 Mapping

- It represents an association in which many elements of the source are mapped to one element of the target.
- If the maximum occurrence value of the target node is equal to one, then the contents of the source nodes are concatenated and written onto the target node.
- If the maximum occurrence value of the target node is greater than one, then this mapping isn't allowed.

O:1 Mapping

You can directly assign a constant value to a target leaf node without defining a mapping for it.

i Note

The following mapping options aren't supported:

Mapping Cardinalities	Mapping Type
N:1	Group to Group
N:1	Leaf to Group

There are two variables which can be used in functions from Group-to-leaf mapping:

- **@position**: This variable gives the position of the current instance in the list of all instances of the mapped source group node.
- **@last**: The variable gives the index of the last instance (equals the number of instances) of the mapped source group node.

Conditional Mapping

You can specify conditions on leaf nodes to control the creation and cardinality of the target group or leaf node instances in a group-to-group and leaf-to-leaf node mapping respectively.

When a leaf node is added as a condition node to a leaf-to-leaf mapping:

- A **Condition** checkbox is provided for the leaf node in the mapping list.

i Note

Conditional mapping is not supported for nodes with

- Constant value mapping
- String Processing type

For such mapping, the **Condition** checkbox is disabled.

- On selecting the checkbox, the **Condition** tab on the mapping pane becomes available.
- The **XPath Code** editor is displayed where you can specify the condition as an XPath expression.

When a leaf node is added as a condition node to a group-to-group node mapping:

- The **Condition** checkbox is selected by default for the leaf node in the mapping list.

i Note

For group-to-group mappings, all additional source leaf nodes are automatically condition nodes.

- The **Condition** tab on the mapping pane becomes available.
- The **XPath Code** editor is displayed where you can specify the condition as an XPath expression.

i Note

You can't add additional group nodes to an existing conditional mapping.

The default XPath expression, `boolean($nodes_in/*)`, of the mapping returns true only if the sequence isn't empty. The sequence `$nodes_id/*` contains all the instances of the conditional nodes, that are relevant for one instance of the target node. You can edit the default expression as per your business needs. For example, if the expression, `$nodes_in/partyRole = 'BUY'`, is specified as a condition on the leaf node, `partyRole`, the corresponding target node is created only if the condition is satisfied.

i Note

For more information, refer to XPath specification published by W3C.

Use the shared code feature to create frequently used code snippets and reuse it as required within your Mapping Guideline (MAG). You can create, share, and reuse the XPath expressions in the same way as the XSLT functions. For more information, refer [Creating a Shared Code](#).

When you choose to delete the last available conditional node in a group-to-group node mapping, the corresponding leaf node and its conditions are removed from the mapping.

You can know the type of mapping used based on the icon denoted over the mapping lines.

Mapping line types

Icon	Description
	Denotes • Function without a source node • Group to leaf mapping • Leaf to leaf mapping
A-z	String processing
	Date Time mapping
#	Constant
	Code value mapping
	Conditional mapping

You can also search for a particular node in the target or source structure using the search bar provided.

If you want to search for a particular entry in the **Mapping List** tab, select the relevant column name and enter the search value in the filter  field. The filter option is also available for the **Code Value Mapping** that allows you to filter values both in the source and target code values.

Duplicate Target Nodes

You can now duplicate a target node and map other source nodes to the duplicated target node. This feature applies for both group and leaf node. To do so,

1. Select the context menu ... of the target node you wish to copy and select **Duplicate**.
2. The **Duplicate Node** dialog box appears with the details of the selected node and a **Duplicates** table. Select **Add**.
3. This will create a copy of the original node in the **Duplicates** table. You can maintain the name and cardinality of this new node.

i Note

By default, to distinguish the duplicated nodes from the original node, the duplicate names are prefixed with a number in square [] brackets.

4. You can create multiple duplicates of the node. Just select **Add** to create another copy of the node.
5. After maintaining the necessary information, select **Apply**. The copied nodes will behave the same as the original node.

i Note

If the original node is changed, then the changes will be applied to the duplicates as well.

Deletion of Mapping Lines

You can delete both mapping lines and mapping entities that aren't required. In **Edit** mode, select the mapping line or entity, and press the key.

Alternatively, you can -

- Right-click the mapping line or entity and choose **Delete** from the context menu.
- Delete multiple mapping entities by selecting the check box of the entries you want to delete under the **Mapping List** tab and selecting **Delete** button provided above the table.

You can also delete an individual mapping line from a mapping entity. You can select the mapping entity and right-click the individual mapping line, and choose **Delete Line** from the context menu.

Value Transformations

These are the special scenarios that you can use while working with an MAG.

To know them in detail, see

- [Date Time Conversion](#)
- [String Processing Type](#)
- [Code Value Mapping](#)
- [Functions](#)

Date Time Conversion

This chapter explains how the date time conversion works.

You can map source and target nodes of type **Date**, **Time** and **DateTime** in the message structure of an MAG. To know more about this data type, see [Primitive Data Types](#).

The following mapping cardinalities are supported for date time mapping:

- **1:1**: Map one node from the source structure with one node of the target structure.
- **N:1**: Map multiple nodes from the source structure with one node of the target structure. The values of all the mapped source nodes are merged into one date-time input and are fed into the target node.

The **Date Time Conversion** feature has the following feature variations based on the type of format:

- If the format differs between the nodes, the format of the source node can be converted to the format of the target node during the mapping. To do so, you need to set the format of the nodes. Select the element and navigate to the **Functions** tab and choose **Switch to DateTime Mapping**.

i Note

Switching to DateTime mapping will delete the existing function. Choose **OK** to confirm.

If you want to implement your own function, you can do so using an XSLT code. To know more, see [Functions](#).

Navigate to the **Date Time Conversion** tab and set the values of **Source DateTime Format** and **Target DateTime Format** and choose **Save**. For N:1 mapping, you need to maintain the **Source DateTime Format** value for each of the source nodes that are mapped.

The format used in these two fields are displayed below them in detail. This helps in identifying the values entered during simulation. Choose **Simulate** and select your relevant option to see the simulation data split according to the DateTime format.

- If source and target nodes are both Date, Time, or DateTime fields then the **DateTime Conversion** tab is displayed automatically. You see the default DateTime format of the source as well as of the target node which is set while defining the corresponding message implementation guidelines.

Handling Missing DateTime Values

When you map a source node with a target node of type DateTime with different format, the source node might not contain all the inputs needed for the format on the target side. Say, for example, a source node with format **MMDD** is mapped with the target node format **DDMMCCYY**. Here, the values for MM and DD can be picked from the source node value but the values for CC and YY are missing. In such cases, you can set pre-fixed values to the missing fields in the **Date Time Conversion** tab. You can set the value as:

- **Current Date/Time**: This option will set the current date/time value for the missing part at the time of mapping execution.
- **Fixed Value**: This option allows you to provide a constant value for the missing parts of the target format.
 - If you choose a **fixed value**, the field is always produced with this value, whether there's an input or not.
 - If you choose a **fixed value for a UTC marker**, you can't define a **Timezone Offset**.

If no values are mentioned, then **Default Value** is set for the fields with the following default set by the system for the respective missing part.

DateTime Default Values

Field	Default Value
-------	---------------

Field	Default Value
YEAR	1970
MONTH	1
MONTH	1
DAY_OF_MONTH	1
HOUR, MINUTE, SECOND, MILLISECOND	0
Timezone-Offset	+00:00
Century	Two-digit years are interpreted by adjusting dates to be within 80 years before and 20 years after the time of the execution of the mapping.

Conflict in using N:1 Mapping

There could be scenarios where multiple source nodes provide the same input to the target node. In such cases, conflict of data could arise. Say for example, you have mapped 2 source nodes with one target node. The source nodes node1 and node2 both contain the format **MMYY** and the target date time format is also **MMYY**. This could result in a conflict. To resolve this, the application displays two separate fields below the **Target DateTime Format**:

- **Month(MM)**
- **Year(YY)**

You can now choose which of the 2 source nodes will contribute to the MM part and which will contribute to the YY part using the drop-down list provided for the two fields mentioned before.

In other case, if node1 is of type **MM** and node2 is of type **DD** and the target node type is only **MM**, then only the data from node1 is picked and node2 is ignored.

Adding Global Parameters for the Source Node

You can also add global parameters for the source node. Suppose let us say you have created global parameters in the **Global Parameters** tab of your MAG that you wish to use in your datetime conversion. To do so , follow the following procedure:

i Note

To know how to add global parameters in a MAG, see [Defining Global Parameters](#).

1. In the **Date Time Conversion** tab, choose **Add Global Parameter** below the source node detail.
2. A pop-up dialog opens listing down the global parameters that you defined in your MAG. Select one from the list.
3. Set the value for **Date Time Format** field for this newly added parameter.

Depending on the format set for the source node and this parameter, the values for the corresponding fields in the target node will get adjusted.

4. To delete the parameter from the date time conversion, select the delete  button next to it.

String Processing Type

You use the **String Processing** type to map multiple source node of type **String** and **Token** to one or more target nodes of type **String** and **Token**.

1. To apply string processing, select the source node and navigate to the **Functions** tab and choose **Switch to String Processing**.

This will automatically consider the properties of the mapped source and target nodes without the need to create any additional functions or post processing scripts.

i Note

Switching to String Processing will delete the existing function.

2. The string processing function considers the length and cardinality of the target nodes to distribute lengthy source message among the target nodes. When the source message exceeds the maximum string length defined for the target node, the message gets truncated.

To process messages that contain delimiters, select the corresponding delimiter from the **Source Delimiter** drop down.

Working with Code Value Mappings

Integration Advisor supports code value mappings and global code value mappings.

Code Value Mappings

Code value mappings represent the transformation of a code value that is set at the source element into a code value that is required at the target element. The strings for each element can be different, but the semantics and definition of both elements should be the same.

1:1 and N:1 Cardinality

You can create code value mappings with both 1:1 and n:1 cardinality:

- A **1:1 relationship** enables you to define mappings between predefined code values from a source codelist and a target codelist. Each code value in the source codelist is uniquely mapped to a corresponding code value in the target codelist, establishing a 1:1 relationship.
- An **n:1 relationship** supports more complex scenarios in which multiple source codelists, each with their own predefined code values, are mapped to a single target codelist. This is referred to as an n:1 relationship, allowing for the transformation of data from various sources into the target source.

You can map more than one source node with a target node for both local and global code value mapping.

Furthermore, in these two cases, you can set a **default value**. This allows you to set a default value for the mapping in case no specific values are configured for the target code value.

i Note

Code value mappings allow you to only map between pre-defined sets of values and doesn't support mapping a non-value. If you want to use your own implementation, it's recommended to create an XSLT code. See [Functions](#).

Code Value Mapping and Global Code Value Mapping

Code value mappings are created within a specific MAG, accessible only by a particular node of that MAG. They're stored as part of the MAG and follow its versioning control. See [Code Value Mapping](#).

In contrast, **global code value mappings** are created independently, outside of any MAG, and can be utilized by multiple MAGs. They have their own version controlling and each version can be reused across multiple nodes within a MAG. See [Global Code Value Mappings](#).

You can convert a local code value mapping into a global code value mapping, whether it's a 1:1 or an n:1 code value mapping.

Related Information

[Code Value Mapping](#)

[Global Code Value Mappings](#)

Code Value Mapping

Code value mappings represent the transformation of a code value that is set at the source element into a code value that is required at the target element. They're created within a specific MAG, accessible only by a particular node of that MAG, and are stored them as well.

1. Go to **►Design ► MAGs ►** and select the mapping guideline you want to work with, then navigate to the mapping editor.
2. Draw a mapping line between the **source node** and the **target node**, both associated with the codelist. This action makes the **Code Value Mapping** tab visible.

Drawing a line between one source node and one target node creates a 1:1 code value mapping, while drawing lines from multiple source nodes to a single target node establishes an n:1 code value mapping. The first source node you select becomes the primary source. The order in which you draw the mapping lines determines their sequence from left to right in the mapping table.

❖ Example

In an n:1 code value mapping, you mapped the source nodes **AddressTypeCode** and **CountryCode** with the target node **AddressType**. The **Code Value Mapping** tab would display a table with three columns: one for each of the two source nodes and one for the target node. The value mappings defined in the tables are processed from top to bottom, ensuring that the most specific rule is applied first during runtime.

Adding Code Value Mappings

You can either add code value mappings **manually** or **automatically using the proposal** option.

Manually Add Code Value Mappings

Choose the source node and select **Code Value Mapping**.

To manually add code value mapping entries one by one, choose **►Add ► One source value ►**. A new line is added to the table, with the source field being (any value) and the target code value left empty, indicating an error state. Complete the configuration to resolve the error.

To add any source values that are not yet included in the list, choose **►Add ► All remaining primary source values ►**. This option facilitates mass operations and makes the process more efficient.

Use **Clear** to remove any incomplete entries from the list.

Once you're finished, the tab displays the code values of the source and target node. The **Source Code Value** displays the list of code values selected for the source node and the **Target Code Value** displays the list of code values selected for the target node. You can change the code value in the source and target node by selecting the value help  and choosing the relevant value from the list.

Automatically Add Code Value Mappings from Proposal Service

If you've used the proposal service on the entire MAG and added a code value mapping from the proposals, the **Code Value Mapping** tab shows the **Proposal** option. This only applies if the code value mapping was added from the proposals.

Select the **Proposal** button for both 1:1 and n:1 mappings to display a list of proposals. You can select the necessary proposals from the list and choose **Add**.

Setting Default Values

By setting default values in code value mappings you ensure reliability in the value transformation in case specific source code values are missing. To set a default value, perform the following steps:

1. Open a mapping guideline and switch to the tab **Code Value Mapping**.
2. To set a default value, choose the value help  and select a default from the list.

You can only set a default value from the rules other than the ones mentioned in the mapping list and you can also select only the target code values for the default value.

The options differ depending on the cardinality:

- o For **1:1 code value mapping**, you can either pass through the source value directly or select any code from the target code list.
- o For **n:1 code value mapping**, you can either pass through the value from the primary source or select any code from the target code list.

3. Save your changes.

Organizing Code Value Mappings

Deleting Code Value Mappings

To delete a code value mapping, choose **Delete** at the beginning of each entry. This allows you to remove code value mappings one by one.

Copy Code Value Mappings to Different Nodes

To copy a value mapping from another mapping element, choose **Copy** and select a value mapping from the list.

Sort code value mappings

Use the sort code value mapping  button to reorder the code value mappings. This way, you can ensure that more specific mappings are applied first and thereby increase the accuracy of mapping each source code value to its corresponding target code value.

Reorder source codelists

Use the reorder source codelists  button to adjust the source code list. In the **Reorder Source Codelists** dialog box, select the codelist and adjust the order using the up ^ and down v arrows. To confirm the new order, choose **OK**.

Resolving Error States

If any of the following entries are present in your mapping, the corresponding rows are marked as an error:

- Duplicate code value mappings
- Empty value mappings
- No target code value selected
- None of the source code values selected

Make sure that you fix the errors identified by the rules listed above.

Switching to Function

A code value mapping is usually made at the semantic level and generated into a transformation (XSLT) function once the runtime artifacts are generated.

To convert your code value mapping into a function, choose **Switch to Function**. This automatically generates a corresponding XSLT function for the code value mapping, under the **Functions** tab.

Caution

This action is irreversible. Switching to function deletes the code value mapping.

The generated default XSLT function is based on `<xsl:choose>`, wherein the source code values are compared against a range of possible values. Each selected source code value is expressed by a `<xsl:when>` clause and the mapped target code value in `<xsl:sequence>`. All source code values that aren't mapped are written into the output, expressed by `<xsd:otherwise>`. You can change the function according to your requirements.

Working with Global Mappings

Create Global Mapping

You can also create a global mapping out of the local code value mapping by choosing **Create Global Mapping**. This global mapping can be used in other mapping guidelines.

Assign Global Mapping

You can use the following options only if you have used a standard code list or a global code value mapping created out of a custom codelist.

Choose **Assign Global Mapping** if you want to reference a global mapping to that particular code value. Once you assign a global mapping, the existing mapping is overwritten with the global mapping values and a link to the global mapping appears next to code value mapping.

To remove the global mapping, choose **Detach Global Mapping**. This only removes the global mapping reference while the values are retained local to the mapping guideline.

Deprecated Code Values

Code values can deprecate, as shown by the icon .

Code values can deprecate for the following reasons:

- The value is still in the codelist but is deselected in the message implementation guideline.
- The value was removed from or replaced in that particular version of the codelist.
- The code value is removed from or replaced in the message implementation guideline. In this case, the domain GUID of the missing value is displayed in the MAG.

You can distinguish between these values using the tooltip provided under the  icon.

If the deprecated values are found under **Source Code Value**, you can delete them using  provided next to each value.

If the deprecated values are found under **Target Code Value**, you can replace them with an existing value using the value help .

You can select deselected values again in the respective message implementation guideline.

i Note

When you create a global mapping out of a code value mapping that has deprecated values due to any of the above mentioned scenarios, those values aren't added to the newly created global mapping.

Also, if your global mapping contains any deprecated values, you can now delete those value directly from the global mapping using the delete  icon provided next to them.

Functions

Functions are used to specify a data transformation between source and target nodes.

You can use functions to transform or compute data within the Mapping Guidelines (MAG) editor. The three types of data transformation available are — Functions, Date time mapping, and String processing. For more information on Date time mapping and String processing, see [Value Transformations](#).

The **Function** tab on the mapping pane is available only when leaf nodes are mapped or when a function is assigned to a target leaf node with no corresponding source node. You can create and assign a function to a target leaf node that isn't mapped to any source node.

The **XSLT Code** editor in the **Function** tab of the mapping pane is used to formulate expressions based on XSLT V3.0 or XPath 3.1. A default code snippet is available when you map a source leaf node to a target leaf node. You can create your own implementation of the XSLT function by editing the default code snippet to meet your data transformation requirements.

You can use parameters to pass values to functions. For instance, you can define a default value for the parameter *PARAM* and pass it to a function as an argument. You can create a new parameter by choosing **Add New Parameter** on the **Function** tab and bind it to a parameter defined in the **Global Parameters** tab. Use the shared code feature to create frequently used code snippets and reuse it as required within your MAG.

i Note

- All XSLT 3.0 and XPath 3.1 functions are supported.

- For more information, refer to XSLT or XPath specification published by W3C.

Example

The following are a few examples of XSLT processing tasks:

- **Source Data as an Input** - An input value is passed as an argument to a function to transform it. Use the `substring` function to extract the first three characters of the source data.

Sample Code

```
<xsl:sequence select="substring($nodes_in,1,3)"/>
```

- **Function as an Input** - The `current-dateTime` function is passed as an argument to a function to extract the year component from it.

Sample Code

```
<xsl:sequence select="year-from-dateTime(current-dateTime())"/>
```

- **Parameter as an Input** - A parameter is passed as an argument to a function. The global parameter *PARAM* defined and assigned a value of 01 is concatenated with the source field and assigned to the target field.

Sample Code

```
<xsl:sequence select="concat($PARAM,$nodes_in)"/>
```

Related Information

[Using Functions](#)

[Creating a Shared Code](#)

[Defining Global Parameters](#)

[Using Functions Without a Source Node Mapping](#)

Using Functions

Use functions to specify a data transformation between source and target nodes within the Mapping Guidelines (MAG) editor.

Prerequisites

- You've opened the MAG in the edit mode.
- You've mapped a source leaf node to a target leaf node.

Procedure

1. Select the target leaf node that you want to apply the function to.

The **Function** tab of the mapping pane is available with a default code snippet in the **XSLT Code** editor.

2. On the **Function** tab, in the **XSLT Code** editor, edit the code snippet as required. The scope of the function is local by default. You can also create a function to reuse it across the mapping. For more information, see [Creating a Shared Code](#).
3. To reuse a function from **Shared Code** in the **XSLT Code** editor:
 - a. Select the **Shared Code** option.
 - b. Choose **Replace**.
 - c. Select the shared code that you want to use from the drop-down list.
 - d. Optional: Choose **Edit** to modify the shared code snippet. The changes are saved globally.
4. Choose **Validate** to validate your XSLT code snippet. If there are errors in your code snippet, the validation will display the errors in detail along with the line numbers for easy identification.

→ Tip

Once resolving the errors, re-run the validation to confirm that your code is error-free.

5. To pass a parameter to a function, perform the following:
 - a. On the **Function** tab, choose **Add new parameter**.
 - b. In the **Create New Function Parameter** dialog box, enter the parameter name in the **Function Parameter Name** field. Choose **Create**.
 - c. **Choose a Global Parameter** to bind it to the function parameter. For more information on creating global parameters, see [Defining Global Parameters](#).
 - d. Select the required global parameter from the **Global Parameters** list. Alternatively, you can also filter for a parameter using **Search**.
6. Optional: Select the **Confidential Function Content** option if your code snippet contains any confidential information such as a specific business partner ID or any personal information.

→ Remember

You can choose this option if you don't want your code snippet to be published to the Global Index Server used by the proposal service.

7. **Save** the changes.

Creating a Shared Code

Create frequently used code snippets and reuse it as required within your Mapping Guidelines (MAG).

Prerequisites

You're in the edit mode of the MAG editor.

Procedure

1. On the **Shared Code** tab, choose **Add** to create a reusable code snippet.
2. In the **Create New Snippet** dialog box, enter the code snippet name in the **Name** field.
3. Optional: Enter a summary for the code snippet in the **Summary** field.
4. Choose **Create** to create a new code snippet.

You can see the list of shared code snippets in the **Shared Code** tab.

5. On the **Overview** tab, enter any supporting information in the **Documentation** field.
6. On the **Definition** tab, in the **Implementation** field, enter the function definition.

7. Choose **Save**.

→ **Tip**

You can also choose **Share** beside the **Local Code** option to create a reusable code snippet of the local code in the **XSLT Code** editor.

Quick Actions

Icon	Description
 (create new copy of this snippet)	Creates a copy of an existing code snippet.
 (Delete Snippet)	Deletes the code snippet.

Defining Global Parameters

You can use **Global Parameters** tab to create and define reusable parameters within your Mapping Guideline (MAG).

Prerequisites

You're in the edit mode of the MAG editor.

Procedure

1. On the **Global Parameters** tab, choose **Add** to create a new parameter.
2. In the **Create New Parameter** dialog box, provide a name in the **Parameter Name** field and enter a default value in the **Default Value** field.
3. Optional: Enter a summary for the parameter in the **Summary** field.
4. Choose **Create**.

You can see the list of parameters in the **Global Parameters** tab.

5. To delete a parameter, select the required parameter from the list of **Global Parameters** and choose  (*Delete Global Parameter*).

Using Functions Without a Source Node Mapping

Create and assign a function to a target leaf node with no corresponding source node mapping.

Prerequisites

- You've opened the MAG in the edit mode.
- You've a target leaf node that isn't mapped to any source node or doesn't have any constant value assigned to it.

Procedure

1. On the target leaf node, choose **Action Menu** > **Set Function**.

A mapping line to the target node with a function icon (fx) appears. The **Function** tab of the mapping pane is available with the default code snippet in the **XSLT Code** editor.

2. On the **Function** tab, in the **XSLT Code** editor, edit the code snippet as required. The scope of the function is local by default. You can also create a function to reuse it across the mapping. For more information, see [Creating a Shared Code](#).
3. To reuse a function from **Shared Code** in the **XSLT Code** editor:
 - a. Select the **Shared Code** option.
 - b. Choose **Replace**.
 - c. Select the shared code that you want to use from the drop-down list.
 - d. Optional: Choose **Edit** to modify the shared code snippet. The changes are saved globally.
4. To pass a parameter to a function, perform the following:
 - a. On the **Function** tab, choose **Add new parameter**.
 - b. In the **Create New Function Parameter** dialog box, enter the parameter name in the **Function Parameter Name** field and choose **Create**.
 - c. **Choose a Global Parameter** to bind it to the function parameter. For more information on creating global parameters, see [Defining Global Parameters](#).
 - d. Select the required global parameter from the **Global Parameters** list. Alternatively, you can also filter for a parameter using **Search**.
5. Optional: Select the **Confidential Function Content** option if your code snippet contains any confidential information such as a specific business partner ID or any personal information.

→ Remember

You can choose this option if you don't want your code snippet to be published to the Global Index Server used by the proposal service.

6. **Save** the changes.

Working with Overlay Mapping Guidelines

Learn how to work with overlay mapping guidelines.

Context

Overlay mapping guidelines automatically include all mapping elements from the underlying base mapping guideline.

i Note

If you change a mapping element in the base mapping guidelines, this change is automatically considered in all overlay mapping guidelines in which this base mapping element is enabled.

Procedure

1. Open your overlay MAG and choose **Edit**.
2. The **Overview** tab displays the general information that you provided at the time of creation such as the name, version, source and target MIGs used in the MAG. The tab also includes a **Base MAG** section, which displays the underlying standard MAG used to create it.
3. Choose **Mapping** tab to edit the mapping details of the MAG. See [Mapping the Source and Target Nodes](#).
4. Overlay MAGs automatically inherit mappings from their base MAG, which are visible in the **Mapping** tab. You can manage the inherited base mappings by choosing to retain or delete them, and also add new mapping elements specific to this overlay MAG. Newly added mapping elements are displayed in green, while inherited elements from the base MAG appear in blue. The following options are applicable when mapping the source and target elements in an overlay MAG:
 - a. **Retain Base Mapping Element:** No action is required if you want to retain a base mapping element. This mapping element is executed as part of the overlay mapping.

- b. **Create a new Overlaid Mapping Element:** Drag the field from **Source** to the field on **Target** that you want to map to.
- c. **Delete a Base Mapping Element:** Simply delete the mapping element in the graphical mapping panel or choose **✕ Delete** in the **Action** column of the corresponding element in the **Mapping List** table. The deleted mapping elements aren't executed as part of the overlay MAG.
- d. **Overwrite a Base Mapping Element:** If you want to replace/overwrite a base mapping element with overlay mapping element, select the existing mapping line and choose **Create Editable Copy**. This deletes the base mapping and allow you to create a new overlay mapping.

See also [Matching Rules for Overlay Mapping Guidelines](#).

- 5. Overlay MAGs automatically inherit **global parameters** and **shared code** from their base MAG. You can't modify or delete these inherited elements.

You can **define additional, overlay-specific global parameters and shared code**. Make sure that they have different identifiers compared with their base MAGs.

- 6. Overlay MAGs automatically inherit **target node duplication rules** from their base MAG. You can't modify or delete these inherited rules.

You can **define additional, overlay-specific target node duplication rules**. If both base MAG and overlay MAG duplicate the same target node, the base duplicates are shown first and the overlay duplicates afterwards.

i Note

The behavior of duplicate target nodes in base MAGs and overlay MAGs changed with release 2601. As a result, all impacted overlay MAGs created before that release were migrated. For details, see [Duplicate Target Nodes in Overlay MAGs](#).

- 7. You can also use the **Proposal** Service to get a proposal on fields that might be most relevant for you.
 - a. Choose Proposals: Get Proposals to activate a proposal indicator that displays which fields might be most relevant for you. This is calculated based on the other available MAGs.
 - b. Choose Proposals: Select **Best Proposal** to automatically select all proposed mapping entities or select single proposed mapping entities by clicking on the checkboxes within the mapping list.

The mapping list displays all the relevant information of the source and target nodes such as source nodes, target nodes, their cardinality and so on. The **Type** column displays the type of each entry in the mapping list. You can filter and view the mappings that are erroneous and mappings that have proposals using the drop-down filter available above the mapping list table.

- 8. You can also search for a particular value in the **Mapping List** tab by selecting the following column headers and entering the search value in the **Filter**.
 - Source
 - Source Name
 - Target
 - Target Node Name

This filter option is also available for the **Code Value Mapping** tab.

- 9. If you want to search for all XSLT snippets and XPath expressions for a certain text pattern and want the list of mapping elements to be filtered by those elements that contain the search pattern, you can use the **Search** field provided above the **Mapping List** table. Use the search field to enter the function pattern and select **Q**. This displays the list of mapping elements pertaining to that function pattern.
- 10. For overlay MAGs, there are search options provided above the mappings table that allow you to filter the mappings in the **Mapping List** table on various criteria:
 - a. **Resultant Mappings:** Displays the mappings that are executed for this overlaid MAG.
 - b. **Overlaid Mappings:** Displays only the mappings created directly within this overlay MAG.
 - c. **Selected Base Mappings:** Displays the selected base MAG mappings inherited by this overlay MAG.
 - d. **Excluded Base Mappings:** Displays the deleted base MAG mappings inherited by this overlay MAG. To restore a deleted base mapping, select the **+** button next to the specific mapping.

11. To delete overlay mapping elements and exclude base mapping elements in the overlay mapping guideline, select one or more mappings in the mapping list and choose **Delete/Exclude**.
12. You can add or manage additional information or functions about each element by selecting the mapping entity line.

The mapping also supports various value transformations. To learn about these features, see [Value Transformations](#)

13. You can combine an Overlay MAG with pretransformation (PTS). For this, you have the following options:

- **Option 1: Base MAG with PTS and Overlay MAG without PTS**

If your Base MAG has defined a PTS, the PTS is automatically applied to the Overlay MAG. If you want to use the PTS of the Base MAG unchanged, you don't have to do anything.

- **Option 2: Base MAG with PTS and Overlay MAG with modified PTS**

If your Overlay MAG should execute a different or extended PTS than the Base MAG, create an overlay-specific PTS.

For the PTS of your Overlay MAG, either create a new (empty) PTS or create an editable copy of the PTS of the Base MAG.

i Note

If your overlay MAG has its own PTS, it always overrules the PTS of the base MAG.

- **Option 3: Base MAG without PTS and Overlay MAG with PTS**

You can create a PTS for your Overlay MAG even if your Base MAG doesn't contain a PTS.

- **Option 4: Disable PTS of Base MAG without PTS of Overlay MAG (Indirectly)**

Currently, you can't disable the PTS of the Base MAG without having a PTS in the Overlay MAG. As a workaround, you can create an empty PTS (PTS without operation) for your Overlay MAG, which then overrides the PTS of the Base MAG.

⚠ Caution

If your Overlay MAG uses a different PTS than your Base MAG (options 2, 3, and 4), make sure that all the mapping elements you want to reuse from the base MAG are still valid in the context of the overlay MAG.

Matching Rules for Overlay Mapping Guidelines

Matching rules that define how identifiers in base and overlay mapping guidelines are mapped to each other.

According to the basic matching rules, same identifiers in base and overlay MAG are mapped.

For example, the segment group **SG1 in the base** is mapped to the segment group **SG1 in the overlay mapping guideline** because they have the same identifier, **SG1**.

If there are **structural differences** between base and overlay mapping guidelines, the mapping line can't be drawn and throws an error. In these cases, the identifier in the base mapping guidelines couldn't be mapped to a corresponding identifier in the overlay mapping guideline.

While errors exist in your mapping, you can't simulate the mapping guideline. Exclude these erroneous base mappings. They're shown as missing elements in the mapping list.

Extended Mapping Rules

Special Mapping Rules for EDIFACT

For EDIFACT mapping guidelines, extended matching rules apply.

If the structure of an EDIFACT overlay mapping guideline is changed, the numbering of segment groups changes as well. This leads to mismatches in the identifiers between base and overlay mapping guideline.

For example, a **base** mapping guideline looks as follows:

- SG1
 - ADDRESS
- SG2
 - NAME

Due to an update to the EDIFACT **overlay** mapping guideline, its structure has been changed to look as follows, with the segment group identifiers no longer corresponding to those of the base mapping guideline:

- SG1
 - DATE
- SG2
 - ADDRESS
- SG3
 - NAME

If the matching followed the basic matching rules, the mapping of **SG1 (base)** would **fail** since ADDRESS and DATE can't be matched.

Instead, the **extended matching** recognizes that the trigger segments match and therefore matches **SG1 (base) - ADDRESS** with **SG2 (overlay) - ADDRESS**. The matching is therefore defined through the **trigger segment** and the mapping line is automatically rerouted.

You can still edit overlay mapping guidelines yourself and disable nodes manually if a mapping is erroneous.

Mapping Segments That Moved to Segment Groups

A segment in a base mapping guideline could be moved to a segment group in an overlay mapping guideline, or the other way around. This mapping rule ensures that the segments can still be mapped.

For example, a **base** mapping guideline looks as follows:

- LIN
- MOA

In the corresponding **overlay** mapping guideline, the segment MOA has been moved and now sits within the segment group SG27:

- LIN
- SG27
 - MOA

With the extended matching rules, the segments can still be mapped, even though it's been moved one layer deeper in the overlay mapping guideline.

Mapping Data Elements That Moved to Composites

A data element in a base mapping guideline could be moved to a composite data element in an overlay mapping guideline, or the other way around. This mapping rule ensures that the data elements can still be mapped.

For example, a **base** mapping guideline looks as follows:

- C002
- 1004
- 1225

In the corresponding **overlay** mapping guideline, the data element 1004 has been moved and now sits within the composite C106:

- C002
- C106
 - 1004
- 1225

With the extended mapping rules, the data elements can still be mapped, even though it's been moved one layer deeper in the overlay mapping guideline.

Limitations

The following limitations should be considered when working with an overlay MAG.

Limitations

Limitation	Description
Shared Code	• Base Mapping Elements can use Shared Code defined in the Base MAG, which will be considered in the Overlay MAG. • Overlaid Mapping Elements can use Shared Code defined in the Overlay MAG. • Overlaid mapping elements cannot reference Shared Code defined in the Base MAG. i Note Shared Code definitions must have a unique name across both Base and Overlay MAGs.

Duplicate Target Nodes in Overlay MAGs

i Note

The behavior of duplicate target nodes in MAGs has changed with release 2601 (March 2026). The following migration considerations apply only to overlay MAGs created before that release.

Integration Advisor has improved the handling of duplicate target nodes in base and overlay MAGs. Originally, target node duplicates from the base MAG weren't visible in the overlay MAG. While you could duplicate target nodes in the overlay MAG, they weren't linked with existing duplicates in the base MAG and could therefore lead to conflicts.

Now, target node duplicates from the base MAG are visible in the overlay MAG. The target node duplicates defined in an overlay MAG are also explicitly stored as overlay duplicates to allow side-by-side duplicates from base MAG and overlay MAG.

With release 2601, all overlay MAGs with duplicates are migrated once accessed. The finished migration is indicated by a warning icon in the Status column in the MAG editor for the relevant nodes. In most cases, the previous mapping is still valid, but in some edge cases, the mapping behavior can have changed. Therefore, it's advised that you **check if your mapping is still valid and produces the same mapping output**.

The automatic migration has the following effects:

- Duplicate target nodes from the base MAG are now visible in the overlay MAG.
- Base mapping elements pointing to the base duplicate nodes are now valid in the overlay MAG and can be used.

i Note

If you ever disabled these mapping elements because they were invalid, you can enable them again now.

- Earlier defined duplicate target nodes from overlay MAGs are automatically migrated to the new overlay duplicates.
- Earlier defined overlay mapping elements by default now point to the overlay duplicates. Therefore, **check that your original mapping still works as intended**

For more information and illustrative examples, see the blog [Integration Advisor - Base Duplicate Nodes in Overlay Mapping and Automatic Migration](#) .

Simulating Mapping Guidelines

Learn how to simulate a mapping guideline.

Procedure

1. Open your mapping guideline and navigate to the **Mapping** tab.
2. Expand all nodes to see if the mapping is finalized and choose **Simulate** to perform an embedded simulation in the MAG itself. There are two different ways to simulate the MAG:
 - a. **Simulate with MIG Example Data**: This simulates the MAG with the existing sample MIG data.
 - b. **Simulate with Payload Data...**: This prompts you to upload a sample payload file (XML or EDI flat file) as **Source Payload** for the MAG simulation. For more information, see the blog [Integration Advisor – direct support of EDI payloads](#) .

Optionally, you can also upload an **Expected Target Payload** (XML or EDI flat file). After the simulation, you can use this file to compare the simulated and expected values.

i Note

The EDI flat file is supported only for EDI-based type systems such as ASC X12, Odette, TRADACOMS, UN/EDIFACT and its subsets, and VDA.

i Note

If there are errors in the simulation, the **Error** dialog displays the error with the specific node and a direct link to access the node for easy resolution.

3. A new field **Simulation Data** appears above the source structure displaying the type of simulation that was carried out.

If you've uploaded an expected target payload, there's an additional field for **Expected Data**, as well as the number of differences that come up during the comparison of the two datasets.

You can re-run the simulation based on the option you chose by using the  *Rerun simulation* button that appears next to this field.

4. A new column **Simulation Data** appears in the **Source** and **Target** structure populated with data related to the mapping for each node:

- The rows with values on the source structure are the input values from the source MIG.
- The values found in the target structure are the transformed input values.
- If there are more than one instance of a leaf node in the simulation data, they are denoted by the navigation buttons  . These buttons allow you to navigate to the other occurrences of the payload values.
- The navigation button  is also displayed for a group node if it occurs at least once in the simulation data.
- The  icon denotes the input value on the source message or the transformed value on the target message is empty.
- The  icon denotes that the leaf node doesn't exist in the simulated data.

5. If you've uploaded an expected target payload, the additional column **Expected Data** appears in the target structure. The target MIG tree expands automatically, and the nodes show the comparison status for group node instances (based on instance count) and leaf node data (based on value).

Use the buttons  *Previous Difference* and  *Next Difference* to navigate through the differences that were found in the comparison of the simulation data and the expected data.

6. Choose a mapping entity in the source or the target structure to view the mapping after simulation.

7. If you've added any conditions on a source leaf node to control the instances of a target group node, you can view the following details of that conditional mapping under the **Condition** tab of that mapping entity.

- The nodes that are involved in the conditional mapping
- The status of the creation of a target node based on the input provided in the mapping

The target nodes that are not created due to the conditional mapping are also denoted with the  icon in the mapping structure.

i Note

Wherever conditional mapping is involved, the creation of the target node instances depends solely on the condition defined under the conditional mapping.

8. For target group nodes with conditional mapping, the **Simulation Data** column only displays the instances of the node which fulfill the condition criteria. If you want to see all the instances of the node, irrespective of whether they fulfill the condition or not, choose .

If you've uploaded an expected target payload, the comparison results are hidden in this case.

9. After a MAG simulation has run, choose  *Download simulation result* to save the target payload resulting from the simulation. The default option is **Download as XML**, in the Cloud Integration XML format without qualifier suffixes from Integration Advisor.

If the target message is an EDI message from one of the type systems ASC X12, Odette, TRADACOMS, UN/EDIFACT or one of its subsets, or VDA, you can also choose [Download as EDI Flat File](#), as an alternative to the XML file. For more information, see the blog [Integration Advisor – Download MAG Simulation Result as EDI Payload](#).

10. If you don't need the simulation data anymore, you can clear it by choosing **X Clear Simulation**. The data from any expected target payloads is cleared as well.

Versioning a Mapping Guideline

This chapter shows you how to activate a mapping guideline

Once you have edited the mapping details in an MAG that suits your scenario, you can activate the mapping guideline for using it productively. There are 3 different types of status set for an MAG and each one of them is defined below:

- **Draft:** When you create a new mapping guideline, the newly created MAG has the **Status** field set to **Draft** and the **Version** set as **1.0**. You can now modify this new version of MAG according to your business requirements.

i Note

You can also copy an existing mapping guideline. The copied MAG will have the **Version** set as **1.0** and also the term **Copy** suffixed to its name. You can later edit the name of the MAG.

- **Active:** After activating a mapping guideline, the **Status** of the MAG is set to **Active**. Once activated, this version of the MAG will be finalised and it cannot be edited thereafter. But you can continue to export documentation and runtime artifacts from the MAG.

i Note

Activating a MAG will also anonymously push the mapping guideline into the knowledge graph to contribute and improve the proposal service.

It is always recommended to activate a MAG to use it productively. By doing so,

- You protect the MAG against any unwanted changes
- Any new requirements for the MAG can be maintained in a new **Draft** version of the MAG without changing the active MAG version currently in use.

If you want to create the next draft version of an active MAG, open the active MAG version and choose **Edit**. You can also choose **Edit** from the action  button of the MAG overview. This will prompt you to confirm creating a draft version. Choose **OK**. Now the next version of the MAG gets created with the **Draft** status. For example, if the active version of the MAG is 2.0, then the new draft MAG will have the version set as 3.0

- **Deprecated:** After activating a MAG, the previous active version of the MAG gets deprecated and its **Status** field is set to **Deprecated**.

i Note

There can be only one active and draft version of a particular MAG.

Procedure

1. Open your mapping guideline and choose **Edit**.

Before activating the MAG, ensure that you have activated the source and target message implementation guidelines used in this MAG. You can navigate to those MIGs using the links provided under the **Source and Target MIGs** section in the **Overview** tab.

To know how to activate a MIG, see .

2. Choose **Activate**. The system now activates the activated mapping guideline.

i Note

Any subsequent edit operations on the same MIG would be saved as a different version.

After successful activation of MIG, a confirmation box appears with the following information

- List of values that get published into the knowledge graph
- List of values that will not get published into the knowledge graph

Deleting Mapping Guidelines

Learn the options that are available for deleting mapping guidelines that are no longer required.

You can choose to delete a MAG including its draft, active, or deprecated versions or only a specific version. You can also mass delete mapping guidelines.

Caution

Deleting MAGs is permanent and can't be undone.

Delete Mapping Guidelines Including Draft, Active, and Deprecated Versions

1. On the **Mapping Guidelines** tab, in the **Mapping Guidelines** screen, choose  *More Options* on the MAG that you want to delete and then choose **Delete**.
2. In the **Delete Mapping Guideline** dialog box, select the **Delete Mapping Guideline (including history)** option.

Delete Specific Versions of MAGs

Using the Mapping Guidelines Tab

1. On the **Mapping Guidelines** tab, in the **Mapping Guidelines** screen, choose  *More Options* on the MAG that you want to delete and then choose **Delete**.
2. In the **Delete Mapping Guideline** dialog box, select the **Delete this (latest) version** option.

Using the Overview Tab

1. On the **Mapping Guidelines** tab, in the **Mapping Guidelines** screen, choose the specific version of the MAG that you want to delete.
2. On the **Overview** tab, depending on the screen size, choose **Delete**, or choose  *More* and then choose **Delete**.
3. In the **Confirm** dialog box, choose **OK**.

i Note

You can't delete active MAGs.

Using the History Link

1. On the **Mapping Guidelines** tab, in the **Mapping Guidelines** screen, choose **History** on the MAG that you want to delete.
2. From the **Mapping Implementation Guideline History Versions** table, choose  **Actions** on the version that you want to delete.
3. In the **Confirm** dialog box, choose **OK**.

i Note

You can't delete active MAGs.

Mass Delete Mapping Guidelines

To delete multiple mapping guidelines at once, perform the following steps.

1. Go to **MAGs** and choose  **Mass Actions** .
2. In the upcoming dialog, select the mapping guidelines that you want to delete.

Use the search and filters to narrow down the list of artifacts and then use the checkbox in the table header to only select those artifacts. For more information on the filter, see [Filtering MAGs](#).

Choose **Next**.

i Note

Mapping guidelines can only be deleted if they're not referenced by any other artifacts.

3. In the **Confirm Selection** step, check that you've selected the relevant mapping guidelines and versions. If you're satisfied, delete them by choosing **Confirm Deletion**.
4. In the **Results** screen, you can see the progress of the deletion. Once the deletion is finished, you can download a results file that contains the details of the deleted mapping guidelines by choosing  **Download**.

Updating Mapping Guidelines

Update a group of mapping guidelines. You can either update mapping guidelines by **replacing the message implementation guideline**, or by **replacing the base MAG version in the overlay MAGs**.

Replace MIG in MAGs

In some scenarios, you need to update a message implementation guideline (MIG) for a set of mapping guidelines (MAGs), for example, by changing the source or target MIG. Instead of manually changing this for each mapping guideline, you can update a set of mapping guidelines.

To update a MAG by replacing the MIG, perform the following steps:

1. In the MAG overview page, choose  **Mass Actions** . The **Update Mapping Guidelines** wizard opens.
2. In the **Select MAGs** step, select the checkbox of the MAGs that you want to update. You can also use the filter options provided in the screen to search for a particular MAG. For more information on the filter, see [Filtering MAGs](#).
3. Choose **Next**. The next screen **Select Action and MIG** displays the list of MIGs available in the system for you to choose.
4. You can choose whether to change the source or target MIG using the radio buttons provided in the screen.

5. Select the MIG from the list that you want to replace the source/target MIG with and choose **Next**.
6. The **Compatibility Check** step does a validation check on the MAGs and displays the compatibility check between the MAGs and the MIG that you chose.
7. The **Compatible** column next to each MAG shows the compatibility check of the MIGs.
8. Select the MAGs again from the list that you want to update and choose **Apply Changes**.

i Note

Don't close or leave the browser screen until the update is complete.

A progress bar indicates the progress of the update in the **Results** step. You can also stop the update using the **Stop** button.

Once the update is completed, a table displays the result of the MAGs update. The **Additional Info** column provides additional information on the MAG update.

9. To save the results as an .xlsx file, choose **Download**.

Replace Base MAG Version in Overlay MAGs

To update a set of mapping guidelines by replacing the base version in the overlay mapping guidelines, perform the following steps:

1. In the MAG overview page, choose **Mass Actions > Replace Base MAG Version in Overlay MAGs**. A wizard opens.
2. In the **Base MAG** step, select the base MAG that you want to update. You can use the search and filter options to search for a particular MAG. For more information on the filter, see [Filtering MAGs](#).
3. The step **Old Base MAG Version** displays the list of MAG versions available for your base MAG. The column **Overlays** shows the number of overlay MAGs based on each base MAG version.

Select the base MAG version that you want to replace.

4. In the step **New Base MAG Version**, select the new MAG version you want to use in your overlay MAGs.
5. The step **Overlay MAGs** displays all the overlay MAG versions available based on the previously selected old base MAG version. Select the overlay MAGs from the list in which you want to replace the base MAG version and choose **Apply Changes**.

i Note

Don't close or leave the browser screen until the update is complete.

Finally, in the **Results** step, a progress bar indicates the progress of the update. You can halt the update by choosing **Stop**. Once the update is completed, a table displays the result of the overlay MAGs update.

6. To save the results as an .xlsx file, choose **Download**.

Import and Export

SAP Integration Advisor allows you to export Message Implementation Guidelines (MIG), Mapping Guidelines (MAG), documentation and runtime artifacts. The application also allows you to import MIGs and MAGs.

Export and Import MIG/MAG

There can be situations where you want to bring few of your Message Implementation Guidelines and Mapping Guidelines from one tenant to another tenant or you might have to migrate all of your MIGs and MAGs when you are shifting to a new tenant. This feature helps you achieve a seamless migration of your MIGs and MAGs.

There are few things to be considered when using these two features.

- While exporting and importing MIGs/MAGs, their dependent resources such as the custom message for a MIG (if the MIG is created based on a custom message), the dependent MIGs for a MAG and the Global Code Value mapping for a MAG, Pretransformation of a MAG are included as well.
- The following overwrite scenarios are supported:
 - **Draft to Draft:** If you have made any changes to a MIG/MAG that is in draft state in one tenant and need to import that to another tenant that contains a previous state of that MIG/MAG, you can overwrite that MIG/MAG using the **Import** option. You can also overwrite a current draft state of a MIG/MAG with its previous state. The import dialog would display a warning stating that there will be a potential data loss with the overwrite
 - **Draft to Active:** If you have draft version of a MIG/MAG in one tenant and you have an active version of that MIG/MAG in another tenant, you can overwrite the draft MIG/MAG with the active MIG/MAG.

i Note

For custom messages, the application only supports overwriting

- Draft to Active
- Draft to Draft
- **Active to Deprecated:** You can overwrite a active MIG/MAG with its deprecated version.
- **Draft to Deprecated:** You can also overwrite a draft MIG/MAG with its deprecated version. But it is recommended to import the active version before importing the deprecated version to maintain the overall consistency.
- **Active to Active:** You can now overwrite an active MIG/MAG again with its active version.
- **Deprecated to Deprecated:** You can overwrite a deprecated MIG/MAG with its deprecated version again. Say for example, you have a deprecated MIG of version 1.0 in the tenant and you have manually changed the status of that active 1.0 version of the MIG to Deprecated. Now this local file will have a different timestamp and you can overwrite the existing MIG with this updated MIG.

i Note

The following overwrite scenarios are not supported:

- Active to Draft
- Deprecated to Draft
- Deprecated to Active
- If you are migrating all of your MIGs/MAGs from one tenant to another tenant, it is recommended to adhere to the following steps to avoid any complications:
 - Stop editing the MIGs/MAGs in the old tenant
 - Complete the migration of all MIGs/MAGs to the new tenant
 - Continue editing the MIGs/MAGs in the new tenant once the import is done

i Note

You can import new versions of MIG/MAG into the same tenant where you exported the previous versions from. You can also export a MIG/MAG, make some changes and import to overwrite the existing MIG/MAG in the same tenant.

- Export/Import feature is optimized for:
 - Linear transport paths such as exporting from Tenant A to Tenant B and then from Tenant B to Tenant C (Tenant A → Tenant B → Tenant C).
 - Natural sequence of import where if ZIP1 was exported before ZIP2 from Tenant A, then ZIP1 shall be imported before ZIP2 into Tenant B.

Import might not fully work for other situations such as:

- Cyclic transport paths where for example, you export from Tenant A to Tenant B to Tenant C and then again from Tenant C to Tenant A (Tenant A → Tenant B → Tenant C → Tenant A)
- Mixed sequence of import such as importing ZIP1 after ZIP2 into Tenant B.

Refer the chapters [Exporting MIG/MAG](#) and [Importing MIG/MAG](#) to learn how to import and export MIGs/MAGs.

Exporting Documentation

You can export your MIG or MAG in the following supported documentation format:

- PDF
- RTF
- Excel

You can use the document file as a reference to implement the messages and mappings in the integration solution also as reference documentation for bookkeeping purposes.

See [Exporting Documentation](#) to learn how to export documentation.

Export Runtime Artifacts

Once you have created the Message Implementation Guidelines (MIG) and Mapping Guidelines (MAG), the next step is to use these artifacts in the integration solution to enable integrations with your business partner/s. Integration Advisor helps you export these runtime artifacts by automatically generating a number of schemes, scripts, and examples from a mapping guideline. This automatic generation is a particularly enormous time saver, because without the application, all these schemes, scripts, or payloads must be created separately and manually.

You can then import these runtime artifacts in either of these integration solutions and use them to implement your A2A/B2B integration scenario.

See [Exporting Runtime Artifacts](#) to know how to export the runtime artifacts.

Exporting MIG/MAG

You can export a list of message implementation guidelines (MIG) or mapping guidelines (MAG).

Procedure

1. Log in to your application and go to  *Message Implementation Guidelines* or  *Mapping Guidelines*.
2. In the overview page, choose **Export** to display the list of MIGs or MAGs available for export.

To view more details of a MIG/MAG, choose  *More Details* under the **Info** column.

i Note

The **Export** dialog also displays the number of MIGs and MAGs along with the number of versions available in the tenant. If you do a complete content move from one tenant to another, you can compare these numbers between the old and new tenant. This helps you verify if all MIGs and MAGs have been moved successfully without missing anything.

3. Select the MIG/MAG that you want to export. You can also choose multiple entries from the list.

i Note

The maximum size limit allowed for exporting MIGs/MAGs is 500 MB. Based on this size, an approximate number of MIGs/MAGs that can be selected per export is displayed on the **Export** dialog box. So, if you want to export all MIGs/MAGs, consider splitting them into multiple export files based on the number displayed.

4. Use the filter option provided under each column header to search for a particular MIG/MAG. For more information on the filters, see [Filtering MIGs](#) and [Filtering MAGs](#).
5. Choose **Export** to download the files in the .zip format.

Exporting a single MIG/MAG

If you want to export only one MIG/MAG, follow the following procedure:

1. Select and open your MIG/MAG.
2. Choose **More Options**.
3. Select **MAG Version (ZIP)/MIG Version (ZIP)** from the list.

That specific version of the MIG or MAG is downloaded as a .zip file.

Importing MIG/MAG

You can now import message implementation guidelines (MIG) or mapping guidelines (MAG) into your application.

Procedure

1. Login to your application and choose Message Implementation Guidelines  or Mapping Guidelines  from the left pane.
2. In the overview page, choose **Import**.
3. In the import dialog box, choose **Browse** to upload the MIG/MAG file that you have exported previously from another tenant.

i Note

Upload file in .zip format.

4. Choose **Check** to validate your uploaded file.

The results will display the MIGs/MAGs along with the validation check details. To view the details of the validation, select the icon displayed under the **Check Result** column.

5. Select the checkbox for those MIGs/MAGs from the list that you wish to import.

i Note

If the zip file that you are trying to import contains many MIGs/MAGs, the processing might not be able to import all of the content at once which could result in an error. In such cases, try after some time to import the same zip file again. This time, the application will recognize the content that was already imported and only proceed with the import of the missing content.

You can only select a MIG/MAG which has passed the validation check.

6. Choose **Import**.

7. The resulting screen displays the detail of the MIGs/MAGs import. To view more information on the import, select the icon displayed under the **Import Result** column.

8. Choose **Close**.

You can now view the imported MIGs/MAGs in the overview screen.

Exporting Documentation

You can export message implementation guidelines (MIGs) and mapping guidelines (MAGs) in a documentation format.

Context

You can export your MIG or MAG in the following supported documentation formats:

- PDF
- RTF
- Excel

After the export, you can use the document file as a reference to implement the messages and mappings in the integration solution, and also as reference documentation for bookkeeping purposes.

Procedure

1. Access the message implementation guideline (MIG) or the mapping guideline (MAG) section.
2. Open the MIG or MAG that you want to export as documentation and choose ... *Additional Options*.
3. Choose **Export**, followed by one of the following options, based on your requirements:

Documentation Export

Option	Description
Documentation (PDF)	A PDF document of the MIG/MAG is exported.
Documentation (RTF)	The MIG/MAG in RTF format is exported. RTF documents can be edited by MS Word.
Documentation (Excel)	An Excel document of the MIG/MAG is exported.

Runtime Artifacts

Explore the functionality of runtime artifacts.

Runtime Artifacts are reusable, configurable objects that store dynamic values, mapping rules, and configuration settings essential for the execution and management of integrations in SAP Integration Advisor. These artifacts serve as the bridge between integration design and runtime, enabling flexibility, adaptability, and efficiency in integration processes.

These artifacts encompass more than just source-to-target schema mapping. They comprise a set of automatically generated, processable schemes (XSD) and transformation scripts (XSLT) that facilitate:

- **Syntax Conversion:** Seamless data format transformations, such as EDI to XML or vice versa.
- **Validation:** Verification of declared nodes and supported codes within a payload.
- **Pre-/Post-Processing:** Addition or removal of information before or after mapping.
- **Mapping:** Value mapping from source nodes to corresponding target nodes, enabling accurate data transformation.

Once these runtime artifacts are generated, they can be leveraged within your integration solution, enabling seamless integrations with your business partners.

To know more, see:

1. [Consuming Integration Advisor Artifacts in Integration Flows](#)
2. [Exporting Runtime Artifacts](#)
3. [Inject Mapping Artifacts to SAP Cloud Integration](#)

Consuming Integration Advisor Artifacts in Integration Flows

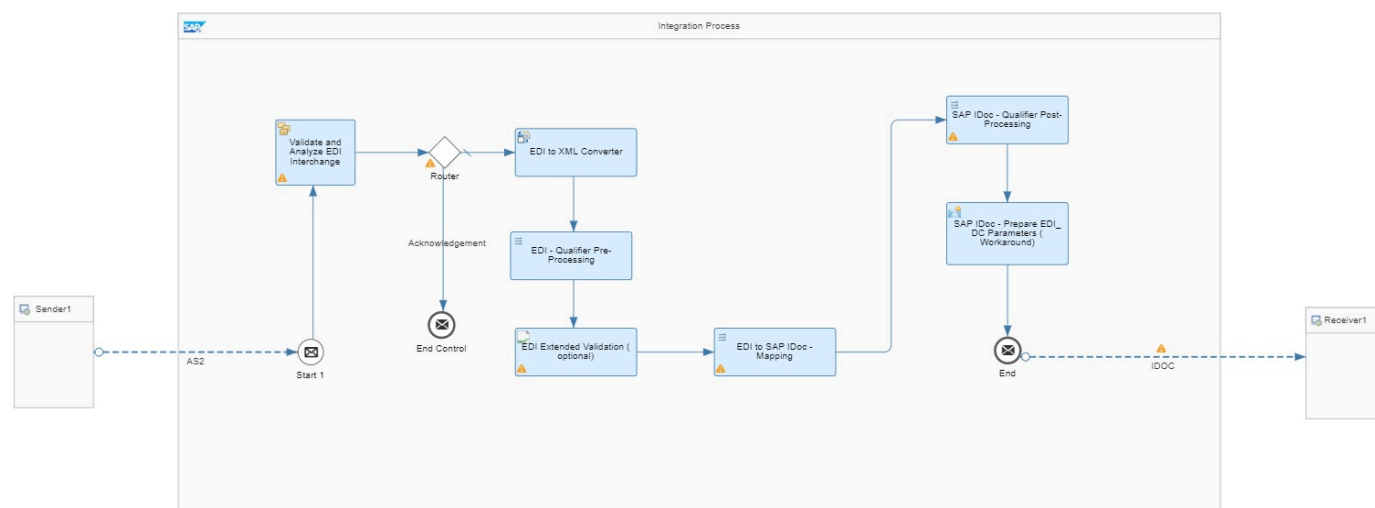
A B2B integration scenario involves many tasks associated with the exchange of business messages between the business partners. The required tasks include defining interfaces for the involved partners, creating mappings between the interfaces, generating the runtime artifacts and, finally, modeling the integration scenario. SAP Integration Advisor and SAP Cloud Integration play a key role in simplifying and streamlining the integration tasks.

Integration Advisor unifies all the required tasks for defining the desired interfaces, mappings, and generating the runtime artifacts. The runtime artifacts generated are used within the processing steps of an integration flow. An integration flow defines how messages are processed and exchanged between the sender and receiver systems in a B2B scenario. The Cloud Integration platform provides a graphical designer interface to configure – the sender and receiver systems, the adapters to connect the sender and receiver systems to the tenant, and the steps that define the message processing. For more information, see [Getting Started with Integration Flow Development](#) and [Configure Adapter in Communication Channels](#).

To get you started with minimum effort in your integration flow creation, Integration Advisor provides a pre-defined set of templates for various B2B integration scenarios. These templates are available in the EDI Integration Templates for SAP Integration Advisor integration package. You can access the integration package via the SAP Business Accelerator Hub. You can copy the integration package into your Cloud Integration tenant and adapt or extend the predefined integration flow as per your requirements. The package provides the required templates for inbound and outbound integration flows. These cover the standard integration scenarios such as EDI to IDoc, EDI to SOAP, IDoc to EDI, and IDoc to cXML. However, the list of templates and scenarios covered is growing rapidly.

The Documents tab of the integration package provides details regarding the templates and the usage of the Integration Advisor runtime artifacts in the integration flow steps.

Let's consider the example of EDI to IDoc integration scenario. The various processing steps and the artifacts involved in this scenario are detailed in the following table.



Processing Step	Description
Sender channel	The sender channel is configured using an AS2 adapter. An EDI message is received by the sender adapter.
Validate and Analyze EDI Interchange	The incoming EDI message is validated, and an acknowledgement is generated. The validation of the inbound message is done based on the XSD configured in this step. You've to use the XSD schema located in the source Message Implementation Guideline (MIG) folder within the exported zip file generated in Integration Advisor (IA) – UN- EDIFACT<MessageType> .xsd. The payload is forwarded for further processing based on the result of validation.
Router	Acknowledgement messages and EDI messages are routed separately.
EDI to XML Converter	The EDI message is converted into XML format using the XSD schema, located in the source MIG folder within the exported zip file generated in IA – UN- EDIFACT_<MessageType> .xsd.
EDI - Qualifier Pre-Processing	The qualifier suffixes are added in the XML based on the source MIG definition to perform a content validation of the message in the next step. The resource <SourceMIGName>_preproc .xsl, located in the source MIG folder within the exported zip file is used. After this step, the payload validation is executed based on the defined qualifiers and qualifier values.
EDI Extended Validation (optional)	The result of the qualifier preprocessing step is validated against the Russian Doll (RD) XSD generated for the source MIG in IA. The resource <SourceMIGName>_RD .xsd located in the source MIG folder within the exported zip file is used. i Note If the source MIG folder contains multiple *_RD*.xsd files, all these xsd files need to be added as a Resource to the Integration Flow.

Processing Step	Description
EDI to SAP IDoc Mapping	This step converts the EDI message into SAP IDoc format using the XSLT mapping generated in IA. The resource <MAGName>.xsl located in the MAG folder within the exported zip file is used.
SAP IDoc - Qualifier Post-Processing	The qualifiers within the IDoc message are removed using the XSLT mapping as they aren't required in the final payload. The resource <TargetMIGName>_postproc.xsl located in the target MIG folder within the exported zip file is used.
SAP IDoc - Prepare EDI_DC Parameters	The IDoc control record is defined.
End Event	The End message event is connected to the receiving system.
Receiver channel	The receiver channel is configured using an IDoc adapter.

Exporting Runtime Artifacts

Export runtime artifacts for integration solutions.

i Note

Currently, Integration Advisor supports exporting runtime artifacts for SAP Cloud Integration and SAP Process Orchestration.

To learn more about these runtime artifacts, see the blog [The Mapping Artifacts and their Purpose](#) .

By exporting the runtime artifacts, the application automatically generates a number of schemes, scripts, and examples from a mapping guideline, including:

- Schemes for **syntax conversion**, such as from EDI syntax to XML or from XML to EDI syntax
- Schemes for precise **payload validation** against the rules and constraints defined in the message implementation guidelines
- **Pre- and post-processing scripts** for presorting payloads or enriching them with additional required information for the mapping
- Main mapping XSLT script
- **Examples** for testing the artifacts in the runtime

i Note

In the <MIGName>_testdata_ICA.xml file, for choice elements, some of the choice options are commented out to be XSD-valid. You can choose to change this in the xml file to have other elements uncommented or commented.

This automatic generation is an enormous time saver because without the application, all these schemes, scripts, or payloads must be created separately and manually.

i Note

In SAP Process Orchestration mode

- Not all artifacts are relevant and therefore only fewer artifacts are exported.

- Exporting MIGs and MAGs based on Tradacoms is not supported (due to significant structural differences of the Tradacoms message structures).

You can then import these runtime artifacts in either of these integration solutions and use them to implement your A2A/B2B integration scenario. See [Consuming Integration Advisor Artifacts in Integration Flows](#).

Exporting XSD File from EDI Type System

You can now directly export Message Types of EDI Type Systems as an XSD file, eliminating the need to create a Message Implementation Guideline (MIG). This streamlined process saves time and effort. Follow the following procedure to know more:

1. Log in to your application.
2. Navigate to the **Type Systems** tab from the left pane.
3. Open the Type System containing the Message Type that you want to export as an XSD file.
4. Navigate to the **Messages** tab and open the specific version of the message type you want to export.
5. Choose **Download XSD** to download this message type in an XSD file format.

Exporting Runtime Artifacts from MIG or MAG

1. Access the message implementation guideline (MIG) or the mapping guideline (MAG) section.
2. Select the MIG or MAG in which you want to export the runtime artifacts.
3. Choose ... *Additional Options* and then choose **Export**.
4. Based on your needs, choose one of the following options:

Option	Description
SAP Cloud Integration Runtime Artifacts	The MIG/MAG is exported in a format where you can directly import them in SAP Cloud Integration.
SAP Process Integration Runtime Artifacts	The MIG/MAG is exported in a format where you can directly import them in SAP Process Integration.

The automatically downloaded ZIP file contains the runtime artifacts as XSD, XSL, HTML and text files.

Consuming Runtime Artifacts

The Integration Advisor works with any kind of interface/API format; its automatically generated artifacts can be used in a variety of runtimes or implementations. Each different runtime has specific approaches, formats, pre-conditions, and implementation instructions. These must be considered, if the generated runtime artifacts are implemented into these runtimes and if the involved applications are connected. SAP's intention is to provide templates and mechanisms that accelerate the onboarding of the runtime artifacts into the diverse supported runtime systems such as

- SAP Cloud Integration
- SAP Process Orchestration

Changing XSD file name for EANCOM

B2B flow steps used in SAP Cloud Integration follow a specific naming convention and Integration Advisor follows the same convention for the exported Message XSD.

There's now a unified naming convention for these Message XSDs for all messages from EDIFACT and EDIFACT subsets. Therefore, Message XSDs for an EANCOM MIG have a new file name which will comply to this naming convention. The details of this naming convention is:

Field	Value
New Naming Convention	UN-EDIFACT_<MessageType>_<Version>_<AssociationCode>.xsd  Example - UN-EDIFACT_APERAK_D01B.xsd - UN-EDIFACT_APERAK_D01B_EAN003.xsd
MessageType	Type of message on which the MIG is based on. (Corresponding to UNH > S009 > 0065.)
Version	EDIFACT version. In case of subsets, the version of the parent EDIFACT is considered. (Corresponding to UNH > S009 > 0052 and 0054.)
AssociationCode	Subset-specific code that identifies the Message Guideline. (Corresponding to UNH > S009 > 0057.) i Note - This is an optional part. It's included into the XSD file name if the field 0057 contains either a Fixed Value or a single code value. Preference is given to the Fixed Value set in the field. - If no unique value for AssociationCode is found in the MIG, the standard EDIFACT file name convention is applied.

i Note

The old EANCOM naming convention is still supported. However, if there is an integration flow for EANCOM with old XSD files and you want to use Message XSD from the new Integration Advisor Export, then you would need to replace the older EANCOM files with the new UN-EDIFACT file in your EDI Flow Steps.

Extended Namespace Support for MIGs

Few pointers to be considered for your MIGs for SAP S/4HANA SOAP type systems and Custom Messages created before June 2023.

i Note

If your MIG is affected, there's a warning icon for the root node in the MIG editor.

The application was extended in June 2023 to support namespace handling. Earlier, namespace prefixes were only allowed at the root node level and these prefixes were removed as part of internal processing. The recent addition of GS1 XML to our Type System library has enabled the support of XML elements and attributes which are a part of the namespace and require a namespace declaration.

To elaborate on the changes, a node now carries its namespace prefix in the internal payload format if it's required by the message standard. In other words, the namespace prefix is no longer removed as part of the internal processing.

The change is relevant for the message types of the following type system:

- SAP S/4HANA Cloud SOAP
- SAP S/4HANA On-Premise SOAP
- Custom Messages (if the message is defined in a namespace)

In case your MIG is affected by this change, you need to make sure to follow the following guidelines (particularly for the first new Export of Runtime Artifacts after the change date). When you update your integration flow, you need to replace all the files using the latest export of runtime artifacts. You can't update/replace only a few files as a mix of old and updated artifacts causes compatibility issues.

You might come across the following scenarios while trying to update your integration flow:

- Your MIG uses XSD Assertions and, the XPath of any of your MIG Assertions uses an absolute path starting from the root node.
 - If this is the case, you need to change the XPath of your Assertion to include the namespace prefix of the root node.
 - Or use a relative XPath starting from the level where you've defined the XSD Assertion.
- Your integration flow has an additional flow step within the internal processing and this flow step defines an absolute XPath including the root node.
 - If so, you also need to add the root node namespace prefix to this XPath.
- Your integration flow has an additional flow step within the internal processing and this flow step executes an additional processing/transformation (like a Groovy or XSLT script).
 - If so, you need to check if your script needs to be updated as well.

Using Full or Reduced Message XSD File

You can decide if you want to use the full or the reduced message XSD file in our message implementation guideline.

i Note

This property only applies to EDI type systems (for example, ASC X12, UN/EDIFACT, ODETTE, TRADACOMS, or VDA).

1. Open your message implementation guideline and go to the tab **Runtime Context**.
2. In the section **EDI Flat File**, you can choose between the following options for property **Use complete Message XSD in EDI Flow Steps**:
 - **true (Default)**: The XSD contains all segments from the message type, independent of the MIG.
With this option, you can avoid unexpected conversion errors in the EDI-To-XML-Conversion flow step.
 - **false**: The XSD contains only the segments from the message type that are also selected in the MIG.
With this option, unexpected segments in the EDI payload automatically fail the EDI payload validation in the EDI Splitter flow step and can be used to produce a negative acknowledgment.

i Note

For all new message implementation guidelines, the default value of this property is **true**.

However, for compatibility reasons, all message implementation guidelines created before May 2025 receive the value **false**.

3. Save your changes.

Setting UN/EDIFACT Message Version (Field 0052) for Older UN/EDIFACT Versions

In your MIG, you can define the expected value for message version (field UNH > S009 > 0052). A default value is automatically applied, but you can change it to a different value if needed, like 1 or 2. This setting influences the file name of the message XSD used in the EDI flow steps of Cloud Integration.

i Note

This property only applies to type systems UN/EDIFACT and its subsets, like GS1 EANCOM or Odette EDIFACT.

In all **releases since 1993**, UN/EDIFACT has been using the value D for field 0052. As a result, the EDI flow steps expect a file name like UN-EDIFACT_ORDERS_**D**96A.xsd. But for **releases before 1993**, other values (like 1 or 2) were used for field 0052. In those cases, the EDI flow steps expect a file name like UN-EDIFACT_ORDERS_**2**912.xsd.

For new message implementation guidelines, Integration Advisor automatically assigns a default value according to the following rules:

Default Value	Versions	Applies to
D	Edifact versions starting with D	All versions starting from D.93A S3
1	Edifact versions not starting with D and having minor version 1	- Version 91.1 S2 - Version 92.1 S3
2	Edifact versions not starting with D and having minor version 2	- Version 90.2 S2 - Version 91.2 S2 - Version 93.2 S3

i Note

For subsets of UN/EDIFACT (like GS1 EANCOM) the same logic is applied based on the underlying UN/EDIFACT version.

i Note

For all message implementation guidelines created before September 2025, D is the default value.

To change this value for older UN/EDIFACT releases, perform the following steps:

1. Open your message implementation guideline and go to the tab **Runtime Context**.
2. Choose **Edit**.
3. In the section **EDI Flat File**, choose one of the following options for the property **Configure EDIFACT Message Version**: 1, 2, 88, 89, 90, D, or S.

Improved and Unified MIG Preprocessing for MIGs without Envelope

Integration Advisor provides unified MIG preprocessing for MIGs without envelope.

Some type systems (ASC X12, UN/EDIFACT and its subsets like GS1 EANCOM, Odette, cXML and GS1 XML (2.x series)) require an envelope structure around the actual message structure. When a payload arrives from the business partner, it typically includes both envelope and message parts.

Originally, if your message implementation guideline was defined without an envelope, the actual MIG preprocessing assumed a payload without envelope parts. For **EDI messages**, the envelope could be removed as part of the EDI flow step "EDI-To-XML-Converter". For **XML messages**, you needed a separate content modifier step in your integration flow to extract the message part from the payload.

For **EDI messages** (ASC X12, UN/EDIFACT and its subsets like GS1 EANCOM, Odette), the MIG preprocessing already included additional logic to remove the envelope part if it was still part of the payload. But for **XML messages** (cXML and GS1 XML), the MIG preprocessing was not working correctly if mismatches between payload and MIG occurred. This logic is now unified.

For any MIG preprocessing XSLT created after March 2026, the following applies to type systems with envelopes:

- If your MIG is **defined without envelope**, the MIG preprocessing automatically removes any existing envelope part because the following processing steps (like validation or mapping) expect the payload to have no envelope.
- If your MIG is **defined with envelope**, the MIG preprocessing keeps the envelope because the following processing steps (like validation or mapping) expect the payload to have an envelope.

i Note

If your integration flow in Cloud Integration implements any **special handling** on these payloads, check if it continues to work as expected.

Inject Mapping Artifacts to SAP Cloud Integration

You can inject mapping artifacts from Integration Advisor to your SAP Cloud Integration tenant.

Context

You need to maintain the SAP Cloud Integration tenant details into which you need to inject the resources under the **Destinations** tab of your subaccount in SAP BTP cockpit.

1. Login to SAP BTP cockpit and navigate to your subaccount.
2. Choose ► **Destinations** ► **New Destination** ►
3. Provide the details of your SAP Cloud Integration tenant under from the left pane and select **Destination Configuration**.

Field	Description
Name	Enter a valid name for your destination. This field is case-sensitive. i Note

Field	Description
	The destination name must start with the prefix <code>CPI_TENANT_</code> .
Type	HTTP
Description	You can provide a description for your reference. This field is optional.
URL	For Neo environment, enter the TMN URL of the SAP Cloud Integration tenant. For CF, copy the url from the service key of the Process Integration Runtime service instance and paste it in the URL field.
Proxy Type	Internet
Authentication	Enter the authentication details. i Note The supported authentication modes are: - BasicAuthentication : Maintain the User and Password fields. - OAuth2ClientCredentials : Maintain the following fields for CF Client ID: Copy the value from the <code>clientId</code> field of the service key created for Process Integration Runtime service instance and paste it in this field. Client Secret: Copy the value from the <code>clientsecret</code> field from the service key created for Process Integration Runtime service instance and paste it in the field. Token Service URL: Copy the value from the <code>tokenurl</code> field from the service key created for Process Integration Runtime service instance and paste it in the field. Maintain the following fields if you are working in Neo environment: - Token Service User : Same value as Client ID - Token Service Password : Same value as Client Secret You need to register the client application as the OAuth client in the consumer account using the SAP BTP cockpit with the TMN node details. In the Security > OAuth section, go to the Clients tab. Under Subscription, enter the VM name of the TMN node(it ends with the node type .tmn). For information on creating OAuth client credentials for Neo environment, see: Creating OAuth Client Credentials for Neo Environment For information on creating OAuth client credentials for Cloud Foundry, see: Creating OAuth Client Credentials for Cloud Foundry Environment.

4. Choose [Save](#).

Procedure

1. Access the Mapping Guideline (MAG) section.
2. Select the MAG from which you want to inject the mapping artifacts.
3. Choose ... and then choose [Inject > SAP Cloud Integration Flow Resources](#).
4. Choose an SAP Cloud Integration tenant from the [Connections](#) list.

5. The next step **2 Integration Package** lists down the integration packages available under the tenant that you chose in the previous step. Choose the desired integration package from the list.

Configure-only packages will not be listed in this step.

6. On selecting the package, the next step displays the list of integration flows available under that package. Select an integration flow from the list into which you want to inject the resources and choose **Inject**.
7. The **Result** step provides the following information:
 - a. **Status** to show if the injection was successful
 - b. Package Name
 - c. Integration Flow

If the resource injection is successful, you can navigate to the integration flow directly from the **Summary** step using the hyperlink provided under the **Integration Flow** field.
 - d. List of **Resources Injected** into the integration flow
8. Choose **Close** after reviewing the details.

Creating OAuth Client Credentials for Cloud Foundry Environment

You can create OAuth client credentials to access your SAP Cloud Integration tenant hosted on the Cloud Foundry environment.

Prerequisites

- You've access to a subaccount in the SAP BTP cockpit.
- You've activated the Process Integration service or the Cloud Integration capability of SAP Integration Suite in your subaccount.

Procedure

1. Log into the SAP BTP cockpit and navigate to your subaccount.
2. Choose **Services** > **Service Instances** on the navigation sidebar.
3. Choose **Create Instance**.
4. In the **New Instance** dialog box, enter the following values:
 - a. In the **Service** field, select **Process Integration Runtime**.
 - b. In the **Service Plan** field, select **api**.
 - c. Enter a name for your instance in the **Instance Name** field and choose **Next**.
5. On the next screen, specify the configuration parameter for the instance.
 - o Enter the following configuration parameter inline in JSON format.

Source Code

```
{
  "roles": [
    "WorkspacePackagesEdit"
  ]
}
```

- o Optional: Choose **Browse** to upload the JSON file containing the configuration parameter.

6. Choose **Next** to review and verify the instance details.

7. Choose **Create Instance**.

8. Choose ... (*Actions*) and then select **Create Service Key** on the service instance created in the previous step.

9. In the **New Service Key** dialog box, enter a value in the **Service key name** field and choose **Create**.

The service key is created successfully. You can view the service key under the **Service Keys** section.

10. Choose ... (*Actions*) and then select **View** on the service key to display the details.

The service key details containing the OAuth client credentials including the clientid, clientsecret, and token url are displayed.

11. Choose **Download** or **Copy JSON** to use the client credentials in the **Destination Configuration** section of your subaccount.

Protecting Artifacts

Restrict write access to some artifacts in Integration Advisor by protecting them.

Prerequisites

Your user in SAP BTP cockpit must have the role `Guidelines.EditProtectedContent` assigned.

Context

If you want to restrict write access to certain artifacts so they can't be edited by every user in your organization, you can add a layer of **protection** to them. Only users with the relevant rights can edit protected artifacts or revoke the protection.

Users with the role `Guidelines.EditProtectedContent` can then protect and unprotect artifacts as well as edit protected artifacts. See also [Results](#).

You can protect the following artifacts in Integration Advisor:

- Message implementation guidelines (MIGs)
- Mapping guidelines (MAGs)
- Global code value mappings (GCVMs)
- Custom codelists

Procedure

1. In Integration Advisor, go to the artifact that you want to protect:

- Message implementation guideline: **MIGs**
- Mapping guideline: **MAGs**
- Global code value mapping: **▶MAGs ▶ Code Value Mappings ▶**
- Custom codelist: **▶Custom Type Systems ▶ Custom Codelists ▶** and expand a codelist entry to see all versions.

2. For the relevant artifact, choose ... *More Options*, then choose **Protect MIG/Protect MAG/Protect GCVM/Protect Codelist**.

You can also open the artifact in the detail view and choose ... *More Options*, then choose **Protect MIG/Protect MAG/Protect GCVM/Protect Codelist**.

3. In the overview and on the detail screen of the artifact, the protection status is now shown through the  *Protected* icon.

To reverse the protection, choose ... *More Options*, and then [Unprotect MIG/Unprotect MAG/Unprotect GCVm/Unprotect Codelist](#).

Results

User Access

Users **with** the role `Guidelines.EditProtectedContent` have the following rights:

- Protect and unprotect artifacts
- Full write access to all protected and unprotected artifacts

Note that if you protect an artifact and then activate it, the activated artifact is protected as well.

For users **without** the role `Guidelines.EditProtectedContent`, the following applies:

- Unprotected artifacts: Full write access
- Protected artifacts: Read and export (runtime artifacts, documentation as PDF/XLS, ZIP); also, you can copy protected artifacts and migrate protected MIGs since this creates new artifacts that are unprotected by default.

Editing, activating, creating new draft versions, and overwriting through import are **not possible**. Drafts created from a protected artifact are protected as well.

Exporting/Importing Protected Artifacts

The following rules apply to the export and import of protected artifacts:

- Protection is a tenant-local setting, so it's not automatically transferred to a different tenant if you export or import an artifact. To protect an artifact in another tenant, you need to explicitly protect it after the first import.
- You can export protected artifacts.
- Users without the required role can import currently protected artifacts that are not updated, but can't import artifacts that need to be updated.

For example, if your unprotected MAG uses a MIG that's protected in the target tenant, you can import your MAG only if the protected MIG hasn't changed. Once you'd need to overwrite the MIG because it was changed, you can't import the MAG.

Settings

Documentation export templates and administrative settings that allow you to customize the appearance of exported MIG and MAG documents and control participation in the proposal service.

The user profile section in the tenant allows you to maintain personalized content such as administrative settings and setting a template for your documentation export. This template is later picked up by the system whenever you export a message implementation guideline (MIG) or a mapping guideline (MAG) in the documentation format.

The administrative settings allow you to opt in or out of the proposal service.

Documentation Export

1. Access your  *Profile* from the left pane in your application.

2. Navigate to the [Documentation Export](#) tab.
3. Choose [Edit](#) to edit your document template. Fill out the sections as follows:

Front Cover Page

Field Name	Description
MIG Document Title	Enter a title that you want to see in your MIG document.
MAG Document Title	Enter a title that you want to see in your MAG document.
Front Cover Line 1	Provide a description for the front cover.
Front Cover Line 2	Provide a description for the front cover.
Splash Image	Use  <i>Browse or drop a file</i> to add a splash image for the front cover.
Logo Image	Use  <i>Browse or drop a file</i> to add a logo image for the front cover.

Content Page

Field Name	Description
Footer Text	Enter the footer text that has to be displayed in the content page.

If you want the administrative data of MIGs and MAGs to be displayed in the documentation, select the checkbox [Show Administrative Data](#).

Back Cover Page

Field Name	Description
Contact Link	Provide your contact details.
Copyright Link	You can add a copyright link to the back cover of your document.
Copyright Text	Provide a copyright text along with the copyright link.

4. Choose [Save](#).

Administrative Settings

Change your administrative settings as follows:

1. Navigate to the [Administrative Settings](#) tab.
2. Choose [Edit](#) and enable or disable the [Enable Proposal Service](#) toggle button.

If you want to revoke consent for data collection related to proposal services, select this option to discontinue data sharing.

→ Remember

Disabling the proposal service has the following consequences:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

- The proposal service stops collecting Integration Advisor data, such as MIG and MAG data.
- MIG and MAG proposals are no longer accessible to this tenant.

The change to this setting takes effect in the future, but it doesn't have any impact on previously collected data.

Enabling the proposal service has the following impact:

- The proposal service starts gathering Integration Advisor data, such as MIGs and MAGs from both past and future sources.
- MIG and MAG proposals are accessible to this tenant.

3. Choose **Save**.

Security Guide for SAP Integration Advisor

The security guide for SAP Integration Advisor provides you an overview of all the security related aspects for the application like process related security information, identity and access management, data storage, protection and privacy, and also what SAP does to ensure the product security standards.

Introduction

Using a cloud-based integration platform imposes dedicated security measures on the software vendor (SAP) that hosts the platform, as well as on those (the customers) who use the platform. This section describes the security-related aspects of the integration platform and shows which measures you can take to protect customer data that is passed through the platform during the execution of an integration scenario.

Customers who use SAP's cloud-based integration platform agree that a significant part of their (and their customers') sensitive data is processed by and stored within an infrastructure owned by SAP.

The core task of an integration platform is to serve as the transit place for messages, which may contain sensitive customer data. First and foremost, these messages must be protected against eavesdropping and unauthorized access.

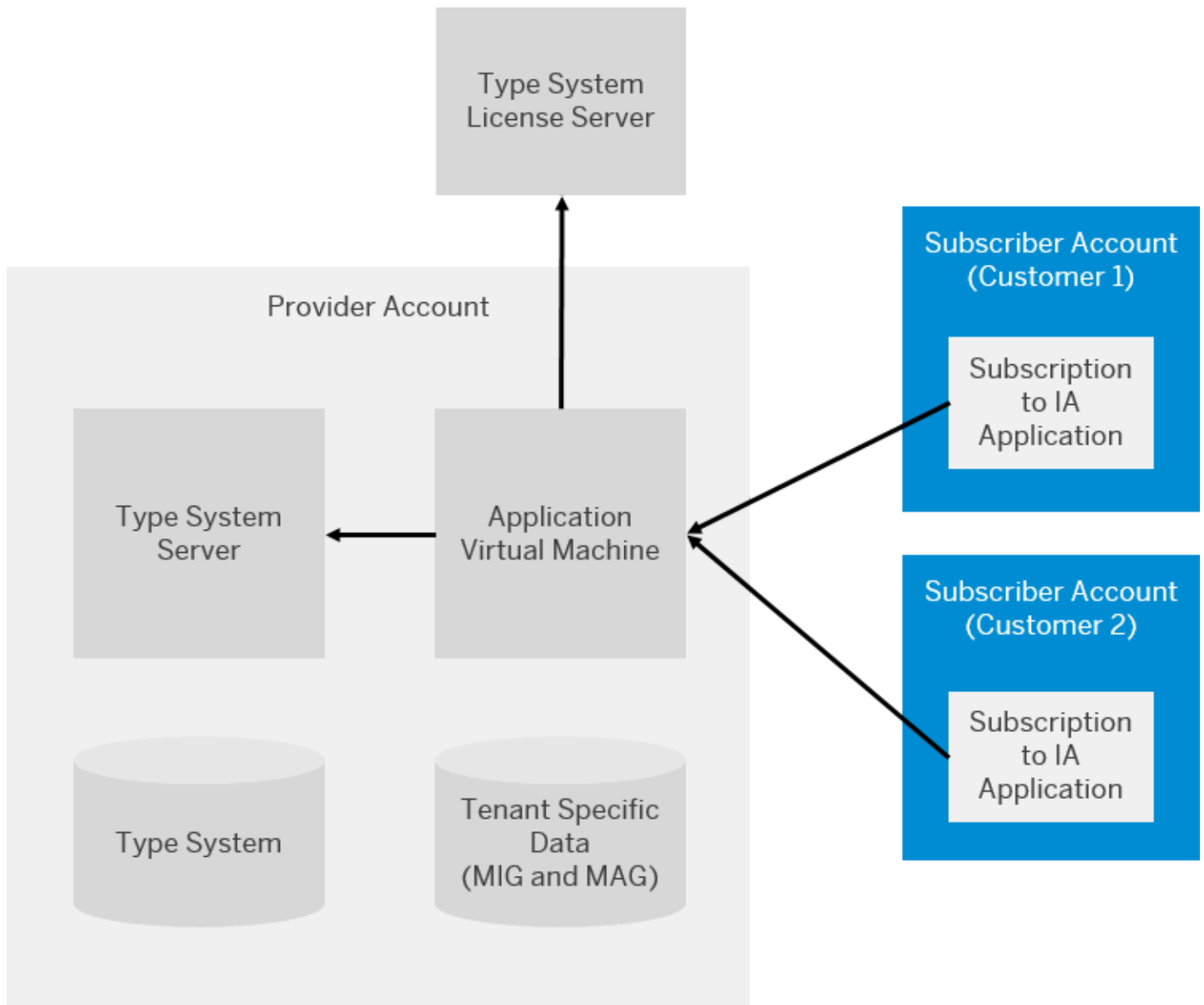
Therefore, the integration platform must fulfill the following main requirements:

- The integration infrastructure provided by SAP is already designed and built in such a way that it meets the highest security standards. It must be guaranteed that the technical system landscape, the communication between the components of the integration platform, and the storage locations of messages are secure.
- The processes related to the usage of the platform meet the highest security standards. This relates to the processes at SAP that are related to the development and upgrade of the platform, the processes related to the provisioning and operation of the customers' virtual environment by SAP, and the customer onboarding process during which customers set up secure connections between their infrastructure and SAP's integration platform.
- Customers have several options to configure how messages are exchanged within an integration scenario so that the involved data is protected at the highest level. When designing integration flows, customers can choose between several options to protect messages by establishing secure communication channels (transport-level security) and by configuring digital encryption and digital signing of messages (message-level security).

Technical Landscape

The technical infrastructure comprises a set of technical components that can communicate with each other and with remote components in a secure way based on HTTPS protocol. In addition, user access to the technical infrastructure is designed in such a way that only users with well-defined permissions can access the different segments.

The following figure shows the high-level technical infrastructure and the main access points and connections. Areas in blue indicate components owned by customers, areas in gray are components that are owned by SAP (in most cases).



Components and Communication Paths

In technical terms, the integration platform is designed as a cluster of virtual machines (nodes) that runs within the SAP cloud. Although all users that connect to the platform through the Internet share the same physical infrastructure, each customer gets access to only one or more dedicated parts of the platform: accounts or tenants.

On each tenant, a clustered, virtual integration runtime is installed (referred to as a tenant cluster). Tenants are strictly isolated from each other with regard to their resources (such as CPU and memory) and to the data that is processed on them. Strictly speaking, each tenant stores data in a separate database schema (this is referred to as tenant isolation).

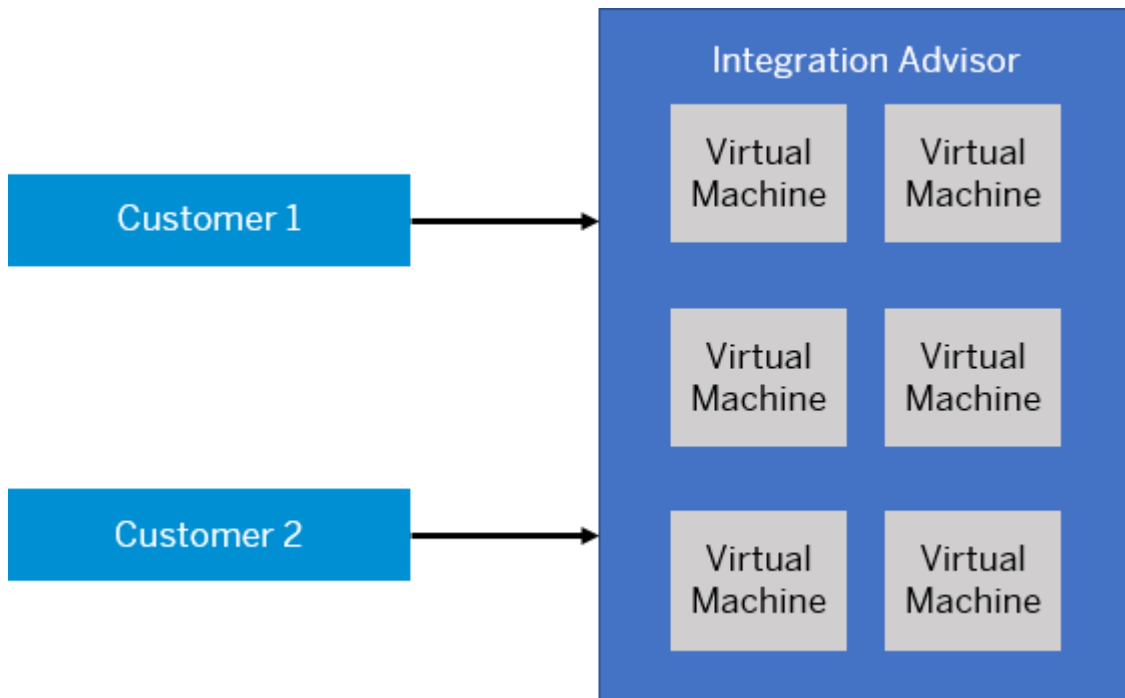
At runtime, the integration platform processes data that is exchanged between the involved participants on a cluster of different virtual machines hosted in the SAP Cloud. A cluster associated with a specific tenant is referred to as a tenant cluster.

i Note

A virtual machine (VM) is a software implementation of a machine that executes a program like a physical machine.

The integration platform is designed in such a way that the involved virtual machines are strictly separated from each other with regard to the related customers. In other words, separate resources (memory, CPU, and file system) of the cloud-based integration platform are allocated to each customer (even though all customers might share the same hardware). In addition, each tenant uses a separate database schema, which guarantees that the data of the different customers is strictly separated. This separation is also referred to as tenant isolation.

The following figure illustrates this concept for two customers communicating with each other:



User Access

In addition to the above mentioned components that interact with each other when messages are being processed and exchanged between the involved systems, additional components come into play when a dialog user accesses the infrastructure (for example, when an administrator accesses monitoring data or when an integration developer deploys an integration artifact).

People with different roles can access the infrastructure – both on the SAP side (as the provider of the integration infrastructure) and on the customer side. Human access points (for dialog users) are:

- Dedicated experts at SAP access the infrastructure to provide a tenant cluster for the customer.
- Experts on the customer side access the infrastructure to design and deploy integration content and to monitor an integration scenario at runtime (integration developers and tenant administrators).

Data Storage

SAP Integration Advisor is mainly a design time tool, which essentially means that the application itself does not have a runtime server where you deploy the artifacts designed using the application. Rather, you deploy the artifacts on SAP Cloud Integration or other applications.

For SAP Cloud Integration security guidelines, please refer to: [Technical Landscape, Neo Environment](#) in SAP Cloud Integration documentation.

Process Related Security Aspects

Processes that are related to the provisioning, update, and usage of the cloud-based integration platform meet the highest security standards.

SAP Integration Advisor is compliant with various SAP-internal technical policies, procedures, directives, guidelines, and product standards. Employees and operators are bound to the SAP code of business conduct and other behavioral security standards such as clean desk and communication.

For example, SAP software is developed in compliance with the SAP Secure Development Lifecycle (SDLC), which helps to implement measures such as test-driven development and threat modeling.

For more information about the certificates held by SAP, see <http://www.sap.com/corporate/en/company/quality.html> .

Identity and Access Management

Access Management

Dialog users who access the platform are authenticated against an identity provider. SAP Identity Service (ID Service) is used by default. SAP ID Service is the central service for the process of managing identities and their lifecycles.

The authentication of inbound calls to the platform depends on the chosen authentication mode. If the client sends a client certificate, the authentication is done by the load balancer. The load balancer terminates the TLS connection; therefore, it checks the client certificate of the calling component against a list of trusted certification authorities (CAs). This certificate is mapped to a user. If basic authentication is configured, the calling entity is checked by the connected identity provider. Besides client certificate authentication, the platform supports basic authentication, OAuth, and Security Assertion Markup Language (SAML).

User Management and Authorizations

Access to dedicated functions of the platform is controlled and protected by authorization checks. Several authorization groups are available to manage the authorizations of dialog users. An authorization group is based on a persona and defines a set of dedicated permissions relating to the tasks that come into play during the lifecycle of an integration project.

Personas and Roles

There are two personas associated with Integration Advisor: Tenant Administrator and Content Developer. Here's a brief description of the responsibilities of each role and the roles required for those personas.

Persona	Responsibilities	Roles Required in Application	Roles Required in SAP BTP Account
Tenant Admin	Manage Consumer accounts and subscriptions Responsibilities: • Enable IA Subscription. • Manage users and roles in consumer account	None	Administrator
Content Developer	Develops and maintains customer content	Typesystem.Read	None

Persona	Responsibilities	Roles Required in Application	Roles Required in SAP BTP Account
	Responsibilities: • Read & consume type systems • Create & maintain Message Guidelines • Create & maintain Mapping Guidelines • Refresh B2B license entitlements.	Guidelines.ReadWrite	
IA Administrator	Develops and maintains customer content Responsibilities: • Read & consume type systems • Create & maintain Message Implementation Guidelines • Create & maintain Mapping Guidelines • Refresh B2B license entitlements • Unlock Message Guidelines & Mapping Guidelines locked by other users	Guidelines.Administrator Guidelines.ReadWrite Typesystem.Read	None
Content Read	View and browse customer content • Read type system libraries, message implementation guidelines, mapping guidelines, and custom type systems • Simulate message implementation guidelines and mapping guidelines • Export artifacts	Typesystem.Read Guidelines.Read	None

Data Protection and Privacy

Various types of customer data are processed by and stored on the integration platform at different times. This data gets the highest level of protection, and SAP takes dedicated measures to guarantee this security level.

General Information

Governments place legal requirements on industry to protect data and privacy. We provide features and functions to help you meet these requirements.

i Note

SAP does not provide legal advice in any form. SAP software supports data protection compliance by providing security features and data protection-relevant functions, such as blocking and deletion of personal data. In many cases, compliance with applicable data protection and privacy laws is not covered by a product feature. Furthermore, this information should not be taken as advice or a recommendation regarding additional features that would be required in specific IT environments. Decisions related to data protection must be made on a case-by-case basis, taking into consideration the given system landscape and the applicable legal requirements. Definitions and other terms used in this documentation are not taken from a specific legal source.

We assume that you have not maintained any data related to an individual in the tools provided by SAP Integration Advisor.

We expect that sensitive personal data can only be included in message payloads. This responsibility lies exclusively with you as the operator of an integration scenario using Integration Advisor and remains your responsibility. If you include sensitive personal data within payloads or message attachments, Integration Advisor may store this information on your behalf. This applies also for data maintained in the tools provided by Integration Advisor, however, data within payloads can be protected by enabling encrypted storage.

The knowledge of sensitive personal data lies exclusively with you and remains your responsibility. The tools of Integration Advisor only use technical users or data without any references to individuals.

User Consent

We assume that software operators, such as SAP customers, collect and store the consent of data subjects, before collecting their personal data. A data privacy specialist can later determine whether data subjects have granted, withdrawn, or denied consent.

Read Access Logging

Read Access Logging is used to monitor and log read access to sensitive data. Data may be categorized as sensitive by law, by external company policy, or by internal company policy. Read Access Logging enables you to answer questions about who accessed particular data within a specified time frame. Such questions could be:

- Who accessed the data of a given business entity, for example, a bank account?
- Who accessed personal information, such as health data?
- Who accessed personal data of accounts or business partners?

A tenant administrator can display audit logs for a tenant using the Monitoring application of the Web UI (under Manage Security in the Audit Log tile).

Additionally, within SAP, audit logs can be displayed for teams in charge of maintaining the virtual cloud environment and to analyze and resolve error situations. Audit logs related to different customers are separated from each other (according to the tenant isolation feature).

More information: [Security-Relevant Logging and Tracing](#).

Information Report

An information report is a collection of data relating to a data subject. A data privacy specialist may be required to provide such a report, or an application may offer a self-service. Integration Advisor assumes that software operators, such as SAP customers, can provide such information.

Safeguard Personal Data

When handling personal data, consider the legislation in the different geographies where your organization operates. After the data has passed the end of purpose, regulations may require you to delete the data. However, additional regulations may require you to keep the data longer. During this period you must block access to the data by unauthorized persons until the end of the retention period, when the data is finally deleted.

Data stored on the Integration Advisor platform is only stored for a limited time period (referred to as retention time).

For more information on the retention times for the various kinds of data stored by SAP Cloud Integration, see [Specific Data Assets](#).

Specific Data Assets

There are four data assets that are stored by Integration Advisor:

- Message Implementation Guidelines
- Mapping Guidelines
- Custom Messages
- Codelists

These assets are stored by the application till the contract for usage of application is valid. After the contract ends, the assets are deleted.

Security-Relevant Logging and Tracing

Audit logs allow administrators at SAP or the tenant administrator to monitor events such as data read accesses or system configuration changes. This enables administrators to take adequate measures to prevent malicious usage of the system.

A tenant administrator can request audit logs for his or her tenant using the Monitoring application of the Web UI (under Access Logs in the Audit Log and System Log Files tile).

The Audit Log section allows you to monitor changes to the tenant data. For example, deletion of message implementation guidelines and mapping guidelines

An audit log records the following type of information for each logged event:

- Type of the event
- Date and time of the event
- Initiator of the event (user)
- Source of the event (IP address of the source that issued the action)

To view audit logs, the tenant administrator needs to have assigned the roles `IntegrationOperationServer.read` and `AuditLog.Read`.

Data Collection for Knowledge Base

Information about the data that is collected by the knowledge base for generating proposals.

To generate and provide better proposals, SAP Integration Advisor collects some data respective to each artifact for its knowledge base. To know more about the contract information, refer to the section **SAP Integration Suite** here: [SAP Cloud Service Description Guide](#).

i Note

To avoid the data collection for proposal service, you can opt out of proposal service completely. You can use this opt-out feature if you don't want to use the proposal service and don't want your content to be collected. To know more, see [Settings](#).

The sections below pertaining to each artifact mention the details of the data which will and which will not be a part of the knowledge base.

Message Implementation Guidelines

The following information of MIGs will not be part of the knowledge base:

- Textual documentation (names, definitions, notes, summaries) of all components, such as messages, nodes, code values, etc.
- MIG codelists
- Example values
- User-defined XSD Patterns
- User-defined XSD Asserts
- User review information
- Administrative data such as Created By, Modified By

All other information including the following will be a part of the knowledge base:

- Node properties, such as cardinality and length
- Selected code values (without textual documentation)
- Qualification details

Mapping Guidelines

The following information of MAGs will not be part of the knowledge base:

- Textual documentation (names, definitions, notes, summaries) of all components, such as mapping elements, global parameters, shared code, code values, etc.
- XSLT code of functions and XPath conditions explicitly marked as confidential
- XSLT code of functions and XPath conditions containing global parameters

- Status and status comments
- Administrative data such as Created By, Modified By

All other information including the following will be a part of the knowledge base:

- Mapping lines between source and target nodes
- Constants
- Code value mappings
- XSLT code of functions and XPath conditions that are not marked confidential
- XSLT code of shared code

Custom Messages

The following information of custom messages will not be part of the knowledge base:

- Textual documentation (names, definitions, notes, summaries) of all components, such as messages, nodes, code values, etc.
- Administrative data such as Created By, Modified By

All other information including the following will be a part of the knowledge base:

- Identification of message and codelists
- Identification of all nodes and types
- Identification of code values
- Node properties, such as cardinality and length

Custom Codelists

The following information of custom codelists will not be part of the knowledge base:

- Textual documentation (names, definitions, notes) of all components, such as codelist and code values
- Administrative data such as Created By, Modified By

All other information including the following will be a part of the knowledge base:

- Identification of the custom codelist
- Identification of code values

Auditing and Logging Information

Here you can find a list of the security events that are logged by Integration Advisor.

The audit log retrieval API allows you to retrieve the audit logs for your SAP Integration Advisor tenant in both Neo and Cloud Foundry environment.

The audit log retrieval follows the OData 4.0 standard, providing the audit log results as OData with collection of JSON entities.

i Note

The examples contained in the table are taken from the usage of SAP Integration Advisor in the Cloud Foundry environment. Unless otherwise shown, the keywords used to identify an event are identical in both environments.

Security Events Written in Audit Logs

Event grouping	What events are logged	How to identify related log events
Message implementation guideline (MIG)	• Create MIG entity • Copy MIG entity • Create draft MIG version • Update MIG version • Activate MIG version • Delete MIG entity • Delete MIG version	Message implementation guideline • created successfully/creation failed • copied from source ... successfully/copy from source ... failed • draft created from source ... successfully/draft creation from source ... failed • updated successfully/update failed • activated successfully/activation failed • deleted successfully/deletion failed Example: When deleting a MIG successfully, the following event is written: ⌵ Output Code Data modification message. Attribute with name "Message Implementation Guideline" deleted successfully. The attribute is a part of an object with type "Message Implementation Guideline" with objectGuid abcd1234 deleted successfully.
Mapping Guideline (MAG)	• Create MAG entity • Copy MAG entity • Create draft MAG version • Update MAG version • Activate MAG version • Delete MAG entity • Delete MAG version	Mapping guideline • created successfully/creation failed • copied from source ... successfully/copy from source ... failed • draft created from source ... successfully/draft creation from source ... failed • updated successfully/update failed • activated successfully/activation failed • deleted successfully/deletion failed

Event grouping	What events are logged	How to identify related log events
Pre-Transformation Script (PTS)	• Create PTS entity • Update PTS entity • Delete PTS entity	Pre-transformation script • created successfully/creation failed • updated successfully/update failed • deleted successfully/deletion failed
Customer message	• Create customer message • Update customer message • Delete customer message	Custom message • created successfully/creation failed • updated successfully/update failed • deleted successfully/deletion failed
Custom codelist	• Create custom codelist • Create draft custom codelist • Update custom codelist • Activate custom codelist • Delete custom codelist entity or version	Custom codelist • created successfully/creation failed • draft created from source ... successfully/draft creation from source ... failed • updated successfully/update failed • activated successfully/activation failed • deleted successfully/deletion failed
Value mapping	• Create global code value mapping • Copy global code value mapping entity • Create draft global code value	Code value mapping • created successfully/creation failed • copied from source ... successfully/copy from source ... failed • draft created from source ... successfully/draft creation from source ... failed • updated successfully/update failed • activated successfully/activation failed • deleted successfully/deletion failed

Event grouping	What events are logged	How to identify related log events
	mapping version - Update global code value mapping version - Activate global code value mapping version - Delete global code value mapping entity - Delete global code value mapping version	
Licenses	- Create new license - Delete license - Refresh license list	Customer license - updated successfully/refresh failed - created successfully/creation failed - deleted successfully/deletion failed License list - refreshed successfully/refresh failed
Artifact	- Import artifacts - Export artifacts	Artifact - imported successfully/import failed - exported successfully/export failed
Authorization and Security	- User authorization failure - License validation failure - Entity access validation failure - Trial license expiration	Authorization - User authorization for resource ... failed License - License validation for TypeSystem ... failed - Entity access validation for TypeSystem ... failed - Trial license expired for TypeSystem ... Artifact - unlocked successfully/unlocked failed

Event grouping	What events are logged	How to identify related log events
	• Unlock artifact (admin only) • Virus scan failure	Virus scan • Virus scan failed for resource ...

To view the audit logging in Neo environment, see [Audit Logging in the Neo Environment](#)

To view the audit logging in Cloud Foundry environment, see [Audit Logging in the Cloud Foundry Environment](#)

Getting Support for Integration Advisor

Create tickets for reporting incidents and errors.

Reporting an Incident

If you find an issue with Integration Advisor, you can report incidents or errors through the [SAP Support Portal](#). For more information, see [Product Support](#).

Please use the following component for your incident:

Ticket Component

Component Name	Component Description
LOD-HCI-PI-SUG	Integration Advisor: Intelligent Mapping Tool

When you submit the incident, we recommend including the following information:

- Landscape information (Canary, EU10, US10)
- The URL of the page where the incident or error occurs
- The steps or clicks used to replicate the error
- Screen grabs, videos, or the code being inputted